# 5th Applied Active Inference Symposium (Part 2, Nov 13, 2025) ~ LIVE

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Welcome back. It's day two of the fifth applied active inference symposium, November 13th, 2025. And the first session will be the future of active inference.jl with Peter, Jonathan, and Samuel. So Peter, thank you to you. >> Thanks, Daniel. All right, back stay. Please mute the YouTube. >> Sorry, I have some sound problems here. >> Thanks. One second. Okay, here we go again. Can you see the screen? Otherwise, I'll just reshare it. And uh there we are. Thank you, Daniel. Sorry for the the sound issues there. I had another tab open that I shouldn't have had. Um, also thanks for organizing this symposium which is incredible the work that's been put into it and worth being grateful for. So just that's worth seeing probably a lot of times during this symposium again and again. Hi everyone. So today uh I and Sam and Yonatan will present on the ongoing and upcoming developments of active inference.jl which is Julia software tool for uh running and applying active inference models to data or in simulation. Um, and I'll start out with just a background and a conceptual thing and then Yonan will take over and Sam will take over and they will go through code and give a demonstration of how to use the package in its newest version that's not yet released but and there's a link in the YouTube channel if someone wants to follow along the code to a GitHub repo which can be cloned and used with this new experimental version of the package. say again okay so um I'll briefly just start with background we're conceptual conceptualizing active inference as a kind of cognitive model and what a cognitive model means is just that it's a formal model that produces behavior of some kind. So all of artificial intelligence etc can be conceptualized as cognitive models and this is broader framework where you have some agent and you have some environment and the environment produces observations to the agent. The agent produces actions that affect the environment. And then the agent has a cognitive state, we call it a theta here, that changes over time. That could be its beliefs about the environment. And um the agent has some cognitive parameters. That could be for example learning rates or its prior that govern how its cognitive states change and govern how it produces actions. And uh in general cognitive modeling and therefore also active inference has three kinds of uses that I know of. One is theoretical modeling. So that's when you run simulations to see what a given agent would do given different parameters. So there you know the theta here and you simulate the environment and you want to see what happens and active inference is full of these. So all this simulation work where you going to show what happens if active infants fish swim together etc could be considered theoretical models and much of the backbone of the field is built on this. Then there's applied cognitive modeling and that's where instead of having the parameters you have the observations and the actions. So you have the behavior for example because you recorded it from a human in an experiment and you want to reconstruct the behavior that govern the parameters that govern that behavior. You want to find out what was the generative model or what was the prior so what was the learning rate of a given participant that's what's used in computational psychiatry but also much beyond that and finally one uses artificial intelligence where again you have the parameters but you don't simulate the environment you have real environment for example you know burning house or whatever and then you want your artifact your tool to generate the right kinds of behavior for example a vacuum vacum cleaner that supposed to move in some ways and not other ways. You can use active inference in all three ways. Our lab specializes in applied cognitive modeling. So the second of these three and last year we focused quite a lot on how to use active inference for that. But this year we will not h because instead we will focus on the new structure of the package and how that makes it more flexible for the kinds of active inference models you can implement whether you want to use them for theoretical or applied modeling or for artificial intelligence. And I just want to briefly mention that there are many other modeling frameworks than active inference. You know, frameworks where you optimize for expected value, frameworks where you uh run symbolic code operations, frameworks where you think the mind is kind of a dynamic system, connectionism and later been renamed to deep learning um control theory etc. And there's also a bigger family of Beijian mind models where you think that the mind uses Beijian inference. So we should have models that use bijian inference to describe the human mind. An active inference which is a kind of variational basian predictive processing model is a type of basian mind model that does three things. So it commits to specifically using variational base which is one way of approximating basian inference. It usually uses message passing schemes to do this and I'll get back to that. And notably it um does not only model perception but also action as basian inference just to place us in the larger field that we live in and you can combine models with these other field these other frameworks if you want as well. Actual inference can be combined with them. So uh as we know basian inference you have some data some observations and you have some parameters or some hidden states of the environment you have a model m and what you want to find so the generative model is just how the data and the parameters usually co- occur could be just some um joint probability distribution can be quite complex also and as usual what we want is we want to find the parameters that are most likely given the data we get and we have base theorem and we all know this. Um crucially we can't calculate the bottom part because we have to the bottom here to the bottom right. Um to calculate that you have to integrate across all possible parameters and if you have parameters that are continuous you have uh so full continuum for each parameter and if you have many parameters hidden states that becomes impossible to do that means that usually you approximate basian inference instead there are generally three methods so one is sampling and I mentioned these because we can actually use them with active inference if we wanted to we'll show that later. So, it's worth mentioning that they exist. In sampling, you just pick values randomly stocastically with um an algorithm that ensures that you're going to pick a value proportionate an amount of times proportionate to how probable it is in the posterior. And there are many ways to try and do this. They can fail. They can succeed. But the main advantage is that they uh can get any shape of posterior in principle. The main disadvantage is that they take a long time and they're not so mathematically beautiful because you don't use equations to get it analytically. Another one is simulation based inference where broadly you generate lots of behavior or lots of data with a model and then you do something to get the posterior out of that. And [clears throat] so one of the classic and the most used method here is amortization where you simulate data with your model and given different parameters and you train a deep network to revert that process and infur parameters given the data. And finally in variational inference you minimize a free energy to uh get the best to get the posterior and notably this is not anything new. So it's been used in machine learning and in statistics for a long time. What's new in active inflence is that is used as a model of human cognition in general and it's combined with action selection as well. Again I'll just briefly remind us that here you have the true posterior which we cannot calculate. We make up entirely arbitrary and approximate posterior Q and then we can calculate the free energy which is just the KL divergence between the approximate posterior and the generative model on the right. We have both of those. So we can calculate it. And then of course the magic is that if we find the approximate posterior with the lowest free energy, we also find the approximate posterior that's closest to the true posterior. And we see this so that we can rewrite the free energy into a divergence and an evidence term where the divergence is the difference between the approximate and the true posterior. The evidence is the surprise. And notably, if you change only your posterior, that only affects the left term here. And that means that you can only reduce the divergence by reducing the um by choosing the approximate posterior that minimizes the freity. That's all very exciting. Hallelujah. Now, uh notably, that does not tell us how to actually find the approximate posterior with the lowest free energy. And there are different ways of doing this. that are all variational based and they're all free energy minimization based. Um and that's you know not necessarily an easy task especially if the the parameters of Q so if you have many um if you have high dimensional posterior because you have many states you're doing inference over it gets quite complex and the way they relate to the free energy might not be very obvious either. So in general, I know of three ways that people find the posterior that has the lowest free energy. And one is just gradient descent. So you just wiggle your posterior around until you find the one with the lowest. And um that requires calculating the gradient. So how the free energy changes when you changes your when you change your approximate posterior. And if you have a complex model, that's usually difficult. But there are, as I'll see, methods to do that even for complex models. Some cases when you're lucky, you can also just analytically mathematically find in one equation the uh approximate That's very fast. That's very nice, but usually not easy. And finally, there are message passing methods, and those are the ones we know the best. So there you do an additional approximation where you take a gent model that's taken from the par al paper. You take a gent model and you split it up. So for each of the components, each of the hidden states or parameters in your model, you split it up and you update your beliefs about them one at a time. Usually you do what's called a mean field approximation where you assume that they're completely independent. But you can do this in different ways. So there's also these basic free energy for example approaches where you split them up into pairs instead of single. So you can retain that two parts of your posterior might uh correlate but they're still assumed to be independent from the rest. Then you update them one at a time and uh because you've split them up and because this is an approximation that won't always work. So you keep doing that until they converge. That's called coordinate ascent variational inference using message passing. And that's the standard method in active inference currently. And it bears just showing briefly the update equations for each of the components. If your generative model specifically is a discrete state transition PO DP, you can have other generative models. But in that case, you you combine two things. You combine a prediction which is your previous belief about the state of the environment plus or multiply it with the transition probabilities. So you take what you thought the world was before you let it go through your model of how the world changes and you now have your prediction and then you add your evidence from the observation. So in almost all message passing schemes you have this prediction plus evidence uh approach and that's why uh we call it predictive processing because when you do message passing you get something often you get something that looks like prediction output. Okay. Why spend all this time reminding us that there are what message passing is but also that there are other alternatives. I'll get to that. Proistic programming languages are [snorts] ppls for short exist. So that's a general technology where you have some system where you can specify all sorts of generative models and then they for free give you methods for doing bijian inference and prediction and there exists a bunch of these. So stan is famous and fur.net for example is also widely used and um they offer different kinds of methods. So Stan for example focuses on sampling but can also do variational inference by gradient descent. Infer.net focuses specifically on message passing. Um and the whole point here is that you're not limited to one type of generative model. You can just write depending on the ppl any kind of gen model and um and get your um updates from that without having to hand write all of the update equations. I spoke to some people from one PPL who said why not would you handr write your update equations all the time which we do. So in Julia there are multiple ppls and they can all be used to do the bijian inference that active inference.jl needs. The classic one is the active inference. Pumpdp. That's the in-house that we develop and that's the one that's identical to the one used in uh PIMVP and in the MTL code and so that's a PPL in the sense that you can specify many kinds of journal models but it's also limited because you can only specify discrete publics so where you think where you get discrete observations you think the world consists of discrete states that change in ch state transitions and that you have discrete actions that affect those state transitions. We're also developing the HDF which I won't spend much time except to say that that's very very similar except that the world is not made of discrete states with discrete state transitions but gian random walls here you also do message passing and uh that's different update equations but exactly the same logic and they can al also be combined which we're working on but there are also other more flexible ppls. So RX infer is one which will do message passing specifically as mentioned before. So that's very fast but it's also an approximation of an approximation of Beijian inference. So that's why some people prefer others. You can also use Turing or Gen which focus more on sampling or gradient descent variation inference. So they are generally more flexible in the types of models you can write in them but are generally also much slower and they don't have um uh as elegant mathematical uh singlestep updates. That's why they're slower and are not they can be but they don't have to be free energy based. And essentially what we want to show the rest of this presentation is that instead of being confined to have to use the discrete PMDP method, it's very nice as long as you are in a discrete environment and as long as you think your actions can only change state transitions and not other stuff for example. It's very nice but uh we want to allow active inference.jl the jail to use any of these options as back ends and therefore to allow people to use a wider range of generative models for a wider range of scenarios but still doing full-on active inference. So the last part here I will just uh briefly remind us of two different ways that active inference like models can be implemented. One is let's call it planning as inference and that's the one where you just put in your possible actions or policies as random variables in your model and you do standard Beijian inference could be variational could be anything to find those actions and so this is a screenshot from the tutorial on RX infer very very nice tutorial where they have a discrete PDP so you can see this A matrix and B matrix as we know it and we have a previous state um and then they write out that generative model. So we have a current state which depends on the previous state and the B matrix and a previous action. And we have our current observation which depends on our current state and our A matrix. So this is specifying the same stuff we actually do in PN DP normally. And then we have a planning part where you go through uh each time step in your imagined future and you um predict forward what would happen. That's what happens here in this discrete decision from previous previous state uh down in the for loop etc. And then finally you have your goal prior or your preference prior in the end where you in addition to predicting what will happen you also have a fixed distribution. Here it's only for the final state. So here the agent actually only has preferences for the very final state. You run this, you can use a sampler version inference and you get out actions that are likely to make that goal prior come true and given your beliefs about the current state and your beliefs about A and B etc. Here there's no parameter learning but you can easily add that etc. And it's just to show you could make many changes to this. It's very flexible. That's the whole point. What this method h does not give us though is that it does not give us the information gain that is in the expected free energy in active inference. Uh so we get the finding [snorts] actions that give us our goal private but we don't give it the information we need and the arson fair people if anyone were at IY are working at ways to extend this. So to have sneaky clever ways of having prior about for example the ambiguity that will kind of get get you the expected free energy anyways while keeping everything local and using message passing and you could do this with other things too. I will contrast this to I'll call it modular planning for now but this is the standard way that active inference is implemented in uh active inference.jl PMDP and in MATLAB where you really have three components to your model. You have your generative model the agent general model of the environment. Then you have a perceptual part where you take that general model and you form beliefs about the environment and make predictions about the next time step in the environment. And this is the that I described before. This is the place where we minimize variation and um then we have a third part where you do your planning and here inference.jl jail etc. You do for example a tree search where you go through all possible all sometimes you go through all possible combinations of actions in the future sometimes you do it in a more constrained way in order to find the um set of actions that have the lowest expected free energy. Um and what's important here is that the planning part is separate from the perception part. So in cognitive modeling that's often called a perceptual model and a response model or an action model and that makes the planning part not local. So it subsumes the perception part. So when you are and especially if you're doing sophisticated inference you're running your perceptual inference forward in time many times. So you do your variational inference many times into the future as you plan. Um the point here is that they're modular. So you can have one generative model say a discrete PM DP with ABC AB matrices and then you can use many kinds of perceptual inference methods with that. So you could use RX in message passing. You could use our housemade message passing. You could use Turing's sampler with the same generator model. And similarly given a general model and a perceptual process you can also uh with different general models and or different perceptual processes you can combine that with the same planning method because in the planning you're just seeing uh if I do that what will I think in the future if you're doing doing sophisticated inference and you could that step what will I think in the future that can be done with sampling with gradient descent based variation inference or with message passing and it can also o work with other kinds of genome models than the discrete pumpd. I'll note that um standard for the planning is to do a tree search and tree search explode if you go very far into the future. So uh they get competition very costly. So you have to um find ways to prune etc. There's a whole field of finding good ways to search through that um large space of possible policies efficiently. But one other option is actually to make the planning stage itself an influence call. So that itself is a model which we want to allow in the future. Um so you have a generative model inside of which there is bas inference on a different generative model. You have a generative model which is a planning and that contains the perceptual part. Um that of course requires us to nest ppls and that's something we're working on. Um but just want to point out that that's usually what's called metacognition. So you have a model of your model and sophisticated inference. That's exactly what you do. You have a model of your future beliefs. If your own future believes in the future and I think that qualifies calling sophisticated inference metacognition and given that it's specifically used for planning one might call it metacognitive planning just a different word for the same thing but one that highlights what's happening you're planning your own beliefs in the future and to do this you need to run inference within your inference and therefore you need to nest ppls and that's what we're building to allow so The excitement here, the hope is that um you can plug in for example RX infer or Turing as a back end so that you can do other kinds of JC models and that we can allow for nesting them and we're building the infrastructure for that. And in the following we'll show you an example where we use uh two different inference methods with the same generative model and the same planning structure. Yeah. Okay. So that's it for me. That's just the background and just to make the conceptual uh point that we'll build on from here. Yonan will take over control of the PowerPoint and in a second we'll show you some code as well. Sure. Thanks Peter. Um yes. So before we actually show you some code, I'm just going to go through conceptually how the new structure is to be understood. um and what kind of needs to be done if one wants to create a custom struct for either transform model uh perceptual process or action process. So Peter if you would be so kind to go to the next slide. Yes. So here we basically see the same figure uh as we saw before where we divide this active inference model up into three main parts where we have a generative model our beliefs about hidden states and how they transition over time and how they generate observations. We have the perceptual process. How do we then infer these hidden states? Um and then we have the action process. Um how we evaluate policies um and then marginalize them into actions. Um so the idea behind this is that sort of like Lego bricks you can kind of uh interchange these parts and so you can exchange as Peter said the perceptual process and then it still works with the same generative model uh and the same action process. So it's it's kind of becomes modular. Um and if you go to the next slide, Peter. Yes. And then exactly what is required uh to make such an extension. Um and so I'll just uh briefly introduce how it's kind of conceptually implemented in the code in the active inference.jl. Um and here we have again these three main parts where you can see the current implementations that we have is the discrete state space pound piece for the generative model.

We have the coordinated ascent variational inference for the perceptual process. And this is going to be uh the part that Sam he's going to show how you can uh create a custom uh perceptual process. So you can use other methods to do the state inference. Um and then we have uh the uh the action process where our default is the just a policy tree search with evaluation over the expected free energies. Um yes, you can go to the next one. Yes. And so here we can kind of see the overall structure of the package and it's basically based on an active inference core module which kind of contains a template for the strcts and functions. Um so the the functions and strcts in the active inference core module are not really used but they are made so that they can be extended into other modules. So as you can see here the module that we have currently implemented is the discrete PP. Um and I'll show you briefly uh shortly in in the code how you can actually extend uh these uh generator models and the the functions like the perception planning and this kind of stuff into the discrete pond so that you can actually create an active inference loop. um with uh the discrete pom dp generative model and with the perceptual process and action process that we have. Um and then you can extend this by creating other modules as well which we will in the future and which is going to be uh where you would add the custom uh strcts. And if you go to the next one again. Yes. And so basically here the idea is that we use uh abstract types uh where we're saying that for a generative model if you want to add another generative model then you have to make it a subtype of this abstract generative model. Um and basically what we allow for here is that in the generative model you can also specify what type of what sort of action or observational state. If you go to the next one, Peter, uh just briefly just showing here that you can for example have a a palm DP uh that can be both continue where you have continuous action or continuous observations and states, but you can also have a palm DP where you have discrete actions and discrete states. In this way we can use um we can use uh this to um to uh for the different functions to um the word is escaping me where we have different functions that multiple dispatch uh on uh the different types even though it's the same uh could be the same pp with the given the different sort of actions or observations we can do multiple dispatches in this way it becomes very modular and very easy um to to work around. If you go to the next one. Yes. And the same one here goes for the perceptual process. If you want to add a perceptual process, then you have to make it a subtype of the abstract perceptual process. And if you go to the next one and the same for the action process. And so if we combine all of these uh if you go to the next one again uh then we have our active inference model which uh is composed of these three parts of the model. the perceptual process um and the action process and and here you can see like it's interchangeable in the sense that you can just and this is what Sam is also going to show you that you can have the same generative model and the same action process but then you switch out the perceptual process with a custom perceptual process instead of the uh kabi perceptual process that we're implementing currently and just a brief note on the action models here the action models is a sister package uh to the active inference uh package which is for doing model fitting to cognitive models uh of which active inference here is one of those models that we're planning to implement and to do model fitting on um or allow model fitting. If you go to the next one yes and so there are four main functions that one needs to be aware of if one wants to implement implement uh a custom uh strct. This is the perception of planning uh selection and stored beliefs. And this is I'm going to show the uh the active inference loop and and you'll see why one has to at least either modify or make sure that these four functions will work with whatever customizable struct that you are planning on implementing. So if you go to the the next one here, we showcase also like the core module, how you're supposed to extend from the core module into whatever module you're working on. So for example here when we're working in the discrete pom DP module we're extending from the active inference core module which has the perception function this the perception function from the active in core module is never going to be used because it only contains or dispatches some abstract uh types uh where it will then give you a neuron you need to create a concrete type and this is what you need to do. you need to create concrete types um for uh each of these uh parts and then as you can see here in the PMDP uh module you then extend it from the active inference where you then have the active inference model where each the generative model the kavi which is the perceptual process and action process are all prefixed with this discrete pom dp and then you have the multiple dispatch uh on the specific uh type of of active inference model that you have and if you go to the next one and you can see this is the name uh for the planning uh so the the perception one creates the posterior state the planning creates a posterior policies um and again we have the the multiple dispatch on the discrete prompp and it's an extension from the active inference core module if you go to the next we have the selection which basically marginalizes over the actions so it actually creates the act or the distribution over actions that we then sample from when we want a concrete action um and if you go to the next one Yes. And this is uh store I believe this is just a like a utility function where we then take all the output from the previous three functions and then we put it into the uh the active inference model for use in the next time step. Um but basically all of these four uh functions need to work uh with whatever customizable or custom uh struct that you're making um and needs to be kept into in mind. And yes, and the next one. Yes. And here you can see this is kind of like the higher level uh active inference loop that is in the um the active inference core um module. Um and and basically because we have the multiple dispatch, this loop can work with any active inference model as long as each of the functions perception, planning, selection, and store beliefs have a multiple dispatch uh on the active inference model type and subtypes that you're using. Um and Sam will show how this is the case. And then the next uh one is just briefly showcasing here that in the perception action loop you have the active inference functions uh in active inference function which gives you probability over actions and this probability you can then use the Julia distributions package to then sample an action that you then give to an environment gives you an observation and and in this sense you have the perception action loop. So it should be high level um high level it looks uh it should be very clean and um now Sam he will show you how this is implemented uh with a walkthrough of a notebook and in the next slide you will see a link to a GitHub page where you can clone it and uh should be able to follow along. All right I give it over to you Sam. the the link is also in the uh YouTube chat and I'll keep an eye on the chat meanwhile in case anyone has questions. And finally, do note I'll just add this. Do note that creating your own alternative version of active inference models is maybe a bit of a power user thing that is good to be able to do. But we also retain the old uh interface where you can do a classic uh active infl sorry no worries. Okay. So let me just share my screen quickly. Okay. Can you at least confirm that you can see it well? >> Yes. So we can see it. Thank you. Awesome. Okay. So uh there's this um little notebook that we have prepared for you um to show you basically like the current implementation. We'll be using the development branch because it has not been um published yet as a as a new version. Um but basically I don't know how familiar people are with Julia but in the beginning there are some formalities and we need to go through like actually the current environment and we have prepared some utility functions just for plotting and um the environment and then we bring our dependencies to um to the scope and so in this example we'll be using a simple multi-armed bandit task um which you can imagine as a slot machine with three arms and um each arm is associated with either reward or loss with some probability. And so in this example um we'll initialize the environment with arm one being the best. So you would have the probability of 0.9 of receiving reward and the second best would be arm two and then arm three would be the worst with only 0.1 probability of receiving a reward otherwise loss. And after each 25th trial, these probab pro probabilistic contingencies will switch. Um, and we will make our active inference agent initially uninformed about the what are the reward probabilities and we'll let let it play the game for 100 trials and we'll let it learn these um these contingencies while trying to stick to the preferred outcomes which are the rewards. In this case, this is just a visual of how it looks. um over 100 trials and the probabilistic switches between arms after each 25th trial. So now we're ready to set up our um generative model. We'll just start with specifying some structural components. So um we have three three hidden states where each hidden state um corresponds to an arm in the in the bandit. We have two observations. So ward or loss then controls those are actions. So we have three actions. Each action basically just means pulls pull arm one, pull arm two, pull arm three. Um, and then we're going to use a policy length of one. So we only we're only planning one move ahead when making decisions. And we're just going to pass all this to create matrix templates. Um, and we're going to tell it to populated with uniform prop distributions. Uh, and this is going to give us the A, B, C, and D, uh, objects. And now we can populate it. So uh for the observation likelihood uh this will give us a 2 * 3 matrix where each column corresponds to a hidden state or an arm in this case and rows correspond to observations. Um and we're just going to leave it uh populated with uniform probability distributions because we want it we want the agent to learn these probabilistic contingencies. And we specify transition likelihood. So we have um three B matrices in this case each B matrix corresponds to one of the three actions. So basically telling the agent that one is in the arm two for example and it pulls the arm one and it's most likely to actually end up in that uh state. And we make it a bit stocastic here just to introduce some uncertainty um to it. Right? So we have our B objects uh created u we've created above and we just take like each slice and populated with with our matrices um each corresponding to one of the three actions and the two last things we need to um specify are the the C vector and the D vector. So the C vector is the preference uh in this case just a two element vector the first element um encodes the preference over rewards and the second one uh the preference over losses. So here I basically just saying that prefer awards over losses and for the priority initial hidden states uh we're going to tell it that um it's going to start in the in the arm one or at the arm one. All right. So now we have the the genative model and we can move on to the definition of the active inference model and so I'm going to start with the default implementation. So sort of the out of the box way of specifying the generative model or the active inference model rather. Um, so we are still keeping the the old way of doing that where we're just passing the individual components of the trainer to the model to the have model constructor and then we don't really need to care about anything else. Uh, we're basically ready to to use our model. Um, however, um, we would like to focus on here is the new modular extensible approach. And so um here we need to specify the individual parts of the active inference model uh separately. So here we'll create the generative model object by passing it by passing the the individual generative model components to it. And this is the default um u discrete pom p uh join to model. Um then we'll need to specify our perceptual process and as Peter and Jonathan mentioned in in our package uh the perceptual process would consist of two main parts which is the the algorithm to perform inference about hidden states. So here uh in the beginning I'll just use coordinate asen variational inference from the discrete comp module which is implemented by default and then we need an algorithm to to do learning. So which basically means to update the parameters of the generative model uh itself. And here by default we are just using the dish categorical model. Um we're going to learn the the a matrix. We're going to pass some parameters to it like the concentration parameter or the forgetting rate because there's some volatility in our environment to make the agent a little more uh flexible. Um and then lastly we specify our action process. So the action process here is really just an uh planning algorithm used to evaluate our policies and this can be standard inference. This can be sophisticated inference tree based methods. Uh by default in the discrete PO DP module here we're using standard uh inference without graph. uh and in this case we're telling it to use the policy cylind which we we have defined above and then we're going to use all the three terms uh to calculate the expected free energy. So stage information gain the parameter information gain and the utility term. Um and once we have all these three parts uh ready, we can just collect them all three and give it to the AF model constructor. And this will generate the model object uh for us. And then we'll just need a couple of more things. We'll need to initialize our environment. So here we have some pre-made environment. Um and we're going to tell it that the best arm has the reward probability of 0.9. And then we're going to switch the contingencies after every 25th trial and some random seed. And then um before we actually simulate, we're just going to set our initial observation, our initial actions, initial action, and we're going to simulate for 100 trials. And here I just have some empty vectors so we can store the beliefs of an agent during simulation for later for plotting. Now we get to the action perception loop. So the main function here is the active inference uh function which takes the model observation and action as arguments and it outputs the uh probability distribution over actions. Now because it's a categorical probability distribution we sample an action from it and then we pass it on to the environment which gives us the observation for the next time step and it goes on like this for 100 trials. And then I'm also saving the first row uh of the of the A matrix and the posterior states later for for plotting here. Right? And so how does this look then in practice? So in this plot the the first part the green dots are the reward observations and the the red dots are the uh loss observations. In the the second one we have the first uh row of the A matrix. Right? So this corresponds to the probability of um receiving a reward for each of the three arms as the agent was learning it throughout the course of 100 trials and then posterior over states which is a little noisy here because we introduced some some stoasticity to it in the transition transition model. So um this was the default out of the box way to do it. Um and now I'm going to show you how to create a custom perceptual process. So here we're just going to import two things from the package that we'll need and that's the type of abstract perceptual process and the learn a type from from discrete pom dp module. And here we're going to create we're just going to call it a container really because um this mutable like it's um like its task is really just to store all the uh important variables that we will need for a perceptual process. Um and importantly uh we're going to make it a subtype of abstract perceptual process so that our functions or the functions of our package can accept it. Right? So we have our inference function which we are going to defi define uh later. We have our aarning uh algorithm and then it stores of course posterior over states posterior over states from the previous time step empirical prior current observation um expected states expected observations and then the plearning and darning we're just going to leave empty here right and uh there just going to create a constructor function and we have our our container for the custom perceptual process So now what we need to do is we need to create the actually the the inference function that the agent is going to use to form beliefs about uh hidden states. And here this is a very simple um generative model. It's just a single state factor which means that we can do um exact basian inference based on these two equations which is basically just a soft max of the lo prior plus lo likelihood. And on the first time step, we just switch the the the prior with the D vector. And otherwise, we're going to use the empirical prior, which is just calculated based on the transition u likelihoods, right? And to put it put this all together, I'm just going to load the softmax function from the log log x functions um library. And we have our exact inference function here. So takes just three arguments which is the AF model itself the prior and the observation at the time step. I'm going to take out the A matrix from the generative model. I'm going to get the index of the observation and then we're going to calculate the log likelihood and the log prior plus some small value for numerical stability and bring it all together in a softix function to get our u approximate posterior. But actually in this case it's an exact posterior. Uh yeah and now we have this ready and we can bring it all together again to the model constructor. So in this case again our generative model is just discrete pom p with the components which we have defined above. But now to the perceptual process we are going to use our custom perceptual process where to the inference function we pass our exact inference function defined here and we're going to tell it to use learning uh as implemented by by this create pomp module and the same goes for the action process uh with the policy length of one and we're going to use parameter information gain uh and the rest is basically exactly the same configuration ation. Uh, give it the first observation, first action, similar for 100 trials, some empty vectors to store the beliefs of an agent. And this one is basically exactly the same. Uh, but the only difference being here that we are using an exact patient inference to form posterior over states. Uh, and this is how it looks. Um so so so far we've talked about uh using our default implementation and then using uh custom basically handwritten uh perceptual process and now I want to move on to how to use ppls as a part of of an active inference uh model. So uh in this case I'm going to use Turing but this can also be an RX infer or gen.jl JL and can be really any any basian algorithm can be sampling it can be uh can be a variational method could be other um uh optimization methods uh but in this case we're going to use sampling via touring. So the only thing we need to do is to again create our our touring model which is going to be our perceptual process function and again it takes three arguments. So our AIF model prior and the observation uh per time step uh we're going to take out the the um a model from the generative model uh get the index of the observation. We're going to tell the function that our likelihood is defined in our a matrix and then we create a touring model and in the touring model we're just going to tell it that our hidden states are coming from our categorical distribution defined in our prior and then our observation are coming from categorical distribution u defined in our likelihood dependent on hidden states and once we have the turn model uh ready we can sample from it so in this case I'm using a metric called hasting inks algorithm and I'm taking 5,000 samples from the posterior. This will give us our chain and then I can just average over samples for for each category because it's a categorical distribution to get to our posterior states and just return it as a factor. Um and that's it. And then the basically the rest is exactly the same. We bring it all together in the AF if model constructor define a generative model and then in the perceptual process for the inference function we set our touring perception function defined here. Um and then yeah the rest is basically same. So we define our observation action number of trials define some empty vectors to store beliefs and the action percent input touring is exactly the same. And when we plot the the result um we see that uh we have made three different models with different um perceptual processes. Uh now um yeah and that's the end of the notebook. If you guys have anything else to say or if there are any questions we're ready for it and otherwise um thank you. >> Thanks Sam. Amazing. Amazing. You know, so the uh maybe the last thing to say to round this off is just that uh we're trying to make this very flexible. So especially maybe some power users may be interested in adding um you know new ways, custom ways such as you use an exact inference method or a sampler etc. but also um make some of these methods ourselves so that they can kind of just be you know you can say I want to use cheering instead. And specifically when we use the discrete PUM DPS generative model we will rarely need to use a sampler because the message passing usually works quite well. But in some other generative models we might need maybe not a sampler but maybe a gradient descent varational inference where we can use the full flexibility of these peak kills and another I mean the exact inference method you showed sim should be faster than the standard message passing method probably because it doesn't have to converge over time and that kind of stuff. So um yes, so we're trying to make it both flexible and able to rely on the huge power of some of these ppls like RX and Fer, but at the same time keep an easy to use interface on the top like Sam showed in the beginning. Okay, we have a few minutes left. If there are questions, I guess Daniel, you can decide um how to spend those few minutes. And uh I've been responding a little bit in the chat, so I don't think any of the questions there. Yes, up to you, Daniel. >> Okay. Yeah, if anyone has a question they can write in the live chat because your defense is tomorrow, Peter, even if it's kind of related, what is the title of your dissertation? What are you defending tomorrow? >> Tomorrow I'm defending it's called modeling the mind and it's on cognitive modeling quite broadly. So we're including this software that we're showing here but also the HF software and the action model software and there um so the whole point of action models is that when you do applied cognitive modeling and you fit models to behavioral data and you can either fit them one at a time to each participant separately but there are much better ways to do that. So you can fit them in these hierarchical models where you assume that there's something common between people. You can even fit them at the same time as you relate them to, for example, questionnaire data. If you want to find out how people's learning rates relate to their schizoph for example, fit that one or even have one model where you fit neural data and behavioral data together if you're interested in how your active inference model relates to brain data, etc. And so, we're making software to do all of that. And since that's compatible with the active imprint stuff we just showed you and that all comes for free as then otherwise is on uh so we use that in to model hallucination uh uh where you model hallucinations as false inferences and in our results they seem to come from having too strong prior and relative to the observation and the theory is that that's because of sensory noise. So noise in the sensory system which makes you learn over time that the observations are noisy and then you trust your prior more your prior your predictions same thing. And we do that with people in different modalities and find out that that's modality specific. So just because you hallucinate you hear stuff that's not there that will correlate with having too strong press in the experiment but um but it won't have effect on visual necessarily. Yeah that kind of stuff and then I think we last year this uh now you ask Daniel is very pleasant but also I'll talk for ages. So I'll just run off by saying that the last thing we also use this to not reconstruct the generative model of a person's underlying a person's behavior but instead using it to reconstruct the generative model underlying a collective's behavior. So you simulate a bunch of little agents you treat them as one agent together. You fit a model to it like this. You know some of the stuff that we can do now that SIM has shown us this. We can actually ask we can use model comparison to test do people use message passing or do they use gradient descent variation inference or might they even use a sampler and might be good theoretical reasons to not think that they use a sampler. There are people out there who do think people use samplers, but but it just allows us to specify these active inference models more flexibly so that they can do so that we can properly compare the the full space of what they can be like. >> Yeah. Is that is that enough response to that question, Daniel? >> Cool. Thanks. >> And then what will be next for the package or how do you see it going forward after this year? So um and you can jump in Sam and Unan but I think um one thing is we want to make this interface as easy as possible and have many have a flexible so that's what we've been talking about today but make that make some good tutorials for that. So in case you want a continuous continuous state space continuous action discrete observation model for your whatever purpose you have that you can do that and still use full active inference stack on top of it. Apart from that one piece of excitement I think is exactly this nested inference stuff where so theory of mind is an example of nested inference where I have a model of Sam's model of me and that means that inside my model when I do basian inference I have to do basian inference so it gets nested and the same with metacognition so I'll have a model of my own model of stuff and the same with sophisticated inference also actually we've tested tested this a little bit and it looks like you can literally put RX infer inside Turing without any problems. Um, and we haven't tested that much but if we can get that to work then we'll get basically theory of mind full explicit theory of mind full explicit meta cognition flexible sophisticated inference for free and that's quite exciting. Um but like I said that's still on the let's say vision space and then there are other things like uh we want inductive inference in there uh one of these faster like we want these bells and whistles to be faster like we want these bells and whistles to be be there but also be easy to add more of them and that's kind of the hope with them the package and then we can rely on something like RX for the heavy fancy you message passing and all sorts of weird and t models. Yeah, lots of excitement and if someone ends up being interested in collaborating or contributing that kind of stuff, they should just get in touch. We'd be very happy to to do this. One point of making it as modular as we can is to make it easy to contribute also. So someone might like to add this specific JSON model and if they do we'll add it to the package so that other people can just press I want that model and then it's there for example. >> Yeah. Awesome. >> Hope that makes sense. >> Okay. Thank you fellows. >> Thank you. Thanks for everyone who's listening. >> Thank you. Bye. All right, we're back with RX Infur in action with Bart Vanerp and Fraser as special guest. So, thank you, Bart. Go for it. >> Thanks a lot, Danielle. Uh, before I start, quick shout out to Peter for the great talk before this. Also, Peter, best of luck tomorrow with your defense. Exciting time. So, uh yeah, go for it, I would say. Uh today, today's today's talk is RX Infer in action. So, my name is B. I'm one of the core developers of RX Infer, also one of the co-founders of Lazy Dynamics. Um so, we have been basically at the cradle of RX Infer. And I thought this symposium would be an excellent moment to share our latest developments with the rest of the crowd, get some new inputs, new opinions. Um if there questions feel free to hop in right I think uh want to make it very interactive interactive as possible. So uh feel free to interrupt when needed. So what are we going to talk about today? So for the people who have never heard about RX infer I want to give a short intro on what RX infer actually is. Um, I'm aware that most of you already are familiar with this technology, but just to get everyone on the same line, I want to make sure that everyone has the same background information there. Um, and from that moment on, I actually want to start building applications. St want to show some cool applications that we've been building over the past couple months. Um, starting with an actual drone demo. We'll also introduce gears.j JL, a new package that we've recently released. We'll talk about our RX in for MCP server that we've been releasing. And at the end, we'll have a sneak peek in stuff that we're currently working on, but I won't tell you what it is. So, if you stay um if you stay like long enough for the end of the video, I think you'll get some cool insights. So, let's get started. So what is RX infer? So in short, RX infer is an open source library for scalable probabilistic uh programming. So what does this mean? Proistic programming scalable. Well, it has to do with active inference, basian inference. We can do planning control. Uh some people call it explainable AI or better AI. But I think at the core of it is uncertainty quantification. We want to create programs, algorithms which are probabilistic in nature, which do not only propagate just values but also uncertainty margins um allowing for more robust programming or more robust algorithms. And what RX infer does very well is that it scales very well over dimensionality. So compared to Turing for example which is mostly sampling based RX infer has a different technology under the hood which allows you to scale to problems with a lot more variables and a lot more dimensions. So how do we do this? So at the core of RX infer we have factor graphs. Um for the people unacquainted with factor graphs basically these are probabilistic graphical models which allow us to describe some model of the outside world. This can be the dynamics of a robot. Uh this can be some assume dynamics of a signal that we're observing for example financial uh data. And what we can do is we can create these probabilistic or generative models and decompose them into smaller terms. Uh each term is here for example one of these nodes in this graph and the edges that connect these nodes these represent variables. So you can think for example that there is a state that progresses over time and each time it goes through some particular transition f and that's kind of these equations govern the process uh on how the state evolves over time and we also make observations. So the observations we plug in as these solid black nodes. Basically once we observe something this information will be captured in the hidden state and will be used for let's say making predictions or doing state inference. The benefit of using these factor graphs is that they're explainable and white box. So we can basically dive into the underlying model and figure out okay what's actually causing the behavior that we're observing. And because these models are mostly generated um by domain experts, by domain knowledge, we can actually figure out okay, which piece of this graph, which small box is causing specific behavior to happen. And it's just like Lego, right? So, uh you can tie together all kinds of different nodes with all kinds of different edges, uh which makes it very modular. But all the computations are performed locally. So although you create a huge model the computations will be on a very local and small scale compared to when you would do let's say all the operations uh at once in the entire large model. So what we have developed in order to create these factor graphs in code is our domain specific language called graph pplpp and it looks roughly like this. So we introduce in Julia the add model macro and this add model macro allows us to specify our model. So here we have for example a gausian state space model where we have observations that we call y. In this model we specify two parameters theta and we say each of these parameters will get a gamma prior with rate and shape uh of one. the hidden state we say okay it's going to be from some submodel and that submodel we can also define in the model macro. So by doing so we can kind of create this hierarchy of models basically nested subm models and this submodel is one let's say time step in our gausian state space model you can see here that the next state so as short for state will be distributed as a normal whose mean is some function over our previous state and the precision is governed by data and our observations can then be extracted from this state through some function G where we model the observation noise with the precision parameter TOAO. So this is one slice let's say and these function FNG these can be very different depending on the application can be drone dynamics can be uh robotics modeling financial data so it can be very diverse and we can basically use this submodel macro to expand our gausian state space model basically putting the different slices after one another and this basically allows us to specify this vector growth in a couple of lines very quite similar to uh the domain specific language that let's say Turing. Also uses but a lot more easier to use than some of its counterparts right there are also a lot of programming languages out there which do not offer such a nice DL. Um, so I think this is very nice, very great to interact with the tool. Makes it easier also for outsiders to actually get started building these models um without having so just some story. So all the way at the start we had to manually create these nodes and then also manually these edges and we had to connect everything together. Uh yeah, that turned out to be very very annoying. So this is basically it's evolved version and in general we're all very happy that this language actually supports this these features but basically they allow us to specify a probabistic model um in which we want to do the computations right the way that we do the computations is through variational inference. So what variational inference basically does although most of you are already familiar with this is basically it allows us to approximate complex often multimodal skewed distributions uh with a distribution that is from the exponential family. So basically instead of doing all these exact uh computations which are very difficult in tractable um what we do instead is we say okay we're going to come up let's say okay we're going to approximate our result with let's say a normal distribution and with this normal distribution uh we'll try to find the normal distribution whose mean and variance best approximate the distribution that we actually the actual true distribution let's say that we can obtain through sampling but sampling is often times too slow. So here we use variational inference basically turning this pian inference problem into an optimization problem. Right? So uh what we minimize is this variation of the energy functional where we try to find okay what Q minimizes this specific functional um for our probabilistic model P and P is the one that we previously defined in the vector graph. This looks all complex but in the end everything is automated in RX infer. So there's no need to worry about these equations. What's nice about variation inference that that skills very well and it's very efficient in comparison to sampling based methods because we u often times have closed form expressions and uh we leverage those to basically prevent having to sample thousands and thousands of times and this free energy functional inherently has a trade-off between complexity and accuracy that's also well known in the active inference community. So basically it's can think of it as kind of an optimal uh approach for actually finding these posterior values that you're interested in. Also for this we have some domain specific language. Basically we allow with RX to give like an easy interface on changing these approximate distributions Q. So uh these distributions Q are the ones that we set let's say the simple normal with which we want to approximate a more complex distribution and we can also apply different constraints on that. So here in the bottom we say okay we want our posterior over s over a hidden state to be normally distributed just using very common Julia syntax and we can also say that well these different states as we have them evolving over time modeling them all as one big gausian is not always yeah desirable because that grows uh of course pretty badly with the amount of states. So what we can say is we can say okay we want to actually impose a mean field constraint. So for each individual state in our model we create a different Q which is not correlated with the following one. So by that we basically can reduce the computational complexity of this simplifying distribution uh in order to make inference run even faster. Then putting it all together. So we have vector graphs, we have aggregation inference. Uh what can we do with it? Well, RX infer as I hinted already on earlier. Uh it doesn't do sampling. It's a bit quite a bit faster than sampling based methods. Instead, what we do is message passing. Basically, what that means is we use the local structure of this vector graph. So these nodes which are very sparsely connected to each other. Um and we perform these computations that basically optimize the free energy. We um we opt yeah we optimize this free energy on a local scale. So each node basically tries to optimize or minimize this free energy uh on a local scale and that allows it to do computations approximations of our posterior distributions. So basically what we can do is we can compute these posterior distributions Q by passing information between the individual nodes and this information flow that's what we refer to as messages. Often times or almost always these messages are distributions themselves. Uh so basically what we are doing is we're propagating distributions across the vector graph across the network. What's special about the way that how RX infers uh works on this is that it's fully reactive. So if there is no data coming in also nothing will happen because it's lazy by nature. That's also one of the reasons why uh our company is called lazy dynamics. So we only really do stuff when it actually is very much necessary and that's also the same principle that underlies RX infer. So if there is no data coming in yeah the model doesn't have any information or need to update its beliefs right um so it then also doesn't do any computations so this is quite adaptive robust so also when there are sensor failures we can adopt to that uh once say one of the parts of the model breaks down like in real world then we can stop passing messages there and because it's reactive it also allows us to put do every computation in parallel uh such that they don't wait for each other but they can just pass the messages long shot in parallel uh making it possibly also scalable to uh new chips that come available on the market. So, RX is for the ones who haven't seen it, it's uh it's open source like the core of RX is fully there. Um on GitHub, you can basically find all the information to get you started. We have a bunch of examples to get started with um ranging from small drones flying around to discrete state space models, bomb DPS, uh gausian processes and basically we're expanding this list of examples like every couple months there will be new like a few new ones. Um I also recently saw that we have new examples on icing models doing some very cool uh projection techniques for approximating posterior So there is a lot basically a lot of information in the example. So if you want to get started I would definitely recommend out uh to check it just uh go to rxifer.com and there you'll find the link to the examples. We also have discussions there where we are very much open for feedback. Uh if you have questions, want to know something, just post it there and one of us will definitely reply to you. Also, if you want to see like enhancements to the tool, um we very much encourage you to post like your opinion, your wishes there. We are currently u looking into the possibility of creating some kind of logging interface where we actually are requesting people to also share their ideas, thoughts um such that we can basically have a look inside RX infer in real time to see what messages are being passed and also how the computations actually happen. So basically just running it in real time uh debugging it in real time. So that's what we are currently requesting ideas for. So if you have them, please share. And of course, we also have issues. I wish we didn't have the issues page, but unfortunately we do. Um, if you want to help us out, you're very much welcome to contribute. I think we have quite some nice ones or easy accessible ones uh out there. If you want to help, I'd also reach uh yeah, just send me a message or uh just post something on on GitHub and we'll definitely get back to you. So good. That's RX infer. So RX Infer, everyone now has kind of an idea of what's what's happening under the hood. Um, and what we wanted to show to you is some drone demos that we've been developing over the past couple of months. So what RX infer basically allows is to create high fidelity control and planning systems. A couple I think yeah a bit more than a year ago uh we also presented a nice presentation here on the active entrance institute uh where we build a drone with RX infer a very simple two-dimensional drone just to show everyone okay how can we actually put these complex uh dynamics of this drone into our RX interface and how can we run it um so I would definitely if If you're interested in this kind of topic into robotics, definitely recommend to check this out because I think it's a very nice video. I won't play it here because I think it's rather long, but uh yeah, I think it's uh definitely nice for people to get started. Uh what we've been doing past couple of months, basically we've been expanding upon this. So in this model stream, we worked on a two-dimensional drone example with simplified dynamics and we went all overboard. Basically, we're now in three dimensions, six degrees of freedom. Um, and we're modeling the drones up to a very very precise uh so we're modeling the dynamics in a very very precise manner. We're incorporating drag coefficients of the body not only translational but also rotational. Um we are modeling the electronamics of the rotors. So basically all the dynamics that we could find of a drone are now basically inside a model and I want to basically share with you the results that we obtained. So we started off with dynamic obstacle avoidance. So we have uh this thrown here in simulation because it's a bit difficult to have these actually these bulbs um with this model modeling turn the dynamics the electronamics everything there and the idea of this drone is that it needs to follow this trajectory. So there is this blue kind of arrow um yeah indicating the target or the goal that the drone should follow and this target moves along this trajectory. Right? What's very cool about this particular demo is that all these spheres that you see flying around the drone doesn't know about them. So while actually flying it doesn't know about them only when the drone gets into a certain proximity. Um it's it sees them and then of course it needs to adapt right. So this drone it will fly around once it sees one of these spheres and if it's in its way of actually following this line it will adapt to that actual obstacle um and also plan around it. So it will try to reroute find another path that at least follows the goal as good as possible in order to still kind of keep track of this target. So I think this is something that we are very happy with with the results. um putting on to hardware is of course a completely different challenge but I think for first experimentation I think we're very far away or very uh we we made quite some progress and this is for the simple one drone dynamic right what we've also been experimenting with is actually having multiple drones just to give you an idea so the drones they start around set is equal to zero so on this ground plane and basically we task all of these drones phones uh like 20 in total to fly upwards. And also in this scenario there is these spheres which are like in the way of the actual reaching the targets within uh within within a straight line, right? So each drone basically operates independently from each other and they also don't know where the other drones are. So it follows a similar algorithm as on the left. Uh but in this case it actually once it sees the sphere it again tries to reroute around it but it also isn't aware of the other drones. So at some point some drones also encounter other drones. And here they'll also adjust their uh trajectory in order to not collide. So you can see that for example here with the pink one. So you can see that the pink actually dodges the green one um by making a sharp turn. So just to show everyone what's possible with RX infer, right? So RX I think uh the examples already are getting more advanced as we as we speak. Um but we can achieve like very cool applications. Drones I think is already one that's very complicated. So we're very happy that we actually managed to get these running also in real time. What we've also been working on is a new package. So RX infer is fully reactive and uh what we intend uh RX infer to work with is basically sensors which operate on different frequencies. Uh we can have for example a camera which runs at like 30 or 60 Hz. Whereas if we have a microphone, it has a sampling rate of uh 48 kHz. IMUs are also somewhere 300 400 Hz. So we were thinking, okay, this is nice. Uh RX is kind of targeted to solve these particular problems. But how are we actually modeling them in Julia? because it's not so easy to create a particular I don't simulation environment where um every sensor runs asynchronously in parallel. So that's quite complex. That's why we introduced gears.jl. So basically what gears.jl allows you is it enables fine grain scheduling for simulations and realtime systems. So just to give an idea what does it mean? It basically allows to create simulations of systems where uh we have different observations coming from them at different sampling rates. So basically in the situations where we want to be in full control of when we observe what or when our environment gives us a new observation uh to process. That's something that gears.gl L has been yeah trying to solve basically. So just to give you an idea of how it looks um once you import gears in Julia you can create a variety of tasks. So in this case we can say for example every 200 milliseconds we want to print the following line that the task is executed at some time step and then we run ouruler. So we can say run the scheduleuler for 1 second. So for the next 1 second update. So basically the task that we previously defined uh these will actually be executed during this 1 second duration and we can make this more complex. Right? So for the drone example, we can say uh process IMU data every 5 milliseconds. Do this or uh perform the replanning or update the control sequence every 200 milliseconds or 100 milliseconds. If processing audio maybe yeah process the audio every 10 milliseconds. So we can have very fine grain control of the actual data sources coming in. And we can run this. What's very nice about this is that the clock that basically runs these tasks and basically does the scheduling um we can also replace the machine clock that gets regularly used with some virtual clock. So basically if you want to do simulations spanning over let's say years like if you're modeling I don't know uh planets in their orbits and you want to do something with that we can actually also add in a virtual clock which allows us to speed up or slow down the simulations if we want to model specific behavior on like um a nanocond scale. Not sure what kind of applications go there, but I guess there are. Then we can of also slow down the virtual clock such that we're not modeling on a second level as our hardware normally would, but on the nancond level. And we have a variety of scheduulers to basically create our own simulation environments. We'll have we have the intervals that we just saw. So every 200 milliseconds do this. Uh we also have some thing called an ASAP scheduler. So as soon as possible. So as soon as something happens uh perform a task. So this can be used for chaining tasks together and also event based. So when let's say an observation comes in. Um I always make the example where we have for example an API request coming from another service. Uh that's also something that you can work. And what's nice is that these tasks they can run asynchronously. So when you have a lot of different task at the same time, it actually figures that out and you can run them asynchronously in parallel. So that's just a short introduction of gears. So basically they allow us to create these complex simulation environments. Um, and in general, I think it allows us to experiment with the core capabilities of RX infer better than just native Judia code would. So, that was one uh one release that we did a couple weeks ago. Uh, something that's a bit more recent and that are also very excited about is RXMCP. So um what we heard from the community is that although we have this nice domain specific language using RX infer is not always the easiest right. So uh it's still challenging. So what we wanted to do with this project is basically what we um we wanted to give the capabilities of RX for a broader audience. We want to have a lot more people interact with the tool without knowing that they're actually interacting with the tool. So instead of downloading Judia which yeah is another programming language people are a bit hesitant if they want to go away from Python um we just wanted to make it easier for people to interact even like senior executives right people who don't do any programming anymore right those are the people that we also uh want to target but what these people often require is some form of analysis uh some form of basically um like processing data with uncertainty because that's where where it shines. So with RXMCB I think it has never been easier to use some uh parts of RX infer. So what it allows is basically grounding LLMs with RX infer using our model context protocol or MCP for short. So effectively you can just chat with RX infer. So if you have a data set that needs processing um you can throw it in ask it a question and it will basically figure it out. So together with the strength of LLMs basically converting numbers into some natural language and using let's say the uncertainty processing of RX infer you can actually get the best of both worlds. We already have integrations now with Mistro with uh CL cursor open AAI. You can find the docs as well. So we added in the QR code in case you're interested. At the moment it's free to use. Uh so we definitely recommend everyone out to yeah basically to to check it uh use it give us feedback and our goal is to improve it further along to give you an idea of how it works. RXMCP it starts with some context. So you can think of oh I have this huge data set. I'm going to throw it in this uh in one of my favorite LLMs and I'm going to ask it to do something. So I'm going to query it. Well RX infer will then basically learn from the data that you have supplied. So it will try to figure out what are underlying relationships between variables. Um and it will also predict. So it will give you suggestions actually it will compute the posterior distributions and those posterior distributions will then hand back over to your favorite which can interpret these distributions. So basically it can give you then recommendations based on the query that you provided and the uncertainty aware results that RX infer provided. So basically you get the best words you get best of both worlds. You get to use the cap the analysis capabilities of RX infer whereas you have your LLM basically explain what it's returned and why this is important just making the audience of RX for a lot broader. Um, so I think we're very happy. We're also getting quite positive results here. A lot of positive feedback of people who otherwise would have never been able to use our tool. Uh, and we're very excited to also engage them. I also wanted to show a short video. It's two minutes of our CEO Albert. He basically recorded a video demonstrating the capabilities of RX infer. um what he did because it's a relative fast video. He basically illustrates the use of RXMCP using um a problem that a lot of companies startups are actually struggling with. So suppose you have a thousand contacts and you want to figure out okay which contact should I actually pursue uh in order to make a deal which one is the most which one has the highest uh return of investment what is the most yeah basically finding the person will get you the best deal in the shortest amount of time. Um he created a data file for this and I think I'll just uh just play it and be silent for 2 minutes. Can let all do the talking for now. Every startup has the same nightmare. You are making critical decisions with limited data. Which feature to build, what price to charge, who handles which lead, and every wrong decisions burns runway. Using just LMS for this, well those don't reason about cause and epic. You need grounded mathematical decision-m that actually learns from outcomes. Let's watch how arixen first solve this. I'll use lead assignment as an example, but this works for any decision, pricing, feature, content, anything. Well, here's a bit of an irony. I'm using lm to interface with arx infer. Uh, and even it struggled to parse with the parse and structure data at first, but well, arix infer underneath pure bijian reasoning. So, we've got five new leads. arix infer assigns them. Jay gets small companies. Liisa gets enterprise. And these aren't guesses. They are mathematically optimal given uncertainty. Now, let's imagine we've got the results back after following these predictions. Well, Jake's sure bets lost 20K. Uh Lisa Enterprise deal made 40K. And here's the really cool thing about Arexenfer. So, I feed back these results and instantly Areenfer updates its beliefs. Um, and turns out that Jake isn't the right for founder deals. Try Lisa instead. And this isn't just pattern matching or token prediction. It's a causal reasoning about actions and rewards. Next, founderly derives models says Lisa, not Jake. Uh, it learned the causal relationship from five outcomes. Obviously, you won't be typing these comments uh yourself. Your agents will call Arix MCP directly. And this works for many things, right? For pricing you can learn price elasticity per segment. Uh for features you can discover what drives conversion and for content you can understand uh the engagement causally. You need decisions that are grounded in reality not just text generation. Uh Arixen per is open source. Are MCP is free with integration docs available now. Uh we invite you to start learning from every decision today. Check out ldnamics.com. >> So wonderful. So that was a very short demo. I uh was thinking about doing a live demo but you always know live demos never work. So um that's how I wanted to let show you. So what he also mentioned and I think this is uh very very interesting is basically RX infer also relearns. So it it learns um based on new data and once new observations come in it will actually adjust its beliefs. This is very different from the LLM chat bots that uh you have with a lot of companies. I was trying to book a flight over the weekend actually and I ended up chatting with this chatbot of which you of course know it's an LLM and despite saying well just connect me through connect me through uh it was quite stubborn. So I think this is also one of the places where our could shine basically update the beliefs of these LLM as data comes in instead of just appending it to the context. Right. So that's currently what's happening. >> Yeah. And then a sneak peek the latest developments. So this is very very rough and I want to kind of warn you that uh functionality is not fully there yet. Um, but we are still want to show it because we're very enthusiastic about it and we wanted to get as much feedback on this topic as possible before we actually further develop it. I think it's very cool. So let's just So what we have been thinking about is doing uncertainty quantification in LLMs. So there is this huge problem of hallucinations figuring out whether LLMs can actually work. Um and on the other hand we have RX infer which is fully probabilistic. It can quantify uncertainty very well. So why not marry both? So we've been thinking about can we actually combine LLMs with RX infer not in the same way that we do with MCP. So with MCP we say okay LLMs go use RX infer but basically what we are thinking about is using RX infer to upgrade these LLMs um basically that it doesn't just provide you with an answer right but that it also provides you with an uncertainty margin instead of saying ah you're right you're fully right definitely uh good good question uh this is the answer maybe it will be better if the LLM's become a bit or new ones so that it says I think this is right with let's say 50% confidence. Um so we've been exploring ideas basically for RX infer to integrate with these systems. Uh we have a very short here on the right yeah visual representation of how something could look like. So next time you're using uh JGPT it ask oh do you want to use RX infer for let's say verifying Q&A accuracy you can say yeah I would like that I would I would want to know how certain the LLM actually is. Um you confirm and you can just then say RX confer will uh will run on the back end and it will figure out how certain the LLM actually is in its answers. So, we're still this is still very much a work in progress, but we wanted to get some initial feedback whether you think this is interesting. Uh what else would you like to see? If you have great ideas, let us know. And if you want to test it out, you're very much welcome to and we highly appreciate it. Um if you want to know how to get started otherwise just send me a message or uh reach out somewhere uh and then we'll definitely give you let's say the API keys to get started. So with that I would actually like to conclude uh and I'm open to start a discussion. Very interested in what you guys all have to say. Thank you, Bart. Fraser, if you'd like, maybe say hello and check in and then feel free to start with any questions that you have. >> Yeah. Hi, thanks Bart. Um, hopefully you can hear me. Okay. Uh, so I'm Fraser. I'm a PhD student at the University of Glasgow. I also do a little a little bit of open source work with the bias lab. Very honored to to do that. Um, yeah, I was excellent stuff. I'm particularly interested in this, you know, gears framework for for scheduling tasks and so on. Um, I'm trying to think about how to build certain active inference agents using RX infer that interface with a world that might itself be reactive and so on. Uh, which as you say is quite difficult right now. I'm I'm just thinking, you know, gears is excellent. Um, in some sense it's a way to it is just a way to schedule tasks. you know, we had RX environments before as a way to kind of talk about actual entities that are out there. Um, I'm thinking, well, if you know, potentially that could integrate with gears, that would be a really powerful way to uh afford those kinds of environments and and things that we could then place agents in um to kind of give them a home. Is there anything you can talk to there about potential cross-pollination between um RX environments and gears or if there's maybe some some other replacement package for environments? I'm not sure I'd be interested in your thoughts there. >> That's excellent. That's a very good remark. I don't think we have actually done any integration there ourselves. Um for the very simple reason that yeah, we didn't get around to it yet. Uh but I definitely foresee that like joining both would yeah we have this package already out there. Um so I think we can leverage both and use both strengths to uh yeah to make it even easier to create these reactive environments and I think gears allows us to u not only make them reactive right but uh to also do the actual let's say precise scheduling when it's needed. So for example for sensor data it's yeah in a sense it's reactive of course uh but at the same time it just samples every so many hertz um so I think RX environments like um these are like fully reactive and that's very nice on the other hand sometimes you do want to have this control over when actually sensor data comes in and I think that's where this gears is really shining. >> Yeah, the combination of both seems like it' be a real winning strategy. Yeah. Okay, maybe I'll think about that myself what I can do there. So, >> excellent suggestion. Thanks. >> I'll I'll read a question from live chat and have some related questions. So, Bert Burkers wrote, "Could RXMCP also work for POMDP models in the future?" >> Uh, yes. uh definitely so these Pom DP models right uh in general I think I'm not even sure whether we maybe already are supporting them so just to give you a very quick overview of what RX infer MCP actually is so we have RX infer which is our software tool we have RX infer server that basically runs RX infer in cloud for some selected models and then we have RX infer MCP which connects the LLM to this server. so for our ser in first server which is also open source it's possible to implement these pom db models by themselves. I'm not entirely sure whether they then also directly become let's say connected to the LLM. Uh so that's something that we need to check but at least we you can send easy a simple API uh connections or requests to RX infer server and have the PDPs running there. Although I do foresee definitely that connecting this with MCP as well could be very valuable when you have indeed some data that you just want to process. Uh you don't want to write all these uh data loading functions and you just want to throw it in there. It's Friday afternoon, you want to finish this up before you leave uh leave to home, let's say. I think that would be excellent. In the end, the idea is for MCP to start supporting as many uh models as possible uh as relevant. Uh but at the moment, yeah, we're prioritizing the ones that we have kind of open requests for uh for the very simple reason that we can we can't put everything there. Uh but we're yeah trying to be a bit smart about our time and our uh yeah time investment there. >> Yeah on on the RXMCP. So you said that it learns from the query and the data. So what structure of graphical models does it start with or sweep over and do you get back at model blocks which you can separately execute? I mean is it doing an end toend RX infer meta programming and then reporting outcomes or what is really happening underneath the call? >> Excellent question. So um what I mentioned so we have this RX infer server where we have some models running for the demo that we just showed. We have a particular model in RX infer the one that yeah created that already. Um but it's still very generic. So basically everything that has to do with uh contextual decisions where you have some data you want to make a decision that's where basically MCP currently is targeted for or tailored for. so the MCP is basically a connection going from your LLM to RX infer server in the cloud. Um it does return quite a bit. I'm not sure whether it returns the full model macro. I don't think so to be honest. Although this is something that if there is a real need to return this model macro, sure that can be made possible. Um but what it's actually returning is all the posterior inference. So basically all the raw results that you could normally obtain with RX infer itself that gets sent back to LLM and then the LLM is used to interpret these results. >> Okay, makes sense. So the MCP facilitates within the repertoire of what the server has access to. >> Exactly. >> Yeah. Okay, I'll ask a question from the chat and Bart and then Fraser you could add any thoughts too. Okay, Rob wrote, "Broad question. Where is uncertainty valuable in applications? I see its benefits in fundamental science, but I struggle to find effective examples in applications." >> There are plenty. There are plenty. But I like I like the question quite a lot because uh yeah, we've been looking through this uncertainty lens already for a couple of years, right? So we can see where it's valuable in almost any application. Um but yeah I can I can definitely give some examples here. Um in robotics we have sensors for example and these sensors they make particular observations but these sensors can also be noisy. So if you move you say the arm of a robot or you drive a cheap robotic kind of around it will give you some measurement of where its next location is. Uh but at the same time you know well the sense readings aren't that accurate maybe it it slips a bit let's say uh on the ground. So yeah there will be some deviation in the end result. So that's one place where you could use this uncertainty basically these margins.

um and you can use that to correct also for it at some point. Another one which is I think very very important is everything concerning financials like financial markets. I wish financial markets were deterministic because if that was the case, we could just pour in money into one stock that we are aware that goes upwards. But in reality, no one knows which stock goes up. Unless you have inside information, of course. But for most, it's just a guessing game, a game where you try to figure out, okay, which stocks are going up, which stocks are going down. And often times, what the term that they use in these financial markets is called volatility. So basically it describes how volatile the stock is. So if you have a stock that's very nonvolatile, its price will be very smooth over time. Whereas a volatile stock will go all over the place and that's something that you want to be aware of. You want to be aware of this volatility or actually uncertainty about how this stock is progressing over time. Um, if you're going to use that investment fund for fund for, let's say, serving in your pension money, that's something you probably want to have stocks which are more secure, have a less of limited volatility. Um, whereas if you're like private equity, that kind of stuff, you want to take a bit more risk. So that's um also an application area where you can think of and I think that yeah makes uncertainty shine. Yeah, like having the point estimate without the variability measure of some kind is just uh half the picture because if there's something that's you expect it to be 1% improvement, if it's an engineering process or financial, but you don't know if that is a 1% plus or - 10 or 1% plus or minus 001%, then you you wouldn't really know if that was actionable and then you would need to go another level okay 1% plus or minus two but how confident am I about two and then that kind of dovetales back into why we want to have these composable flexible basian models where we can interpret those different uncertainty structures >> yeah exactly for some occasions uh propagating no uncertainty itself is also fine right so for some applications you don't need Um however what you see is for like very critical applications it's actually good to have some kind of quantification of margins uncertainty. so yeah definitely definitely >> pretty sure want to add anything. Yeah, just maybe for a really speculative question now. Um I I attended IY earlier in the year, the international workshop on active inference and uh laser dynamics and the team from the BI side you guys presented. there was some work on um kind of recasting the EF for planning in active imprations as the VF um so that we just have one free energy marginal well one free energy functional that we're minimizing in order to do active inference and you know that'd be quite nice because we have two sort of canonically floating around in the active inference world and maybe that's a little bit annoying um there obviously this is the fruit of ongoing work at the bias lab so it's all very speculative but Is there anything you can comment on with respect to the way in which this might eventually land in RX infer? You know, is is there a way in which you can see how the beth free energy might change in consequence or a way in which users might be able to plug in? Oh, you know, they [clears throat] want this kind of unified functional thing for message passing to happen here as opposed to not. Is there is there anything you can say about that at all? It's very very broad question. Sorry. >> No, it's an excellent question. Um now indeed so uh our colleague W about has been working on this particular topic trying to frame the expected free energy as variational energy minimization procedure by doing some smart tricks basically on this factor graph and in this message passing procedure I think once so the expected free energy has compared to let's say variation of free energy some perks right so you get I think one of the most important ones is that you get this information seeking behavior. At least that's one of the one of the perks that variation of the entry by itself might not always have. Um so for some applications this is very critical critical um it could help you a lot in actually trying to learn let's say the parameters of your robot or the parameters of the financial markets although for those you don't really have an influence yourself. So I think robotics would be a better example. Um but indeed it's it's ongoing work. Um how we see it is we see it kind of kind of as a different flavor different flavor to what we're already doing. Uh it's also still uh very much under development but I think yeah progress is very good. The results are very encouraging. So um I can't wait basically for the results that will be popping up in half a year or a year from now. And uh once they do this there's definitely incentive to integrate this in RX infer as well right so it's not that we are thinking oh this is better free energy or variation of free energy it's kind of the holy grail um for some applications expected free energy is just more interesting more nicer it gives you properties that other other targets wouldn't give you let's say so that's definitely something that once Once this research kind of matures a bit, once we have a cleaner idea of how we can integrate it in a improved manner uh on a larger scale, then that's definitely something that we are planning to work on. That's also why we're of course performing this research because yeah, we want to keep on improving the toolbox not only for uh yeah, but also for actual industry problems. Excellent. Cool. >> What are you most looking forward to or what do you think will be able to bring it into most people's hands in the coming year? >> Oo, that's a that's a tough one. I think we're making very very nice progress with um MCP. So, as I mentioned, the motivation behind MCP was really to get RX infer into the hands of as many users as possible. I think providing it through an LLM basically just having something that everyone can access even if you don't have like university background or uh you don't know mathematics you can still use it you'll still understand that there these uncertainty margins can be useful for specific cases I think getting it out in as many hands as possible is what they're really striving for and that's why this MCP uh in our opinion it's a very first step right it's it's kind of first step in getting RX infer out in the public and we are currently looking into applications where we might benefit most so basically maybe specific application areas of this MCB where we can put it in as many hands as possible so not just selecting these leads as we did as always showed in the demo but maybe there are also other application areas out there which are very much relevant um but that we currently are overlooking. So yeah if any if there are any suggestions regarding this feel free to reach out because we would be happy to extend the functionality of RXMP with these kind of new applications. So that's basically one thing that we're looking very much uh out for. Another thing is what I already mentioned is so we're kind of not reviving but we always had this uh idea of creating this amazing debug tool for RX. Um so from a developer point of view I'm very excited about that as well. So what you can do now is basically it's very hard to debug because everything's reactive. Uh we do have some very deep developer options to actually figure out what's causing what. But we wanted to make that process simpler. So there is now an open discussion on GitHub. Uh basically requesting ideas, enhancements from everyone on how we can make this debugging or interface experience better for a lot more users, a lot more developers. and from the lasic side, we have already been experimenting with some of the tools. let's say we haven't integrated them with RX infer um but we have starting to get yeah a better understanding of what's happening um what would be a nice interface what could help I don't think we want to go let's say all overboard and let's create our own uh full UI ourselves so probably we want to leverage on existing technologies um but yeah we're very much open for ideas and suggestions. If you think, oh, this would be cool, let us know. Definitely any feedback there is recommended or appreciated. >> Thank you. And it's been really fun with Frasier facilitating the RX and for learning group and and a lot of cross-pollination. And I think compared to where we were a year ago, you've really highlighted this birectional relationship with the LLMs. On one hand, there's the MCP direction, which is the LLM using RX infer as tool use, and then on the other direction, there's kind of two flavors. There's the uncertainty quantification of LLM outputs. So, you can just ask the LLM repeatedly, um, what's the GPS coordinate of this location? And then use those estimates to get a posterior distribution. And then the next step there is to choose the prompt to the LLM with active inference to make informed sampling of those latent spaces. And so that is kind of building out that two-lane highway going both ways between larger models and how they're increasing in power and then all of the interpretability and benefits that graphical models and factor graphs bring. So that's very fruitful and it's a very pluralistic exciting time for building those kinds of hybrid models. Definitely I think uh like from our perspective LLMs are very good with natural language and we are all using them uh even people who claim that they're not using them we're all using them right uh drafting emails from time to time and uh well well you know right um but it does provide an easy interface and I think that makes it very accessible for people to use it um so definitely would be personally I'm in favor of leveraging on the let's say the basically using both both tools I think uh combining them and using best of both I think that's uh what we really want to strive for. >> I'll just ask one final short question from the chat. Jay Graves wrote, "Several projects including AI knowledge Sidekick are scanning selected information sources to create personal curated summaries. Could RXMCP be applied to monitor such a feed? Excellent question. Um, yeah, I just need to think a bit on how this. So in general, proistic uh models are very much useful for doing personalization. Um just to give some some context, it's a bit out of scope from the actual problem described here. Uh but what we were designing RX for from the very beginning is actually high-speed applications and most specifically hearing aids because that was the one the company basically was funding uh RX infer from the start. um what we were using RX infer for uh one of the one of the parts right not only the signal processing but also for incorporating user feedback so when you do an interaction with the human being and I'm speaking for myself right so I had this uh was struggling with this chatbot for booking a flight over the weekend if I say well this answer is not listed I want the model to kind of learn from this I want don't want to keep interacting I don't want to say it 20 times like connect me through connect me through after two times I think it should be very clear that I want to be connected through um what we used RX infer for personalization in hearing aids was basically figuring out okay what specific sound sources do we want to listen to so normally these algorithms hearing aids or earbuds they have this uh active noise cancellation So they cancel everything or they have this hearth through mode where you basically hear everything again. What if you only want to let's say listen to one specific sound when you're maybe in a car you just want to listen to sirens. Well, not that you don't you want sirens to be all over the place, but once like an ambulance drives by you want to listen to that and you don't want to listen to the car engine. At least that doesn't make sense. um when you're out in the park for example, maybe you want to listen to the birds and you don't want to listen to the people who walk past you and have these uh conversations. So train station is also an excellent example, right? Um when you want to listen to basically all the announcements, but you want to zone out the actual sound of the trains. So figuring out what a human person would want to listen to is something that requires user feedback and it's also context very much context dependent because if I want I'm at the bar I want to have a different experience than when I'm working in the office. Um so so we have used RX infer it's very long detour uh but what I wanted to say is we have been using RX infer for preference learning where we have interactions with human beings and creating these personalized reports I think that definitely falls within the scope of what RX infer can uh can achieve that having having said that I'm not sure whether it's directly applicable um to or that sorry that it's not applicable that RXMCP directly supports his personalized um summary generation. I think that was uh the original problem, right? So, but definitely there are possibilities there. Definitely we can bridge this gap. It's not that we aren't able to do it. It's mostly we just didn't do it yet. I think that's uh yeah, that's the current status of it. >> Any final thoughts, Frasier? I I could stay for for hours and hold hold bar hostage for for for a long time. But uh no, I mean I think I that was that was really excellent. I I enjoyed everything. I'm looking forward to everything that we can do together. Um there's a lot of leads uh there and what you presented. So yeah, thank you very much. >> Yep. Thank you to you and the team and for engaging with the open source and next year we'll have another spiral of learning with RX and so thank you guys. Bye. Definitely. Thanks a lot. All right. Next is going to be a 2 minute and 30 second video by Michael Lennon. Here it goes. Hello, I'm Michael Lennon, a member of the active inference scientific advisory board and delighted to be presenting today on um reflections on some economic myths that I've been encountering around AI over the last year. And uh however, as I prepared this slide deck, I noticed that there was one point in particular that I wanted to draw attention to first. So this is a rapid clip about a specific opportunity that I wanted to put in front of the community and um and speak about and if there's more time than to go into the uh formal fuller presentation offered. So the um the rapid headline is about uh the formation of a new consortium of major social impact investors in the United States who have pulled about half a billion dollars in funding basically aiming to influence the present future of AI towards more pro-social um social impact. And so uh they are uh as you can see uh an impressive group. They have three explicit goals that they've set to expand the group of people who are involved in the decisions of AI to strengthen the organizations that are bringing people together to shape AI in the future and supporting the creation of public goods. So this is very well suited to what the active inference institute has been up to. And so I wanted to uh start here. I'm uh drawing attention to two of the slides that I have later on in the second clip that speak about both bio regionalism a form of um local environmentalism that is organized around natural ecosystems rather than political or uh jurisdictions. And um how uh in addition to having networks of allies that we have um established methods and specialized tools that can make us really stand out um uh in this space and uh we have been for some time been working on this but that uh as a community would like to draw attention to how might we be building additional collaborations and how might we be building additional um synergies to really um make a collaborative difference in the world. When you All right. Next up is going to be a threeminute video from Joel Deetsz. Here we go. Showing you a tool that I made originally for 11 Lab that I've sort of been tending to active inference. Um but first I will show um sort of how it works and the concept behind it. So the goal over here is to show the summary of a research and field and extending showing you a tool that I made originally for lean lab that I've sort of been extending to uh active inference. Um but first I will show um sort of how it works and the concept behind it. So the goal over here is to show the summary of a research in field in a field um on a time series to show how well various hypotheses are validated. Um and if you kind of roll back the sort of clock within a field um you'll be able to see you know over what um time series you know you start validating various hypotheses. Uh this is ideally to cause people to make more and better hypotheses and to sort of automate this. Uh it uses the probabilistic knowledge graphs. So the concept here is that everything is you know extracted sometimes from peer-reviewed papers. That's mostly what we're working off of um into various evidence and assertion things. The uh weight of evidence is done by an LLM. Um and so um all these things are kind of bundled together and then uh published in an publications format uh which is on an immutable ledger. So, it basically will give you uh you know a a record. Um and it also helps if the LLM, you know, you don't totally trust it or it's flipping or whatever else. Obviously, it's still an issue and it may be more an issue for more contentious topics. Um so that you can basically have a different LLM score or have multiple different versions or you know gradually improve it. So certainly [snorts] this technology is technology that's in flux and so it's not always perfect. Um this is a sort of importing function. And so you can easily kind of import the various RDF files. Um so you get these kind of nice sort of summaries and uh you know sort through the various aspects of it. Um so yeah originally again this was really just for some 11 things. has also had an economic calculator which I'm not going to show but it shows the sort of downstream uh economic impact of various things including projections and it has a bit of a comparison feature to see um for people who are interested in investing or putting more money uh into these things uh what that might evolve with. So we kind of have a similar sort of concept here with um active inference. Uh more or less replicate the same model but obviously the hypotheses are not the same um as they come through. Here you can see the kind of evidence accumulation um and uh sort of here what it thinks are sort of validated hypotheses um including just a sort of variational uh free energy um kind of concepts uh with sort of with respect to sort of neural systems that are providing Brock basian inference um decision-m control and reinforcement learning passes inference um down to the things that are like not entirely validated as it says to basic basically psychopathology decision-m and things like that. So, we have some um work to do I guess in a way because uh this has been a bit of effort to sort of build uh software in between. Um but I just thought I would give it out as a sort of discussion point um that could be useful way of uh doing new forms of meta analysis and uh also um for things that are maybe in the more contentious category to have a sort of m multiple different versions. Besides the fact that this can kind of be automated and potentially used to uh drive different types of experiments. Uh thanks for your time. If anyone wants to use this software uh please get in touch with me. Very happy to uh make this available to other researchers. Okay, next up is going to be a 5 minute lightning talk by Alexander Sabine. >> Hi, my name is Alex and here are some ants. Uh, these ants are all mathematically driven. I think of them as marovian agents. Um, and the memory is actually stored in uh the actual environment itself. So these are marovian agents exploring their environment. And you can see um them starting to build little defense mechanisms here. And you can also play with different variables and different parts of the um the script. You can add different uh layers to the network and so you can see what's going on. And I've done this with lots of different biological and physical systems. I'm really interested in what happens when variational free energy reduces and I see it as coherence increasing that is uh coherence is memory density through time. Um here we have a mycelium network and you can see uh the coherence field there growth activity network. So all of the scripts are all mathematically driven um using the same uh formalism which I'll show you in a moment. Um here we can see myelium network underground glucose exchange and moss starting to exchange nutrients uh in the network there too and some L system trees up there as well with the photosynthesizing sun uh hurricane. So on my talk you'll see that that has a higher omega value uh cuz it's more in an exploratory campaign. Um so it's a higher temperature. Um here we have thunderstorm and I'll just explain as coherence increases in the cloud charge eventually it reach a rupture threshold and we get thunder and lightning. uh the sounds from these scripts where their sound is all diagetic from within the script itself. Here we have an atom and change the electrons and it will change the uh obviously the atom that we're looking at. And just to explain that uh CR or coherent structure regeneration uh which is the um formalism I'm using is a coarse grain formalism. So I'm not saying that this uh does anything microscopic within paradigms, but it's a good uh I find it a helpful way to to think between paradigms. So here we have the sun again modeled with the CRR. And here again we're looking at how coherence builds through time, how we reach these rupture thresholds where exploitative campaigns can no longer operate and we need exploratory new generative model um which reweights from the past states of coherence. Here we can see ice and change temperature. Watch ice forming. So I've done basic modeling as well. This one has a lower omega value. Um it's sort of less exploratory, more rigid as a physical system. And here we can see a hologram. So here we have perturbation of the coherence field. And this is exactly modeled uh in the script. Um here we can see a maze uh or sort of um rim solver. So the agent is a marovian agent and the memory itself is stored in the field in this in the environment. So what is CR or coherence structure generation? Well, here we have coherence um formalism. So coherence is building memory density through time. It's essentially all past states matter in a non-marovian sense. And we have this lx towel here which is how memory density is built through time. uh coherence reaches critical thresholds where the environment provides new free energy spikes and can no longer the exploitative campaign can no longer work. So we require an exploratory campaign or creative campaign and this is where the rupture or the direct delta um leads to a new kind of generative model. And you could think about this as scale invariant moment to moment or coherence built through time and then rupture being like ecosystem collapse or something in the body collapsing um or a psychological shift or an ontological shift through time. Here we have regeneration. So here we have the historical states here picking up from a coherence of memory density here and we have a Boltzman like regeneration operator. So regeneration is exponentially building on what mattered most in those past states through time. So this is an exploratory uh playground. Think of it like a free energy playground. Um exploring coherence based approaches. I put various videos on here, explainers, uh other bits and pieces for people to play and explore. And I'm just looking for people that might be interested in this. So I'm really happy to be part of the symposium. And thank you. All right. Next, I'm going to play a video from Jana. And Jana also requested that I read this GitHub post, which I'm also going to put the text into the chat. Okay, here we go. In a quiet village on the edge of a sprawling forest, there lived a peculiar gardener named Andre Marov. Unlike other gardeners who planted their crops in neat, predictable rows, Andre saw his garden as a living mathematical puzzle where each plant had its place not just for beauty, but for balance and harmony. One misty morning, Andre stood at the center of his garden. To build a marov blanket around his beetroot, Andre carefully selected plants that would support and compliment it. He planted onions nearby. Next, he chose broccoli. Not just because it stood tall and proud, but because it loved calcium rich soil, the beetroot, by contrast, on either side, he placed Brussels sprouts and cabbage, their sturdy stocks forming a protective barrier while their leaves shaded the ground, keeping it cool and moist. When the planting was done, Andre stood back and smiled. His beetroot now lay safely within a circle of companions, each plant playing a role in the dynamic Markoff blanket. Wiping his hands on his coat, Andre whispered to the garden, "Together you are more than the sum of your parts." And as the seasons passed, the Marov Garden thrived, balanced, resilient, and harmonious. Welcome to the collaborative co-making session where we explore active inference through companion planting, groupings, and relationships. Okay, next is going to be a 4-minute lightning talk by Rishu. Here it is. >> Hello everyone. This is Rishu, PhD. I'm a mom and a nerd running a business solo, which means that some days I feel like a CEO and other days a vending machine dispensing snacks and emotional support. That's why I would be wearing a soft and warm mom nerd type shirt during this talk. Hi everyone. So, have you noticed that when we move a toddler's toy to a new place, it becomes a new toy for them? The object doesn't change, but the context changes. And yet, it feels completely new because their brain learns through context. And it's playful prediction. And [clears throat] adults actually do the same thing just with jokes. For example, how to refresh old routines by setting them in new contexts using the same punch line but new premises. And your relationship will thank you. And by the way, punch line is a line that flips the normal expectation for the situation premises which is the scene topic situation example gym culture graph school and parenting. And somehow it feels fresh every time. And that's because humor like play is a little prediction game. Your brain expects one thing, gets another, and rewards you with laughter. In parenting, the premise changes every 5 minutes. Say, my toddler is manipulating the system, aiming for something she wants, but endlessly generating new premises, that is in grandma's room, in the bathroom, in the living room, in a car. This is one of the best jokes I wrote with chat situation. My toddler is basically running a multi- aent reinforcement learning experiment. First, she goes to grandma for 5 minutes of Xiaoongu short videos. Grandma says, "You have to pee after." Okay. But it's free paid screen time. She ignores the contract. And the second situation, then she comes to me. No words, just takes off her pants, sits on the potty like we both know the protocol. Because in my lab, she only gets the phone if she starts by sitting on the potty. I didn't teach her manners. I taught her API integration, which means knowing the exact steps to unlock the system without understanding the system. So, it looks like she's obedient. She pees grabs my hand and goes, "Let's go dance." Which means, "Let's go watch dog dance video on YouTube." Punch line. At this point, we realize she's not learning politeness. She's AB testing our parenting models to maximize screen time and we are both failing QA. This is Rishi PhD. I'm a science writer and AI strategist. What I've learned during all my projects. We don't have time to become an AI engineer. But we need an AI as an appliance like a microwave oven. You push the button and something helpful comes out and you go back to life. But the fun part is we can use AI to make our everyday life efficient and everyday routine more playful. We can use chatbt to think more deeply, not less. And we can use it to understand our kids and ourselves with more softness, letting it expand what we can imagine, analyze, and express. And I'm sharing these experiments through my new podcast that's called Struggling NerdCast and the Chaotic Mom Nerdcast. You can also find how I use chat GPT to write this conference talk script that you are listening to with all the behindthe-scenes struggles. I'm also working on how AI can support parenting psychology, why kids meltdown, how emotional regulation could develop, and how to respond with empathy instead of panic. Okay, there will be a couple minute break and then we will be back with the next session. Okay, we're back with Robert Warden, consciousness and the free energy principle. So, Robert, looking forward to your presentation and anyone watching live can write in the chat. Go for it. >> Okay. I I'm talking as Dan said on conscious and the free energy principle. So we get straight into it. the FEP is very closely involved with consciousness because what I'm talking about is phenomenal consciousness which is the the what it is like to be alive if you like rather than access consciousness which is what we do with the contents of our consciousness. And a very large part of phenomenal consciousness is spatial consciousness. And that is having in our heads a conscious model of local space around us. And that is the main part of our conscious experience. Really is just understanding things in front of us, things around us, our own bodies all in one model of local space. And this model as we well know is constructed from sense data. And it is a maximum likelihood model. It's the most likely model that we can make given our sense data of the current world around us. And so it's a minimum free energy model. So that's why the free energy principle is very relevant to consciousness because the the conscious model central conscious model that we have is an FE model. And there are several FE theories of consciousness around. Uh the one I'm most aware of is the pro projective consciousness model by Rudra and others. But there is a problem about them because they are all neural models. They all say that consciousness arises from neurons firing in the brain. And as I shall describe all neural models of consciousness have a very serious problem. What is that difficulty? uh this is about to appear a paper in neuroscience of consciousness but what the liberty is that the brain is a neural computer it's obviously computing what to do given sense data and any computer all the information on inside is encoded in the brain it's encoded in neural firing and the point about computers is that they have encoded information inside them but the key which you need to decode code that information is outside the computer not inside the computer. I'll give several examples examples of this as we go on. So all information in the brain is encoded as neural firing rates and the neural events inside the brain don't decode them. The decoding is done outside the brain as I shall describe. But spatial consciousness, the consciousness of space around us contains unencoded information. And I'll describe that in a minute as well. Uh for instance, any information about the boundaries between things, the edge of your screen or the edge of your table or whatever it may be. And so [clears throat] that unencoded information can't be got from the encoded information in the brain without a decoding process. But the key we need to do the decoding is not there in the brain. So the result is that neural events inside the brain can't be cause of consciousness because the information you need to decode them is just not present nowhere to be found. So to go into a bit more detail detail about this, the first fact about conscious the main fact is that it's a model of reality and there's a kind of onetoone correspondence between the world out there and our conscious awareness of it. And the conscious the onetoone correspondence is the boundaries like a boundary between that little branch and the air is there in reality. It's there in our consciousness and there's we have qual inside our heads [clears throat] which [laughter] as it's well known you can't actually tell what the outside world is doing about quality your qualia of green might be green or blue or red but it's very consistent in other words if you got a yellow qual here uh and a yellow qualia there the two qualia here and there in your consciousness are the name the two qual in the to there's a correspondence between things in the real world. So all this amounts to is that the geometry of consciousness has a very close match to the geometry of the real world and it's a very detailed match. There's a lot of information in that match and that's the most informationrich fact we have about consciousness. So that is the fact that a conscious theory ought to explain. But the key point is that the consciousness of of space is not encoded. There's a direct mapping between this boundary here and that boundary there. So why can't neural theories do that? Because when you have a neuro theory of conscious, you have this the idea that there's loads of neurons in the brain and firing patterns in the brain encode information. And that is heavily encoded. That information in the brain is a highly encoded version of the information in consciousness. Here's consciousness. And so there's decoding information IBC called here which is very large. There's lots of information you require to decode what's going on in the brain to give you consciousness. Consciousness itself is very like the real world. So there's very little decoding goes on between consciousness and the real world. And the problem for neural theories is that that decoding information is very large and no neural theory tells us where that decoding information comes from and it's [clears throat] not actually in the brain. It's it's almost as though information in the brain was encrypted and somehow we got to decrypt it to give consciousness. Nobody knows how you do that decrypting. So I shall step back and talk about computation in general now not just in the brain and there's been a recent realization which actually a very is a very old realization by horseman and company by Stephen Kendon that computation itself is not on its own a physical process. It is a way we interpret a physic process. Uh and the interpreter is always in external to it. This information inside any running computer is encoded and the decoding key is not inside the computer. It's in an external interpreter like a person. And so without any decoding the physical events inside the computer have no meaning. They're not about anything. And this is not a new insight. Um it's been rederived recent by set in Kender and Courseman. It goes back as far as livets which I will describe in a so to make this point clear I'll draw an analogy between books, computers and brains. These three devices they all have information inside them which is physically encoded. Without encoding that information has no meaning. The decoding is always done outside the device. Now this point so inside the device there is no meaning. There is physical events going on but you don't get meaning from them without interpreting without decoding them. So this fact is actually quite obvious for books and the question is why is it not obvious for computers and brains? So meaning in books physically the meaning of what you've got in a book is squiggles on the piece of paper here. And there's a load of decoding we do to get meaning out of it. We decode the ink shapes as letters. Encode strings of letters as words. Encode strings of words as sentences. We decode the sentences as meanings. And we definely decode the meanings as stories. So we're all the time when we read a book and get meaning out of it, we're doing a load of decoding or interpretation. You get a similar picture for computers. Books are static, but computers are dynamic, but that's not a big difference. In digital computers, physically, what is going on is voltages in circuits, electronic binary circuits. And so to get meaning out, we interpret. And again there are several layers of interpretation. Interpret the voltages and transistors as bits. Interpret strings of bits as numbers. We interpret the numbers as solutions to algebraic equations for instance. And we interpret the algebra as models of physical things. That's what we do when we run a simulation the computer. And we only get meaning of the real world by doing all this decoding. This goes back as far as limits. uh in 1714 his book menology he is a famous piece and he says we must confess that perception and what depends on it is inexplicable in terms of mechanical reasons that is through shapes size motions if we imagine a machine whose structure makes it think sense and have perception we could conceive it as enlarged keeping the sames so that we could enter into it as one enters a mill assuming that when inspecting its interior we will find only parts that push on one another and we will never find anything to explain the perception. So he was saying essentially the same thing without using the language of computers a machine whose structure makes it think that is a computer and he was saying without decoding um we don't get information out of it. not meaning inside it. And Liners was saying this not just as a philosopher but as a mathematician and an engineer. He designed one of the first calculating machines called the step renner which is shown below. What he was saying was there aren't numbers inside that. Outside we have numbers and we interpret the numbers. We encode the numbers as turning the wheels and then these little pointers on top turn around and we decode those pointers as numbers are the result of calculation. The calculation is not going on inside the machine. It's us that's interpreting the results of the computation of of the machine as a computation. So li said it first. So if we finally come on to brains particular what's happening neurons are firing and again there's loads of decoding going on in our theoretical models of the brain. We interpret neural spike trains as sense data or we interpret them as probabilities in the basian models of the brain probabilities of things and events or we interpret them as geometry and synaptic connection strengths we interpret as learning and memory. We and the interpretation that main interpretation that goes on in the brain or outside the brain is that it puts out spike trains and they're interpreted by the muscles as motor commands which make movement. So that's the interpretation which happens outside a brain. So what we've said so far is that if you just have a computer, information inside the computer is does not have meaning on its own. It only has meaning when you interpret it outside the computer. So this is direct computation contradiction of a doctrine a philosophical doctrine called computational functionalism. And this is a thesis that performing calculations of the right kind is necessary and sufficient for consciousness. So this is saying that thesis is wrong. It's been a very popular position in philosophy for a long time but it's wrong. And that means that all the theories of consciousness that are based on it most of them are wrong. So if we go through some familiar theories of consciousness, most theories assume that it's caused by neural events inside the brain and that certain computational processes are taking place and I'll call those processes P1, P2 and so on. And the assumption of most theories is that some of these processes, some special processes cause consciousness and different theories have different special computational processes. uh it might be representing the things in the visual field or it might be in a FE model. It might be building basian models of reality or the higher order theories that say the the key thing for consciousness is the process of reflecting on the brain's own computations or it might be computing emotional response or so on. But the point about all these models is that those processes P1 and P2 and so on don't actually happen in brains. They happen only in our interpretation of events in the brain. So without that interpretation they could be completely different processes Q1, Q2 and so on going on. So neural theories which say the physical things happening in the brain are the cause of consciousness can't be right because those are the theories of the brain have lots of encoding. We have lots of interpretation of what we think the brain is doing but physically the brain is not doing that brain is just doing neuron firing. So how do we get out of that? And and [clears throat] here's the positive proposal. Suppose there was an analog model of space inside the brain. A model that's not heavily encoded. And this is the physical picture here. Suppose we have neurons firing away in the brain. And inside the brain, there's actually a physical analog model of space. So there is a lot of decoding of the neurons to create that physical analog model of space. And the analog model of space is as it says very like real space. So in that idea that analog model of space could be the source of consciousness that is a theory of consciousness which I've developed and which I believe has a great deal to do with the free energy principle. So you say why does the brain have an analog model of space in it? What use would it be? It could be very useful because it could help to store a basian model of what's where in space integrating sensitive. It could be a short-term memory persisting that spatial information for short times. It could be used to classify things to know what things are around the animal. So what things need to be something be done about them. It could be used to simulate and plan physical actions. And it could be used to compare the present moment with previous situations. So if you have analog model in the brain, it could actually in engineering terms be very useful. There are reasons for it to have evolved and there is a possible physical form for such a model in the brain. And the proposal for the analog model in the brain which is in an archive paper here is that there is a wave excitation in the phalamus and as is well known a wave excitation a three-dimensional wave excitation can hold 3D spatial information as a 3D for transform in space just as a hologram does a hologram is a physical example of an analog spatial model it goes through through a FIA transform. But that's not an extensive decoding operation. Doesn't require a FIA transform to get back from the hologram to the real world. Doesn't require a a huge amount of decoding information. It just requires a scale for the FA transform. Now, why does that fit the data? Uh how could there be a wave inside the phalamus? The phalamus as is well known is a site of integration. Lots of sense data of different modalities. They all pass through the phalamus and the phalamus has a very well preserved round shape which is very suitable to hold a wave in three dimensions. Now the other piece of evidence that I won't go into is that if you don't have a wave that the anatomy of the phalamus a small round thing in the middle of the brain doesn't make sense. it would do much better to disintegrate and move near the cortex. And another piece of evidence is that lesions of the phalamus tend to interrupt consciousness whereas lesions of the cortex for instance do not. And so the hypothesis is that the wave in the phalamus is the source of So I'm doing quite well for time here. We're going to undersshoot time quite radically. So, how does that relate to the FEP? Current FE models look a bit like this. A very crude sketch. There's a brain which is a neural computer and it has an outer markoff blanket. Inside the brain's encoded information in in neural pulse trains. There's a decoding process which converts strains into muscular action and that goes around to affect the world and through the senses sense data comes in is encoded. So there's an encoding and there's one particular kind of decoding which is decoding spike trains into muscle actions but there's no decoding of spike trains into a model of space. So there is no consciousness in these How do the models have to change to accommodate the proposal that there is an analog model inside the brain? It looks roughly like this. The same picture but made a bit more complicated. There's an outer markoff blanket between the brain and the external world. There's an inner markoff blanket between the neurons of the brain and the analog model in the phalamus. So there's a lot more encoding and decoding goes on. Sense data comes in is encoded as neural spike trains and is decoded into the analog model. There's a little feedback loop here between the analog model is encoded as spike trains again and it's iterated until the analog model is a good basian model of the external world. So there's an encoding and decoding cycle there. Also for planning actions there's a encoding and decoding cycle here. The model is encoded muscles neurons work out possible planned actions and they are decoded to compute the impact of those actions on the world. So there's more encoding and decoding goes on there. And the proposal here is that consciousness arises from the analog model. So the key thing is that we can carry on building FE models. We need this extra ingredient in them and is a very necessary extra ingredient because without them the FE cannot account for consciousness and consciousness is probably the key factor about the brain that we really need to account for. Uh so I'm coming to my conclusion now well [clears throat] ahead of time. Um the conclusion is we need to understand consciousness in the FEP but we can't do it with a neural theory because neural theories of consciousness can't be right neural complication it's computation is heavily encoded but conscious itself spatial consciousness is not encoded so we need something else in the brain the proposal is that the something else is an analog model of 3D space held anthalamus and so we can extend the FP in that way to have an analog model as well as the neural model of the outside world inside it. So uh there we are that's the proposal I would very much like to try and integrate this model in the FE and if anybody else is interested in working on building FE models which have analog models of space inside I would be very happy to to work with them. Okay. So, I finished way ahead of time. Uh, I hope there's a bit of discussion. >> Thank you, Robert. Ahead of time, but you can't be ahead of space. Um, okay. People in the live chat can write any questions. I'm just going to write down a a question was you you said physical events don't have meaning without interpretation. So then must interpretation be something other than physical or is it that the act of interpretation is inaccessible but it can hold the access consciousness of the interpretant. Well, some interpretation is done by us as people in our minds when we sit outside a computer and we we say the computer is computing this or that. There is also interpretation that isn't done by minds. For instance, the main example I gave was where the brain puts out spike trains and muscles interpret those spike trains as physical commands to contract. And so that's an interpretation which is done outside the brain but is not uh done by any other mental entity. It's just purely done by a physical thing, a muscle. So then is interpretation the consequence of the stimuli or a sense of the relevance [snorts] because >> um the interpretation is is the consequence of the stimuli. In other words, brain spike trains have a consequence that the animal moves. >> What would make that >> real action in the world? What would make the possibly waveform implications of like visual stimuli be accessed in an aware sense as opposed to the ostensibly unconscious or at least not perceived by a central consciousness of a muscle contraction. Uh well the the hypothesis is that for some reason which we don't understand the analog model held in the wave is a thing that causes consciousness and consciousness exactly mirrors that that analog model. So that's if you like that's David Chararma's hard problem of why something physical in the brain causes a mental thing of consciousness and we don't have an answer to that hard problem but I'm personally not worried about hard problems because in any science all of physics you come down to hard problems eventually so if you have a decent theory that explains most of the data and there's a hard problem somewhere like the hard problem of the mind is how do physical events in the mind make consciousness and the hard problem of physics is what's the fundamental particles you go down through atoms molecules protons quarks and so on and the hard problem is where is the fundamental particle I'm not bothered about the hard problem of physics I'm not really worried about the hard problem of consciousness there has to be laws which say physical events cause mental cause consciousness and we just have to postulate those laws and keep them as simple as possible. The law here that's postulated is that when there's a certain kind of wave going on in the talamus that is the cause of consciousness but it has to it you can't go down deeper than that. What do you think the architecture of this model that you'd like to build would be? And then what would it um reveal and how would it what would be the implications of what we could learn from it? >> Uh very good question. Um I'll just go back to the previous slide. Um and the architecture of the model is essentially that picture. Um so it is a fundamental addition to the FE to active inference models. Um I think we have to explore its properties in detail to start asking ourselves what are the implications. So it's a proposal that we build models along those lines with the inner markoff blanket [clears throat] uh as a boundary between the wave and neural firing. Um I think we just have to explore the consequences. They will be unpredictable till we've dug into them. Sorry, that's not a very exciting answer, is it? >> Yeah, I'm just looking into the center of the decoded analog model. And that is I mean, does this just regress another layer into a sort of opaque nested onion model? is if if [snorts] >> it's not at all >> it's not onion because onion you would say will go down to a deeper layer and a deeper layer and so on and so the a lot of the ideas of the FP like sort of scale invariant markoff blankets within markoff blankets don't apply you have this markoff blanket then you got the wave and that's the end um no inside markoff blankets no onions Is is what's relevant the statefulness of the center of the donut with the decoded kind of private key exposed? Because if so, then why would the statefulness um be the experiential element? >> Well, you're coming back to the hard problem again. We don't know why. [clears throat] I mean, it's just consciousness is there as a part of nature. We know it's there. And previously we've had to invent laws of physics to relate parts of nature. For instance, many years ago we had matter [clears throat] like hard things that you could touch and we had electricity and the two were very different and force and electricity were very different. So we invented bridging law to bridge force and [clears throat] matter and electricity. force and electricity. Similarly, we have to invent a bridging law to bridge consciousness and a material wave, a physical wave and we just have to postulate that law and don't worry about it anymore. Basically it's a hard problem relating to completely different phenomena like force and electricity or like uh waves and consciousness that is you have to just propose that there is a law then the law has to succeed or fail by fitting experimental data. you don't worry about the philos philosophical um foundations of proposing such a law if it if it gives you a good theory that fits the data. Yeah, it's it's a very interesting um I don't know if meta-cience is the right take, but definitely proposing a synthesis doesn't need to be constrained by repeatedly the fact of the matter that prior to the synthesis the phenomena are fragmented. That's kind of that's the starting point of the proposal, >> not the not the um conclusion of of the execution of the proposal. And then where it becomes the research ahead in the development is what are those implementations? What kinds of models do we need to sweep out here and what measurements and what empirical data sets to connect to >> of which the data sets wouldn't necessarily be of a different kind than data sets that could even already exist. Like it could be um propriceptive data or spatial awareness data. It it could be uh developmental information on when children or different species have awareness like object impermanence. That's right. I I think the test of a theory is the fit of data and as you say the data is all kinds of data relate to consciousness. the the main piece of data that relates to consciousness. I'll go back to the slide is this one. The fact is that we do have this thing inside our heads that just very like the real world. That is a huge important set of data points with lots of information at any moment of the day. We've got this whole conscious model of the world as a a very tight onetoone correspondence with the real world. That correspondence that matching is a huge experiment fact. And so I think that's the main thing theory of consciousness has to has to account for. So we need to just build those theories and test them against that fact. It's a kind of private fact which just makes it a bit awkward but um nevertheless I I think it's a private fact that we have >> what is the private fact? >> The private fact is that I'm aware of the world right now as a three-dimensional model. You're aware of the world world right now as a three-dimensional model. There are things we can't talk about like I can't talk about my green qualia in a way that persuades you that you're seeing green the same way as I see green. So there's private parts of it. But we can discuss public parts of it. For instance, if we're in the same room, we could look at the same thing and say that's a straight line or that's a flat surface or that's a bump and we could agree on that. And those things are the correspondence between my private model of world and the real world or your private model of world the real world. And the fact that they match up in this neat way is a big experimental fact. >> Yes. >> Um Good. The >> main thing that people I think will find uncomfortable is that there been so many neural theories of consciousness and there continue to be there's there's a website with 350 theories of consciousness on it and most of them are neural theories and all neural theories have to be wrong. There is something they're missing out this decoding step and I'm saying the decoding happens physically in the brain to create the wave in the phalamus. I I'll read a question from the live chat. So B sulfam wrote fascinating and provocative theory. Could you comment on the possible mutual illumination between this conception and of the semiotic thinking of Pierce and the connection with introspection? well uh probably semiotic I I can't actually comment much but introspection particularly I will comment on. I think introspection is the brain is the mind thinking about its own thoughts and that I think is not necessary for consciousness. In other words, plenty of animals have no introspection as far as I know. But in this theory, we would expect those animals to have spatial consciousness. So the theory says spatial consciousness has nothing to do with introspection. It's not a higher order theory of consciousness is it's a base order theory of consciousness. Yes, some people to the to the question, some people have explored Majid Benny and and colleagues have explored the Pierce connection specifically and I think where it connects is is this question of primary semiosis and um higher order theories, metacognition, introspection, these kinds of inner seiosis that don't deal with the primary like redness of red, the onioness of onion, the the atandness of of an affordance, but rather the the introspective awareness of that primary semiodic event. >> Yeah, I I think primary semios is is is the way to put it. Primary as a as opposed to high. Yeah. >> So, what are your next moves? what what would make it a good productive research holiday season in 2026? well, I'm not an expert on FE implementation and I would really like somebody to step forward and who does know how to do FE models and say let's build an F model. Let's take the F fe tool kits we've got or new toolkits if we need them and let's build an FE model complete with variational minimization and so on and let's apply it to apply this and incorporate the theos wave in it see what the model looks like. >> Yeah. And as we've discussed also in the insect setting, it might be more accessible to integrate a a biohysically based uh fly body or insect body simulator with a full brain emulation. >> Yeah, absolutely. I mean, I haven't talked about that at all, but this theory in that mammals have the phalamus, which could be a possible site for the wave. Insects have the central complex and the central body. You know, the fan shaped body and the lipoid body, ellipsoid body, which are very well preserved across all insect species and seem to me to be a likely site for the sensory integration, the same sort of sensory integration, which could lead to the same wave exertation and the same spatial consciousness. And I think insects are a great place to look for it because their brains are so much simpler to think about. >> Yes. Lot lot to uh think about. A lot of good motivation for for um tool development and for how we uh keep working on these questions. Anything else you'd like to add? Uh, not really. I'm sorry I've undershot left you some fair time, but >> it's all good. >> Okay. Thank you, Robert. Farewell. >> Okay. Thanks a lot, Dan. >> Bye. >> Thanks. Bye. All right, next up is a 9-minute lightning talk from John Graves. Here we go. Many automated systems take data and apply some kind of reasoning called an algorithm to help make a decision. The more advanced ones can even plan and execute a sequence of actions to reach a specific goal. But the most sophisticated systems evaluate counterfactuals. Judea Pearl and Dana McKenzie called this the ladder of causation in the book of why. Seeing is the first rung. Doing is the second rung. Imagining is the third rung. The type of questions the system can address become increasingly complex as we climb this ladder. At the first rung association which is unfortunately about as far as traditional statistics can take us. We can ask how are the variables related? What if I see X? What then? At the second rung intervention, there is prediction about the world as it exists and feedback from actions taken in the world with the possibility of the difference between the predictions and the outcomes driving a learning process. So doing enables learning and here the questions become active. What if I do X? Does the action I take have the result I expect? At the third rung, counterfactuals, there is a prediction about the world as it might be or might have been. What could I do? What could I have done? So these second and third level systems don't just ask what is the answer. They ask what can we expect when we do or imagine doing things. This is the essence of having a causal world model. The theory behind these systems is presented in the book active inference by Thomas Parr, Giovani Pizzoulo and Carl Fristen. A learning system operating in the context of a world model can reason about hypothetical situations and potentially use that information to map out a sequence of actions which is optimal within the limits of the systems current knowledge. That means truth is subjective from the systems perspective but empirically verified. Feedback from the actions taken can drive learning expanding the knowledge available to the system recursively improving future predictions and driving more optimal actions. Alternatively, the book Everything is Predictable by Tom

Shivers presents the historical development of basian logic. Shivers makes a number of very wellphrased observations. First, the great insight of probability theory is that we should look at the possible outcomes from a given situation, not what has gone before. The probability of an event is the number of ways we know of that that event can occur divided by the total number of things we know of that could occur. Probability thus gives a quantitative measure of the things we don't know for certain. It may require a subtle shift in mindset, but think of certainty as a continuous rather than a discrete binary value. One can represent complete certainty and zero can represent complete impossibility. But the points in between can reflect levels of confidence and skepticism with 0.5 representing an equal possibility of true or false, a state of complete uncertainty. The key insight of the basian method is that you must add any new information that you get to the information you already have to make each subsequent prediction. Intuitively, think of learning by trial and error. When we learn from the errors, we don't make them again on the next trial. Sampling or statistical probability is different from inferential probability. What can we predict about a sample given that what we know or think we know about the whole hypothesis testing? We don't really want to do that with inferial probability. We're really asking what can we conclude about the whole given a sample or samples we've taken. Beige showed how starting with an estimate of what is most likely to be true to your best knowledge, your model, you can make inferences and draw conclusions about what is likely to be true in the world. Consider, as a practical example, the doctor who looks at a patient's data, particularly their symptoms, and has to reason to the most likely conclusion, a diagnosis. Diagnosis guides a decision about treatment. and treatment may or may not result in a cure. Since the world is messy and often ambiguous, data is never perfect. Systems must deal with uncertainty, changing circumstances, and incomplete knowledge. This is where genius agents can help. They're designed for decision-making under uncertainty. The following comes directly from the genius documentation. Genius agents tackle ambiguity through probabilistic modeling and basian inference. This is very different from the approach of large language models which are built on a foundation of neural networks. Genius agents use a probabilistic graphical model to represent the world. Specifically, a directed asyclic graph which links the model variables frequently from cause or prerequisite to effect. We'll focus on two related types of graph. The first, discrete basian networks are a great starting point for introducing the core logic and themes behind how these agents reason and make predictions. Here in the sprinkler example model, the cloudy variable has arrows to both rain and sprinkler while the wet grass variable is the end point of the arrows coming from rain and sprinkler. The arrows make the graph directed [snorts] and the fact that there are no loops makes the graph a cyclic. The next step is to add probabilities to each variable. Cloudy is shown as having a 50/50 split. Then rain on the left is shown as having an 8020 split when it is cloudy but a 2080 split when it is not. Sprinkler on the right is shown as off nine times out of 10 when it is cloudy but split 50/50 when it is not. Finally, wet grass is shown with eight different probabilities depending on the four states possible for the two variables sprinkler and rain. For example, in the first column, when the sprinkler is on and there is no rain, there is a 0.9 chance of wet grass. The notation for expressing these conditional probabilities uses a vertical bar to denote given. When a probability has multiple dependencies, they are listed separated by commas. A factor graph replaces the arrows connecting variables with these probability tables, which are then referred to as factors. This sets up the system to make two types of inference. First, a forward inference, also called conditioning, where, for example, an observation of either the rain falling or the sprinkler running in green, allows for the inference of wet grass in red, or the inference can go in reverse. So, observations of wet grass with a sprinkler running in green can be used to infer the chance of it being cloudy in red. The next lesson will cover how the variables, states, and factors come together in Genius. right, Pablo, how you doing? >> Hello. >> Hi, Daniel. Uh, doing good. Uh, are we live? Not yet, right? >> Not yet. Not yet. >> Okay. So, Okay, welcome back. Hello Pablo. This will be a little different session. I have put a link into the YouTube live chat and it's available in the program as well. And Pablo is going to share this space and then we're going to just run around in that space and you will share whatever you have prepared. >> Yes. Awesome. Okay. Um I think I might be sharing my screen now. >> Yep. >> Okay. Awesome. Um hi um Pablo. um going to be talking about uh what could be the future of internet. Uh we think the next inter iteration of internet is going to be telling stories in uh 3D environments and with of course agents hopefully uh agents uh with active inference uh systems and we have prepared this environment for the community and for uh the the symposium. someone already. Paul, [laughter] don't worry. You will be back. You're okay. And uh the idea of this um space is explore the space and collect the uh four different um uh symbols. uh uh that we are are around. So we should be like exploring and sharing. It should be very educative. Um we have been uh involved with the uh community and the active inference institute for I think already three or four years and it's uh yeah it's an honor to be doing this test uh take this as a first um approach of a 3D internet and hopefully next symposium we will have more to share. So, we should be exploring the different islands. This, as you can tell, is a kind of a brain and an interpretation of [gasps] what could be a decentralized uh symposium. Uh you can follow me. My name on the top of my avatar is Pablo FM. And we can go and start exploring a little bit. So here is the program of the symposium. So, we can uh tell how it it goes, the different days and the different uh persons that are going to be talking. Yeah, this this is not working. I'm going to try it any um anyways. Oh. Ah, [laughter] so many levels there. We're we're screen sharing. >> Yeah, >> screen share yourself. >> Yeah, cool. So, the idea is that uh we can move around and go for the popups and read here the adventure. H interesting. >> Yeah, it's fun how you pulled out those quotes that are a little bit like a a doorway between the technical content in the textbook and just these bigger adventure and mystery questions. >> Yes. Um not sure if it's behaving as I expected. Welcome to the adventure. Here is a cognitive map of the active inference but it's incomplete. Um the four uh states of the system are missing. So find the four uh ellipses gems and send them here uh to the correct state. The books will give you clues about where the states of the of each gym. And yeah, so we should be like looking for Wait, I'm going to fall. I'm a little bit nervous because I think it's not behaving. Yes, it's working. No, it's working. Okay. here I it here we have some friends we can go all together to explore. Yes, you can join me and let's go first to this island of the drift. the island of the drift. So here is the book. So this state uh alters the environment and with it per perception itself it does not uh arise from impulse but from the need of reduce uncertainty not only transform what exist it transform what the system believes. key to move is just uh to displace it is to test a hypothesis with the body and re reshape a world to match uh your model. Okay. >> so we're reading kind of a riddle and then we're picking which of the four states of the particular partition it corresponds to. that's right. Which one would you say it is? Let's go with active states. Yeah, [laughter] there we go. So, for someone that uh knows active inference, it should be easy, but I think it could be a cool tool to explain uh newbies or people curious about uh the ecosystem, how how this works. So let's go and let's find the next island which is uh island of the rema island. So uh one thing that we did here is that you can also infer which state it is without even reading which is not the perfect but uh for kids and people that don't really is getting the this level of of complexity it's uh I think it's a good idea. So um in this case uh is not part of the system uh nor it is under it control even so it saves every uh signal that could be interpreted. It's uh ex excitance must be in inferred um street can never be observed. A key some causes can't be touched but without them your perception would be noise. What reals is happen uh what real is what persist even when unseen. Okay. Sorry for my English. Um, so yeah, I think it's uh the external estate but um okay we got uh two out of three of four. Whoops. You can

can see already there that we are completing the riddle or the treasure hunt. Uh, let's go for that one on there. I think we need to teleport to the island of Bastiga. There we go. Yes, we use that teleport. And here is the fourth state. uh this state is not uh directly uh perceived yet it guides every decision the system make. It is a probabilistic uh representation of the world built on experience and expectations. It does not respond to the present but to the model the system manifests uh of reality. Key predictions is not what you see it's what you believe in you are going to see and that's already changes what you see. Okay. [laughter] So as we can infer is the blue. So we already know that this is going to be sent to the internal state. Okay. Awesome. We need to go back to the main square and it's only more one more to go. It's already it's out [snorts] there. Iceland of haze. So, let's find Yes, this is the teleport. Whoop. Here we are. And the last one is um this is the only state directly accessible. It is uh discontinuous changing arriving like a rust. It can inform but also the safe if taken as truth is the first thing that happens and also the easiest to break. It's only useful if you knew what you expect to feel. Okay. sensory state. There we go. So now we have complete the the adventure. Um [clears throat] we have some Easter eggs and you can also download the active inference book from here. Uh, it opens um the link straight to to the download. And here is an 3D image of something that could be familiar for most. And then because we have discovered everything we can get our seed of knowledge. um now we have managed to do the this small adventure and this is going to lead us to another adventure that we did uh last year. I'm going to I'm going to enter into the into this world. We have built three different spaces for the active inference institute and having a blast uh during the the the building of them and trying to explain as easy as we could uh for new uh persons that would join the active inference institute. We are not uh scientists. Uh our team is mostly uh studyers and tech uh lovers. Um and this is the adventure of curiosity and this is another riddle that tried to explain uh how the active inference um system it works. So we have the internal state uh road and here we have a small um in instructions and here we have also oh an outside. So we have our brain and this is a representation of uh what we already have been uh seen as the main uh explanation of of the complex maths. uh this is the represents the mark of blanket and this is the sensory uh state where some of the notes are missing on each section. So we should take notes of what are the missing notes and with uh this information we can work walk work around and try to figure out how how all this goes. Here is the basic music lesson. Everything Okay. So these are the missing notes that we were already inferring. We can talk with the teacher. >> It could is it is it intentional or possible that the missing notes are the matrices that we need for active inference? Uh yes that I mean it's intentional that uh you need to infer uh to finish the the adventure. I'm not sure if I understand properly but uh the idea is that you have here uh like to walk through the different notes and let's say this one is the first that was missing but if I fall I have not taken notes of the second note so I going to fall Um yeah. Oops. Yes. Here. So the idea is to fill the gaps uh and reconstruct the the the piece of the that is missing the paralysis. And yeah, we keep improving this uh proof of content that uh or very early alpha. Here it starts. >> Playing the song. >> Yes. And second the last ones do on the second pray. already me. So, we go back. I'm not going to finish the venture. I'm going the the other part. So to go to do which is C E [clears throat] and here oh I don't recall no [laughter] so yes we just need one more it should be should be done. And then you can get another piece of uh seed off of uh knowledge. So I'm going to go back to the symposium space and keep a little bit with the exploration because it's another few things that around here. So um if you're new also and you want to saw what's been uh how we build things on the active inference institute uh you can you can come here and probably through a more visual way uh uh get um the links where the lessons and the courses and the different um information that we are building uh would be would be available. The idea is that we already have uh five different uh years of the symposium and it would be have like one island for each year or something similar. So you can you can go through the uh different information that is uh is being uh served. Yeah. I I have saw something like the active inference institute ecosystem. So go to the ecosystem. This is uh part of the ecosystem that most uh that we have. So we have built and the other uh space is inspired on this uh and it should serve to um explore on a 3D. Oh, this link is broken. No, it's working. um uh the the third space that it should help the community to um to join these 3D spaces. Um the idea is that on the near future we could speak, share screens here, have uh meetings, um meetups and and pretty much interact on a 3D environment which we believe is part of the future of the internet. Um so here it should be like the information to be an a volunteer. Um if you are more about learning stuff you would uh uh have there the different links. uh if you want to become an intern or you here would be like the the teleport to the intern's uh space and it's it's just um um much funnier gamified way of understanding the internet and the learning and we we're pretty much working on on all this. So yeah, I don't know if we have any questions or someone has some c curiosity. It's something that we are going to keep building. So, and it's uh evolving and hopefully we we we do something very fun for for the institute and we believe that the education is way better with a gamific gamified and storytelling approach. So here this should be the space for the board of directors. And if you look this um from the top which I cannot fly yet but uh on the future we will be able to fly. And what we what we uh will see is the pretty much the same structure as we we saw on the previous picture. Yeah. Very cool. Yeah, it's it's it's a different mode of of learning and playing. It's really fun to see it when and and just it's as we kind of break into the digital and these kinds of spaces. Um we can have all of these linked spaces, different rooms and adventures. And right now are those characters just procedural dialogues or they could have LLM or they could be other kinds of things or what? >> Uh yeah they can have LLMs. We have already tried uh that in the past and it's very cool because then uh you can speak to them in any language. they can be um very uh knowable about one specific topic and if we have one LLM that only is for on boarding it would help people to uh get a better experience and they can explore and and be helped by these LLMs and the idea is uh yes uh during this year start uh having not uh whenever because I think it's useful to have like a writing or or previous um uh NPC as speaking and telling you what's what's going to happen and at one point it can become an LLM and and it could be also very fun and and and cool to to have them move them moving around this basis and and serving uh to the uh to the institute and to the to the ones that have curiosity about how everything works and and all that. Uh anyone that want to participate on this project is an open source project. Uh it's all about experimentation, learning and having fun and imagine a better digital future. uh we really believe that and have hope on building something that is not very dystopical. So um we we we try to do our best and put our grain of of arena to to build a better future for the ones to to come behind. So, anyone that's curious or can add value or have any ideas or like or dislike anything, any feedback is welcome. And yeah, we are around the discord channel mostly yeah very passionate about uh education and what active inference is bringing to the world and how we are uh on open science uh learning and yeah it's been a blast being around this this so yeah >> thank you Pablo definitely and I'll ask some questions from live chat. So these will be some fun things to discuss. Okay. B Sulfam wrote, "Will the link work after the symposium?" Yes. Uh the idea is the uh internet uh 3D internet is going to be live and we could be improving and using these spaces to uh uh share maybe the YouTube uh recordings or streams that we do. >> We'll try to keep that link up. But then more more generally whether it's um adventure.active inference.institute or some other URL that we'll be developing over the coming years. We'll we will have these spaces be very uh clear. Um okay next question. Is there a separate URL for whatever the PI logo refers to? What does the PI logo refer to? I don't know Daniel you know [laughter] uh we we [clears throat] have we have this logo on 3D we will we will give this assets to uh we will put them so anyone can use them but yes please let us know. >> Oh yeah. Oh yeah definitely a lot of things they could be but that's the institute logo. Um where's the sign up? Well, for this there's no sign up needed. This just runs in the browser. >> And for other institute programs, we we always try to communicate it clearly, do our best so that it's available without any barrier and you can just email blanket at activeinference.institute with questions and as you enter other kind of programs or courses or experience then those are signaled clearly. >> Yeah. >> Yeah. The idea is that to keep it uh open and as low barrier as possible and then of course on the future it's going to be a sign up mode but it's not going to be necessary for most of the content. Yeah, it's it's fun. It really does feel like a new kind of journey and chapter in what even just technical education can look like and also these broader work and professionalism questions. Um I mean it wasn't so long ago I was in grad school and the idea that science would be carried out in a 3D space with chat bots and all these kinds of things. I I just didn't even have it on my radar and now it is a delight and it's just we have all of these options for chat and text and code. We can kind of generate the personalized textbooks for for broad fractions of material. Maybe at some cutting edge level there's some um poor performance but for core background topics where many of us will spend many years the materials are there and so I think it's really important and forward looking about what is the experience of those and how do we stay in this for the community and the the long run and the marathon and and the balance and health. Not just um propagating a sort of narrow view of what it looks like to learn in just kind of one style or just one type of material when the the real technical material are available in these fun formats too. >> Yeah. And it's a a layer uh ver easier to bring more people in and then you can go deeper on different layers of the knowledge. And yeah, for for us it was like the the main idea to make a top down layer of uh learning gamification and exploring. And this is still a proof of concept, but uh yeah, we believe that uh it's it's going to be something on the future. Already we have like Minecraft and Roblox and a lot of experiences that uh are very cool. [laughter] Yeah, it it's an interesting connection with the earlier talk of Robert Warden and the spatial consciousness because in these digital spaces, it's often hard to have awareness of like how many other people are in this channel or in this live stream or like what resources are there? And so we have folders of files and all of these infinite branching coded documents. And then once you have that inner map, exploring them is very streamlined. But there can be a big learning curve with just getting the first situational awareness of what is even there. It's like a big library with kind of branching adventure chambers. And so having a browserbased top level spatial navigation visual form helps put it all in front of you with what ways that you can move. And then that's the kind of learning by doing and and ultimately the active inference itself is like what is off to the right? It's like I do have to look. Oh, that's epistemic value or this is not what I expected for that. And so just even in the game play and in the awareness basically there's all of the touch points that we could want for going into these technical topics just from this gameplay. And we could talk about perception and cognition, action, all of that. just with what's happening here. So, it's it's a great combo. Um, okay, one more question in the chat. Do you plan voice interaction with nearby neighbors in the 3D space? >> Yes, it's uh we we did actually test it on last week and also sharing the screen. The idea it was for this year uh at least during this day uh sharing the screen on the on the main stage and and audio it would be a available. So if you are watching together all together this movie we could you know stop and discuss and talk uh on on real time on on what's going on. Um it's a sad it's sad that it didn't uh work for this year but we are super close of of doing that and in other spaces it's already working. Um also it's an Easter egg uh anyone interested on popups or 3D uh objects uh um sorry uh NFT objects or digital goods. Um it's uh it's uh this uh this uh this thing I will share the the link later and if you have uh if you write down this word you can have a collectible and that's also a po fun part that uh I think next year we will share more about digital goods and the future of digital goods and how we could gamify better the learning experience through these objects that um uh ensure that you at least have uh a degree of um familiarity with uh the concepts that are being talked or told in the different areas. So this is yeah very cool also. >> Awesome. So thank you Pablo to you and to the rest of your team. >> Thank you Daniel. Okay, till next time. See you. Bye. Bye. All right. Thank you, Pablo. Farewell. Good luck with everything. Thank you, man. Talk to All right. Next will be a 29minut pre-recorded video by Anna Pereira. Here Uh so I'm going to go ahead and um talk about making sense of uncertainty designing experiences for adaptive interface. And I actually want to start by taking a moment to explain what this title is because we're going to be bridging two two uh historically sort of separate worlds. one of um really sort of technical space of what's called active inference and free energy. And the other is we're going to be looking at okay well what does that mean for those who might not know those exact same words um or might not be intimately familiar with them but are either practitioners or individuals looking to better understand themselves and how to make changes um within themselves or create content um or applications for others that are net positive. Um so I'm Anna Pereira. Um I'm a research fellow here at the active inference institute. I also have a set of projects that you can see at actincycle.com. Um I'm a research scientist and also have a recent book uh navigating uncertainty simple scientific insights for everyday life. So let's go ahead and jump into it. So by way of introduction you can talk about okay why does this matter? So we have these things um that are called active inference and free energy and they tend to look like the image at the top of the slide here but they're largely inaccessible. So, if you haven't heard about it, that's pretty common. Um, it's been around for 20ishome years. Usually takes about 40 or to 100 for something like this to become common vernacular, but it eventually does, right? Like a lot of us didn't know what DNA was, but now we conceptually understand it. Um, so that's some of the hope behind what's happening here. Um, and why I think it's especially important is that this helps explain how humans can overcome uncertainty. And we have a high degree of that in society right now. um in design thinking. So more than just colors or fonts uh but the overall experience and understanding offers a bridge between this abstract theory and our lived experience. Um and so that's we're what we're going to explore in our time together. Okay. So in this context and we're going to slow down a little bit to make sure we're bringing everyone along. So if you're technically seeped um in the space perhaps thinking about it from a different perspective is of interest to you. If you don't yet know what active inference is, don't worry. Um, we're going to talk about that as we go. So, active inference is like the science, the very physics of us. So, who we are, we're constantly inferring what's happening within us and around us. So, you're doing that right now and listening to me, um, processing with the felt senses that are coming up. And what you're doing is you're working to reduce the uncertainty and the dissonance. So every thought, feeling, and action ties back to this fundamental process. So through understanding this science, we gain the insight to more effectively manifest our intentions, right? Because it's not my place to tell you what to do with your life. Um it's really up to you how how you want to do that. But part of my design thinking might be to encourage you to remember that you this is your single precious um sort of remembered lived experience of life. uh and then to put this in a bit of context. So over the past uh year and a half there's been a series of presentations from uh preliminary research to a book proposal around how this work could be done to create accelerate active inference in the free energy principle in a way that was mutualistic for the larger community. Um leading to a preliminary book release. it ends up some formats like audio actually take a really long time but hopefully by the time you're seeing this it's actually available and then finally we have a deeper discussion around some of the science and principles that are in the book which is happening here uh so thank you to the active inference institute and hosting this symposium uh so we can have these conversations okay so for outline we're going to talk about design strategy then we're going to go into active inference and free energy and conceptualize them, make them easy to understand. Then we're going to talk some time about um one small sort of um activity around it, right? Active inference and free energy is really large concept and we're going to narrow in on a potential activity, strategy and design. Then have a discussion and and finally close. So design strategy um again thinking about more than just fonts and colors um but thinking about translating complexity into clarity. So complex ideas like active inference and free energy they're um the downside of doing this is that they're going to lose some of the nuance when they're simplified. Uh and what we emphasize or omit really depends on what we're trying to do. um in this place in this case the cultural and contextual factors um but it really depends on what your your design goals are. So the particular design goal for this was to distill into clear relatable ideas that were meaningful to contemporary western audiences. And so that means a lot of simplication simplification, right? If you um look at some of the stats, about 28% of American adults can understand the science section of the New York Times. So effective communication requires a very high degree of of simplification. And one of my favorite places to to draw inspiration from is um Krisazout. They do a really great job of this um for a particular media and form. Um an additional goal is to think about breaking down our perceived reality, right? um sometimes I almost think of it like a a neutral omnipotent perspective of what we're thinking and experiencing as a way to help us kind of break down some of the biases that that we might have. Uh so as part of this work we kicked off with a preliminary survey. It was a sample of representative US population. We saw a substantial interest in the intersection of this larger wellness space and science. It's very interesting to um note that people were especially interested when it was giftable or under $20 US. Um on the right I have an image of what it looks like at the best case uh to get uh the sort of content out. So this would be a book with printed illustrations that you could have at home that you could refer to over time, you could mark up, you could come back to. So not something that's in a digital format. it's away from your devices, but it's also not at the library, right? Because it's more personal and more accessible and marked up. Um, in the best case scenario, you're more than double that. Um, this is not funneling money my direction. This is simply the cost of printing, shipping, and um, the third parties involved in that. Um, and I want to highlight this because it's a very interesting design constraint. Um, and it has really um, it's really interesting to reflect what does this mean from a societal perspective? We're not going to get into that now, but it has interesting implications implementations. So, um, about the design strategy, people are frequently more receptive to information that is visually inviting and emotionally resonant. So, thus, attention is applied to aesthetics as a central construct um because it helps build trust and engagement. So, it's a tool that we use in that regard. Um it's not something personally uh that I um devote a lot of time to outside of a strategic sort of lens. Um yeah, like a design lever um to help do this work. So let's think about conceptualizing active inference. So historically, if you're familiar with it, um it looks something like this. If you wanted to know what active inference was, you put it into Google right now. You'd probably um land on the Wikipedia page and this is the image you would see. Um it's complex, it's abstract, and it's technical. But then again, so is the math that's going to explain everything that's sort of behind us. Uh so one way that we can break it down is to start to think about domains, right? And we're remembering our particular design objectives. Um so we considered what it unfolds across the historical representation of threes and normative co cognitive capacities right you see a lot of threes in math xyz you see it um in [snorts] social contrast uh constructs like religion like trinity um and so the goal was to con contextualize these key principles in a way that helped people see new possibilities so we ended up landing in environment mind in body. Um, and you know, I think after they're learned, they're great to start reminding people, well, you know, these sort of like the Marov blanket, it exists, but it doesn't. It depends on how you're looking at it. Uh, and then the next thing you can do is start to play with color motifs. Um, and trying to think about, okay, well, what's distinguishable and what's memorable? So, for environment, you end up with green invoking the sort of nature. with body you end up with red orange sort of balanced and inclusivity um and things like blood and internal warmth and then with mind you often um we found ourselves with blue thinking about calm and thought and reflection. So the result is this visual language that is grounded in science and it's structured to be accessible designed to be repeated and elaborated. So here you see um a description of the active inference, an early description we use in the book, kind of breaking down these processes, but because we're thinking about pedalological methods and slowly introducing people to more complex methods, at this point we only allude to the predictive brain. Um so you can see arrows moving in the opposite direction. For those who are less familiar with what the the predictive brain is, it's the idea that what you're probably aware is that you can see the world, you can understand it, you can make actions. So that's potentially happening in this direction. But at the same time, your brain has already learned about the world and is making predictions about what it's going to see. So what we're experiencing is always these two things happen happening and has a lot of interesting implications um for the human experience uh that we end up getting to getting into in later chapters. So if we think about conceptualizing free energy now so we talked about which is the how the ongoing cycle the predicting the sensing the acting the updating we start to think about the why like why is the cycling happen in the first place and how do we reduce this uncertainty so we end up illustrating free energy um by adding gray for the unknown. So building on our existing uh visual motif. All right. So this is just one example pulled from one um chapter where you see active inference start to be laid on some summaries from the social sciences especially psychology. So with fragmented cycles or cycles with more free energy. You see things like rigid beliefs, resistance to new insights, sort of closed dialogue. You see the body having increased stress and disconnection, limited regulating practices, the environment, there's perhaps more conflict between science and religion, increased social division. And then in cohesive cycles, you think see things like more like reflective beliefs, openness to new insights, open dialogue, increased coherence in the body, connection, regulating rituals. um in an environment. And environment we mean more than just um say nature or pollution. We're talking about anything that isn't you. So this could be communities. For example, this moment I'm in my home in Switzerland. Um so my current environment is in my house. Um so in the larger environmental context in a cohesive cycle with less free energy you might see things like respect for science and personal beliefs increased social sort of um cohesion. So activity strategy. Yeah. So now that we understand the principles you could imagine we've been sitting at um a whiteboard sketching out what it means to ride a bike and how you ride a bike. uh we can start to think about okay well what does it mean to actually learn to ride the bike? Um so we can start to think about some of that strategy. And so one of the things we were thinking of doing is how do we increase people's capacity? Um and again it could be towards whatever intentions they have. It could be um how to adapt to rapidly changing technology. It could be changing behavior patterns they've been stuck in. Um, you know, in um in more extreme cases, you might have people who are really entrenched in things like addiction who might be um um might be looking to break free in uh in assistance from others. Um, and so some of the ways that we can start to think about this, and obviously these activities would change kind of depending on what your intentions were and and where you were looking were in larger in much more of the larger like wellness space. Um, and um, yeah. Uh, so you can then think of things like prediction, action, and learning are all deeply interconnected. All right? And so we have these suboptimal inference loops. Um, like maybe some uh, change behavior patterns. I know in my past I would find myself arriving at the same destination even after I made a series of changes and it confused me as to why. Um and then how intentional practices and again this is a subset of the larger active inference space but how intentional practices can support flexible updating greater coherence essentially breaking out of some of those worn in behavior paths uh that we might have or the ability to learn a new skill set um that we would like. And so, um, this work, it is, um, kind of the more in-depth exploration of what chapter 27 is in the book, uh, navigating uncertainty. Um, and what we're going to do is we're going to take a look at, okay, well, what does it mean to update ourselves through activities and we're going to compare that to algorithm training. So, we're going to borrow from machine algorithm training combined with active inference. What's interesting here is that humans train through embodied both mind and body experience. So you'll see how we kind of um expand what we do for normal algorithm training um to understand the science of humans and what that means for the human experience and updating ourselves. So ground truth for human systems perhaps unsurprisingly we landed in a space of mind, body and environment. Um, one possibility. Um, and so for for mind really landing in the intention, you know, is it like reflection, regulation or focus, you know, with the body, what are you doing with it? Are you walking? Are you sitting in the environment? Like what's the context? Um, is it social and the larger environmental conditions? Um, and when aligned, it reduces free energy or creates a sort of ecological coherence. um promoting adaptive clarity, effective learning and potentially meaningful experiences. Uh so we could take a running example um or you could almost put this to any sort of fitness activity. Um and so if and it this again would depend on the larger context of what's happening. But in this case, we could consider it as an opportunity to think about um increasing adaptive capacity through practicing um reduced free energy. So increased free energy more of this like lack of coherence and would be like a busy city landscape, songs with words, wandering thoughts. But if you were to use the same time and same activity to decrease free energy and increase adaptive capacity, you might run loops in nature with reflective soundsscapes. Um, borrowing uh from musical playlist used for uh psychedelics can be really great here. I know John Hopkins has a really great playlist um that's just really great for introsceptive sort of reflection and awareness. And then the focus might be somatic awareness instead of wandering thoughts. Um but again really pulling in that indiv individualization really matters right and it needs to be flexible to meet the goals and realities of the individual. Um that might mean age. We've already talked about what economics might mean for some cognitive capacity live context. These are just a couple examples of things that might play a role. Um and really emphasizing supporting this sort of self-generated exploration with enough structure to reduce uncertainty. So this very delicate sort of balance. And so essentially the goal is just to use the science to help humans today. Um and this kind of communication can help people begin to shift their experiences. Now I also hope and think that the science can help others create um for others through a lot of different means. Um and that's um one of the reasons why I'm choosing to communicate in this way because we're at a point in time where there are there is more work to be done than there are people to do it. So one of the hopes from this work is that this can spill out through different avenues. Okay. So let's think about actually designing the activity. Uh so the additional details especially those that are more technical and extend the sort of algorithm metaphors are available at the paper on actinccycle.com/paper. um just in order to streamline and make this condensed. Um we went ahead and just focused um a little bit more on the human side. And uh we landed in again for this particular example in this particular chapter um a framework that looked at a four quadrant model defined defined by the two dimensions of body low intensity, high intensity and mind quieted and highly stimulated. Um, and this offers opportunity for increased psychological flexibility and adaptive learning. [snorts] Um, and I'm but I'm going to keep saying this over and over again, right? Um, people have agency and um, I just like to remind myself how important that is to to remember and uh, people get to choose, you know, where they want to focus and some, uh, may choose to narrow their focus depending on their goals, their context and their current capacity. Um we focus on broadsp spectrum updating um because that is supported by active inference algorithm development in the broader um psychological literature. Uh I can't help but wonder those who do choose to to sit in one quadrant. how many might um choose to change and kind of diversify a bit more if they understood what they were uh what they perhaps had to gain by perhaps visiting some of the other quadrants and what that might mean for their main quadrant. Uh so sometimes I just wonder if it's an aware decision or just um uh one that that has been made less cons consciously um and what people might decide if they have the opportunity to uh okay so let's start with quiet mind and body. So mechanistically this is supports the uh sort of intraceptive awareness. Paris thatic regulation can function as a form of dnoising and det uh detect subtle persistent signals masked by everyday noise. Um so for this particular example we have restorative yoga uh to nature sounds and meditation to no music. Um these are just two examples of many um selected because they fit well in the presentation and they're well supported in the literature. But really there there's a lot to choose from. If we step into active mind, quiet body, you see things like uh cognitive stimulation without physical exertion. This is linked to improved memory consolidation and cognitive flexibility. So things like journaling and reading. Then you have things like active mind and body. Uh so you have embodied practices that require both movement and mental strategy. So it supports sensor motor integration, social coordination, rapid prediction updates. Um the examples we have here are martial arts in team sports and then finally we have quiet mind and active body. So this supports down reggulation of mental activity while maintaining somatic engagement. This is associated with things like reduced rumination, improved mood and enhanced default mode network regulation. Uh so we end up uh with this framework that um a number of people could play with either the individual or um a practitioner um who is uh perhaps working with somebody um to explore. Okay. Well, how how might a person use activities as one of the ways to uh create a more uh adaptive sort of um set of capacities within themselves. Um and again, this would potentially be one tool among many um as the human system is is complex with a lot of layers. So, some additional considerations. Um, yeah, this one I find pretty interesting that it might be beneficial to intentionally include activities with higher levels of unpredictability or informationational load. Um so especially because we're living in a time of so much uncertainty really training um this part can help cultivate resilience by reinforcing strategies for self-regulations and adaptive inference um as a sort of form of adaptive training. Um since I know there's a lot of um academics on this particular forum I'll mention one of my favorites is actually to go to unconferences. So imagine instead of showing up in a sort of domain specific conference and on conference is across domains and you show up and there's a large whiteboard and essentially you're going to fill in your talk and um so you're going to attend talks across a lot of different domains um and then in the evening spend time together as a community say like by doing a massive cooking project um with complex social dynamics and loud music. Um yeah, I just wanted to leave that here as um I imagine I if some of you are looking for ways to sort of stretch those quadrants uh given this is an act um an academic community. I I wanted to share that that potential one with you all. Okay. And then it's also interesting to think about the balance of variation. you know, which activities might want to stay consistent, but also, you know, does it make sense to bring in some variation as a a way to extend the sort of adaptive training and to explore more nuanced patterns of engagement and build resilience within particular cognitive or physical modes. Great. So, let's step into the discussion. Um so this body of work is meant to be this initial step that really uh contributes to translating active inference and free energy into forms that are conceptually regul rigorous and acceptable accessible. Um the framework induced here is intentionally adaptive designed to involve in dialogue real world real world use and ongoing research. This value is also intended to lie in the conversations, applications, and refinements it inspires. Um, if you have any feedback, um, you can leave that at actin cycle.com/feedback and I'd be, um, delighted to receive it. Uh, so this work, it relies on multiple sources of rig rigor. Um uh perhaps the biggest is the existing uh research and foundational literature. Um it's been truly remarkable to step into um that body of work and really see just how robust it is um across different domains and methods. um dialogue with leaders in active inference in the free energy communic uh communities as well as collaboration with experts across other domains pulling in other types of knowledge and wisdom and feedback um from users and peers. It's also interesting to uh really think about the fact that humans often update through discomfort but many avoid this tension right making it one of the greatest challenges for implementation. So helping people update more effectively with less suffering may be a really crucial lever. Conclusion. Okay. So this is essentially a starting point for bridging complex theory and practical application. um accessible vis visuals and experiential framework strategy uh based on robust science aim to foster learning engagement and adaptive action. You know, how can we make these changes? We might be looking in ourselves whether it's desired to learn a new skill set or break an existing pattern of behavior, increase capacity, be a better leader, develop a new tool for somebody else. Like these are all all intertwined. the same science stance um and meant to evolve through feedback, continued dialogue, and interdisciplinary exploration. Um the references, they're listed in the the paper at actton.com/paper. And I want to take a quick moment for the acknowledgements. I'm super grateful for the group that came together around this. um Carl Princeton uh for being the academic adviser and um taking the time to really engage with the community even uh when seeped in the more um technical world. Al Ludo who just was such a great designer to work with and iterate on some of these illustrations and motifs to really make sure we got those things right. um and the colors and just really built such a helped build such a beautiful design strategy for this. Um Zach who was the editor um we uh of course used AI as a complimentary tool. Um everything originated with humans and was a final stamp by humans and created by humans in the middle but Zach really helped make sure that that communication um was very human. And then Sylvia, which I imagine you'll hear from um other times, uh was um a much appreciated reviewer and also did the initial lift on the 500 references uh in the book. Um I personally had a chance to go through them and again we'll just um I know many of you watching this have probably contributed to that space. Um it's truly a remarkable body of work um of which to to get to intersect and work with. So um thank you for that. And um I hope it's not too much for me to um to sneak in this last thing, but we saw in the earlier slides that it's really hard to make the economics of this work. So please consider supporting the work um either through um reviews, recommendations, um acquiring a book or um providing feedback. Um any of the above are very much appreciated. So, um, thank you so much for listening and I look forward to continuing the dialogue. Okay, we're back with Matt Brown, adaptive robotics for the real world, a new paradigm for autonomy, and a repeat symposium guest. So, thank you for joining again, Matt, and to you for presentation. >> Thank you so much, Daniel. Uh, and glad to be back. Um, I hope I'm not giving everybody whiplash. Uh that was a great talk by Anna and this is kind of a a big shift in topic uh and probably a bit of a provocative title. Um so I'm Matt Matt Brown. Great to meet be here. Uh I'm I'm with a company called Thought Forge. Uh and we're doing something uh a little unusual. Um Thor Forg began as uh a research project back in 2007 exploring uh alternate biologically plausible uh approaches to machine learning and we're applying it to adaptive real-time dextrous control for robotics. Um, just a little background on myself. Uh, I've been working in AI for almost 25 years now at this point. Um, including time in the defense industry, video games on franchises like Halo, Bioshock, Splinter Cell, stuff like that. Uh, and then about 10 years ago, uh, was an early pioneer in deep reinforcement learning, uh, at a company called Bonsai, which was acquired by Microsoft. Um, and the last five years, I've been really kind of, uh, focused on taking this interesting research technology and applying it to real world applications. um so just a brief agenda about what I'm going to talk about. Uh last year I gave a talk around the low-level mechanisms uh about what we're doing and the um kind of the approach from the bottom up. Uh and this year I'm starting from the top down. Uh I'm talking more about um the current motivations and goals for the technology. Uh how it's being applied and a little bit about the methods. Um, and I'm trying to keep it to more interesting aspects, more of the philosophical stuff. So, I'm going to start starting out by talking about robots and dexterity, why it's hard, and why I think a new approach is needed. Um, then kind of contrasting current AI approaches uh with u deep learning. Um, and then kind of just talking briefly about the the techniques that we're using and our implementation. and then I'll show off uh some videos and applications that we've been working on over the last couple years. Um so why are people trying to apply AI to robotics? And I'm apologize for anybody in the audience here who's uh very familiar with robotics because a lot of this is going to be super obvious. Um traditional robots operate in controlled environments doing repeatable things over and over and over again. That's kind of what they've been really been good at for the last half century roughly. Um, and they haven't really been applied outside of that because the real world is messy, uh, unpredictable, uh, variable. Um, you know, everyone's aware of of of these challenges. I think um what AI is looking to solve or what people are using AI to try to solve are, um, three main things. Um, perception, understanding the world, you know, understanding when things are not uh, in their controlled um, state. Um, adaptive manipulation. um people really would like robots to be able to do a lot of the the tedious man uh manipulation work that's done by humans um around the world on a daily basis. And then generalization, being able to um adapt u to learn and adapt without having to be explicitly reprogrammed for every every situation that they're in. Uh and at least right now, AI seems to be the only viable path to unlocking uh robotics in these kinds of challenges. Um, so let me just make sure I didn't skip a slide. Okay, so these are the kind of cutting edge approaches in AI that are are currently being explored. Uh, and where there's tons of money, you you probably hear about, you know, bunches of companies getting funding to go after AI and robotics, these are generally the techniques that they're they're using. Uh, visual language models. Um, basically use LLMs to kind of ground vision and actions in uh through language. And there's some examples here with RT2, Aurora Gen, OpenVLA. Um, and they none of these solutions have yet been proven to solve all the problems or gotten to the point where they can be used in factories reliably or in homes. I should probably start start out by saying that. And so, you know, each of these have their own challenges with visual uh language action models. They they have poor action fidelity. So, they're they're very kind of poor quality motion. Um, and al also language is not necessarily the best way to understand the physical world. uh and they're also uh very non-deterministic and so it's it's very hard to know with confidence that they're going to be able to accomplish a task reliably. Uh diffusion based control and planning. This is basically using techniques from um image generation or video generation um where they use kind of diffusion um from noise to generate images. They're doing the same thing for robotic uh trajectories for generating motion. Uh and there's there's there's sometimes this is combined with visual language action models and with foundational models. So people try to mix and match. Right now this is very very ripe for a lot of experimentation. Uh the problem with diffusion is that you need really really high quality data and that's really hard to get um you the simulations usually not good enough. So people are teleoperating the robots. So you'll see a lot of that uh to generate the data but that's also very tedious way of of getting data. Uh it's also very expensive to to actually use um in the real world. Um and it's brittle to distribution shift. So anything changes, anything about the task changes or their objects change um and then the the policy stops functioning well. Uh and the last one is is kind of a a broader goal which is the foundation models for embodied intelligence. And this is generally people trying to create a chat GBT moment for robotics where if they get enough data um about physical manipulation in the physical world uh you'll get to some some threshold and you know these AI models will suddenly understand how the physical world works and then you can kind of use that as a base model to go after uh individual um individual tasks and and the problems here are are somewhat obvious. The amount of data that you need is enormous. The amount of compute that you need is enormous. Um, and I don't know if I talk about it in the next slide here. Um, yeah, I I was going to talk about how how how challenging the data problem is. Actually, I'll just move on to the next slide. So, um, so the the these are the main problems with the traditional approaches in a uh AI applying to robotics. Um, and for the most part, data is the biggest problem. uh there's just not enough um there's not an internet of data, you know, of of robotics data that to train on the way that Chapet GBT might have. Um and it's there's a lot of different embodiment, you know, different numbers of actuators, different kinds of actuators, different kinds of sensory input, vision, tactile force, uh propriceptive um and every robot is different. And so getting enough data, um that can generalize across all different kinds of robots in the physical world, we just haven't gotten there yet. Uh, and the the example I like to give is um self-driving cars. It's, you know, self-driving cars are finally getting to the point where they're they're they're usable and they're they're functional, but it's taken over a decade of data collection, billions of miles of road driven, and that's just to handle one use case, driving a car, uh, as opposed to the expectations for robotics are that they're going to be able to do everything, you know, uh, clean your house, uh, work in factories. Uh and so the the uh the mismatch here between the data set sizes and the expectations are are really really quite large and there's you know same problems that you have um with self-driving with the longtail variability. Um it's very hard to simulate reality accurately. There's just anything can happen. Um and uh similar to self-driving cars in fact um when one of these models fails in the real world it has very expensive uh and often uh dangerous uh consequences. Um the other problem is compute. uh the amount of compute to actually run one of these these new age models is absolutely astronomical. Um and robotics have real time constraints. Um they're latency sensitive. So if chatbt takes 5 seconds to give you an answer, that's not a problem. But if a robot's in the middle of manipulating something and it takes, you know, more than a fraction of a second to make a decision, uh you you probably have broken something or dropped something. Uh and so the the performance constraints are much much more challenging than in language. Um, and this is one of the reasons why um, AI in robotics is so challenging. This also means that your AI model can't live in the cloud. That means that your AI has to live be small enough to be on a computer right next to the robot. Um, because you just can't um, deal with the latency uh, that you get from working in the crowd. So, um, the the biggest biggest challenges here for a robotics still exist in that, you know, they don't really adapt to real world variability in the way that they need to. Um, and so what we've been working here at Thought for Forge is a kind of a new way of looking at the AI problem in general using um uh cybernetics as kind of the the original body of of theory that we're working on. And obviously there's a lot of connections between cybernetics and active inference. And I'll go into that a little bit um later in a couple slides u a little bit more thoroughly explaining that that relationship. Um but you know when I when I started kind of looking at these these problems and in fact the learning problem um I was very c curious about how nature solve these problems because obviously uh we exist in the world animals exist in the world how do how do animals kind of deal with these problems uh and they don't do it with huge data sets and huge computers training on that data o over a long period of time um and they don't just train once and then hope that they they can generalize. um and in particular animals don't separate learning from acting. You know, it's kind of um conccomatant and it happens at the same time. And so this is just kind of contrasting here. Um the kind of paradigms for what people are looking at with AI currently and what we should be looking at and what we're kind of um at Thoughtworks are kind of attacking um continuous feedback loops for perception and action simultaneously. adaptation at the point of contact. Uh and learning um shaped by physics and you know active inference and the free energy principle are are kind of foundational you know principles and and theories around that. U and the idea here is to leverage learning in the moment versus uh learning in the past. Um, so just to kind of dig into that a little bit deeper, um, you know, most AI systems today are are focused on prediction, um, and are are giant prediction systems, you know, and so the pro the approach we've taken is a huge shift over these oldfashioned deep learning approaches. Um, and uh, I'm I'm terrible at naming. So deep learning uh, is you can look at most deep learning networks as universal function approximations. So what you're doing is you're taking huge piles of data and then approximating that into a function that an input output function. Um I've called what we're working on inactive synchronization. I'm terrible at naming. I guess there's a quote, you know, there's two hard things in computer science, caching validation and naming things. And um uh we may have mostly solved cash invalidation, but I haven't really quite figured out naming things. Uh this is my current favorite name for for our approach uh in active synchronization because I think it's detailed enough and specific enough um to be useful. Um other things I've thought about calling it is uh but are related but maybe too generic and don't really give much insight is u you know uh generalized synchrony cybernetic learning metastable adaptation things like that um homeostatic intelligence uh or manifold synchronization learning these are all accurate but perhaps not as specific and what I mean here by inactive synchronization is that um it's a synchronization process um with kind of an inactive tint to it and in in that inactivism is is um views cognition as a dynamic interaction between environment and an organism. And um the world model that's created through this process through the synchronization process is co-created through interaction. Model and control are kind of intertwined. And so the world model itself is constructed of sensory expectations and and motor uh control um uh percepts essentially. Um and so these are kind of intertwined. And so this this world model is created through interaction and created by the control dynamics themselves um or they emerge from the control dynamics themselves. So other ways to look at this is um universal function approximation you know current deep learning systems are retrospective you know they basically take past data and to approximate functions and what we're doing is um pro perspective equilibri equilibriation uh basically continuously adjusting to maintain predictive coherence and maybe this will make more sense once we get a little bit more into the details um but traditional approaches are very data dependent time agnostic um whenever and feed forward meaning um they're input output functions so once you have an input, it computes and generates an output. Uh, and that output is directly related to the input. Uh, and there are ways of making it with recurren neural networks and things like that where it has some history. Um, but for the most part, every time there's an output, it's a direct u response to an input. Um, this is is very very different. Um, the these are time aware which means input signals are coming in, output signals are going out. Um, but the output signals the motor controls are not directly um a computation on the inputs. they are uh you can think of the model both generating you know uh uh predictions about their sensory experience and motor controls at the same time. Um and then where those uh where the sensory input uh the sensory expectations are um are are incorrect kind of filter through the network and modify future outputs of the of the network. Um so very different way of looking at the learning problem. Um, and the goal of what the the inactive synchronization here is to build up this model of the world that can kind of predict how the world is going to react to your your motor actions. Uh, and then the goal of the motor actions here are really to get the um change the world to to become uh easier to predict or or to put you into a preferred state. Um, and a lot of this aligns very very well with active inference obviously. Um, but this kind of gives kind of a high level um philosophical view of what we're doing. um the top down approach. Um so this is just a very very quick slide to connect the kind of cybernetic approach with active inference and obviously these are very you know closely tied together. Um the free energy principle [clears throat] is you know I I probably don't need to go into it too much uh but it talks about how uh all living systems act to minimize their free energy. Um basically pulling from physics a lot of a lot of these these um really great ideas [clears throat] about how the natural world works. Um and then active inference is a framework within free energy principle explaining how organisms actively gather information to reduce uncertainty and maintain internal stability. Um this is is um is kind of the kind of connective tissue into cybernetics uh because cybernetics uh was a field you know that's kind of disappeared in in in recent history um but built upon the same ideas um where it provides kind of the theoretical foundation for how feedback um and self-regulation leads to this uh minimization of of predictive error and you know uh uh this kind self-regulation, self-maintenance process. Um, digging a little deeper, our our specific techniques are modeled around um, a cyberneticist known as Ross Ashby. Um, and he had some really kind of uh, foundational ideas. Uh, he's most known for the good regulator theorem and the law of requisite variety which are are both related to each other in in some sense. And the good regulator theorem is you know basically says that if if one system wants to regulate or control another system that system has to you know have or be a model of that other system. Uh and the law of representive variety is somewhat similar. If if you want one system to kind of control another system it has to be at least as expressive or it has to have at least as much variety as the system it's trying to to regulate or control. Um and he saw homeuh homeostasis as kind of the goal of stability as kind of the the universal mechanism for adaptation in the natural world. Um and so he built a device in the 1940s called the homeostat very small and I I'll talk about it in the next slide a little bit more um where he kind of demonstrated um adaptation as stability or stability as adaptation. Uh and it uses feedback between the units to kind of generate a distributed homeostasis through a a local property called alter stability. Um so basically he had these um these nodes that uh in a in a network. These were built out of World War II bomb parts. So he was this is pre-digital computer. He was using um you know doing pure analog computing at this point um using recycled bomb parts. Um and each node is basically trying to is performing this property called ultra stability and when networkked together generate distributed homeostasis amongst the group. Um and you can kind of look at the network as a self model where each node is trying in this network is trying to predict its neighbors nodes. Uh and if all the nodes in the network are doing this you kind of reach this equilibrium this homeostatic equilibrium um over time through this ultra stability process. Um so just to talk a little bit more about the homeostat because it is incredibly important um to our techniques and what we're doing. Um it was the uh first device kind of capable of adapting itself to the environment. I I I guess you know depending on on how you want to look at it. Uh it exhibited behaviors such as habituation uh reinforcement and learning through its ability to maintain homeostasis. Um this he kind of viewed homeostasis as you know the fundamental mechanism for adaptation in complex systems. Um and he basically built this device to kind of show it off kind of what what he what he was thinking and it was you very kind of primitive at the time. Um but and I think for the most part people didn't really really understand what's going on here. Um but um at the low level in terms of what he was trying to solve, you know, there's a lot of um a lot of what they're trying to solve with reinforcement learning right now was kind of exemplified in the system. And um and uh I to contrast this with uh going back to kind of the 1940s, you know, models of the mind and things like that. deep reinforcement learning now a lot of people don't know this uh is based on loosely based on BF Skinner operate conditioning and so um really kind of old-fashioned ideas kind of put into computer science is what we're dealing with now and I would like to think this is a little bit more um forward-looking in terms of concepts for for how learning works um this was also described as an isomeorphism generating machine which um which is obviously useful when you're building a world model and you wanted to kind of abstract uh concepts in the world as groups and things like that um and I think I've already talked about how ultra stability generates homeostasis. Um, here's just a more in-depth look at what it was. Um, you can see a picture of it right there. By the way, this is this is Ashby himself posing in front of his homeostat in case anybody was curious who that person was. Um, that's a picture of the homeostat there. You know, these big clunky things made out of recycled bomb parts. Uh, on the upper right there, you can see just kind of a network diagram of what was going on. So, basically each node in the homeostat was connected to all the other ones and they also had a self connection. Um and uh the diagram on the lower right there is meant to describe the mechanism of what's going on with ultra stability and how these these um these nodes update. So uh in deep learning the nodes in the network are perceptrons kind of actually probably five years five or six years before the homeostat. Um you know it's kind of funny that all these attractor definitions came out of the same decade in human history. Um but so this was just you know a few half a decade later than the perceptron was this kind of model and uh and how they differ is that chaotic attractor definition um and so you you know it's very similar to the perceptron in the sense that you have incoming signals incoming connections from other nodes um with weights um and but how we actually update the network is very very different um these obviously don't use back propagation instead they use this um ultraability and so another way to contrast uh deep learning with um this kind of um homeostatic network is that in deep learning this back propagation process is a global update. You have to freeze the network you know not use it um do a a global update of all the nodes through back propagation. Uh with ultra stability the update step is local. So uh you don't have to freeze the whole network. The the nodes are continuously performing this ultra stability process uh in parallel. Um and so this is this this works a lot better for large networks. if especially if you're trying to implement this on computing hardware, you can kind of distribute your network across computing uh resources uh and you can kind of update them independently of each other. Um uh because of the way that the connections are all you know um recurrent um you can kind of think of this network as trying to predict itself. If each node is trying to predict its neighbors the the kind of group itself is building kind of a self-prediction model. um uh ASBY never actually connected this to uh external control systems or sensors. Uh and so the the base homeostat was really just trying to show how this uh equil uh could be reached um in kind of a me mechanistic way. Um so I just like to include this letter in here in in my talks just because it's it's a fun little piece of history. Um at the time Ashbby and Turion were were I wouldn't call them rivals but um Ashb was very much you know pushing for analog computing this homeostasis way of building um computers and computing machines and Turing was very much about digital computation uh and um and this is a letter from Turing to Ashb um basically saying that uh you know his universal computing machine is more general than the homeostat because you could simulate a homeostat on a universal computing machine. And and while this is true, I think he's they're missing the point. This is kind of the original analog versus digital argument. And at the time, um, you know, analog systems didn't really, you know, uh, complexity theory hadn't been developed and so stabilizing analog systems hadn't really been figured out in terms of how you can use negative feedback to make them reliable and things like that. Um, and so I think they're kind of conflating two things that I I think are still being conflated here where um computation is not the same as um, for lack of a better word, cognition um, and or universal adaptation is the different than pure semantic computation. And we've kind of I I feel like this paradigm that we're seeing in AI with you know uh all these different approaches are really just pushing further further down that semantic computation direction as opposed to understanding what's actually going on in intelligent systems or complex systems. Um and so I feel like this is the beginning of this this uh this divergence in in beliefs about how we should build a computer a computing future. Obviously ASP lost this argument at the time Turing everybody knows who Alan Turing is. Um, but it's it's a fun it's a fun little tidbit for I like to include. Um, so how do you actually build one? You know, um, the there there was there was one physical object built. I think it's still in existence somewhere. I think maybe the Ashki family has it. I'm not sure. Um, but I basically took Turing's advice in the last um letter here and simulating. So I'm using a modern tearing machine and simulating it on uh and simulating the homeostat as an analog system in there. is based on techniques. Um I before I got into all this stuff, I I was an early pioneer in what's called software radio. Uh which basically was I I don't know if I would call it a science, but a field around how do you how do you properly simulate analog systems in digital hardware um in kind of accur in physically accurate ways. And so we kind of used I use a lot of those techniques to build the first prototype. Uh, and one thing that's interesting here is because of the chaotic interactions between these these nodes in this in in this in these systems and the way that the homeostasis generates is that it's you can't um you can't you can't analytically calculate the network and so you kind of have to simulate it step by step and let it evolve over time. Um, and that's what I mean here by you can't you can't account for the interactions algebraically. You you the circularities and and the recurrence really have to be kind of play out iteratively over time. uh and there's a great quote here about uh from pri about complex systems uh don't forget their initial conditions they carry their history on their backs here and and what I'm trying to say here is that the structure of of how the system evolved itself uh stores within it a memory of the history of how it got there. It's a lot like you know how a snowflake itself is structurally uh a history of the the temperatures and um pressures that it it went through with during its formation. In the same way, these systems are also very complex and unique. Every time you run one of these, the way that it stabilizes is always unique, kind of like a snowflake, but the the end result that you get there kind of contains the history in in a way in a very kind of compressed way. Uh, and that's kind of a property that we want in a uh a world model. Uh, a dynamically generated world model through interaction. Um, so I started with kind of ashb's in in terms of how to actually build this. We I started with Ashby's kind of physical descriptions and analog hardware here. Um some of this came from his technical journals which were digitized almost 20 years ago. Um some really great stuff.

It's all handwritten so uh be prepared to to read his handwriting. Uh he's got some amazing illustrations in there, but it's about 7,000 pages of technical journals and so there's some gems in there if you're patient. Um so I started by pouring through stuff and building kind of a very um accurate simulation of the homeostat and then over a long period of time abstracted it down and simplified it down to something that looks a lot more like modern AI networks you know where you can kind of build these into abstract shapes and topologies. Um and each node is doing the same thing. Uh it's basically trying to predict its neighbors as the system as a whole reaches equilibrium. But you can think of this as kind of a loosely coupled set of dynamical systems. you know, each one of these is is a homeostat. Um, and an abstract homeostat. Um, and uh, yeah, I'm trying to think how much I want to dig into the the dirty details on this slide here. Um, you know, you can kind of imagine that this is still disconnected from an environment. So, we haven't yet in in at least at this point in the slide connect this to an environment with sensors and motors or anything like that. So, really all it's doing is when it's reaching equilibrium is finding a set of positive and negative feedback loops. You can imagine um these connections in the network here forming um a large set of overlapping feedback loops and depending on what the the value of the weights some of these are positive some some of these are negative and what's going on through this this local ultra stability process is that they're searching a state space here um of of positive negative feedback loops such that um they all are self-balanced. So all the you know once you re reaching equilibrium it's not a quiet equilibrium in the sense that um it's very much active um in that all these feedback loops are very perfectly balanced so that uh so that they're stable. Uh and and this you can kind of think of this as as a as a critical state. Um and this is really important because it means that the slightest change or the slightest new piece of data can completely destabilize um one of these feedback loops and cause cascading changes throughout the network. Um, and as long as you can kind of restabilize the network quickly, th this [clears throat] is great. That means that one small piece of data can drastically change the dynamics of of the model. And that's what you want in a system. That's the opposite to back propagation based systems where you need a lot of data in order to change the behavior of a model. Instead, you want one surprising piece of data to drastically change how the model behaves. Um, that's much more data efficient. You need a lot less data to get to a a useful world model this way. Um, let's see if there's anything else I want to talk about here. um in terms of how this relates to active inference um uh when it's re reaching equilibrium here, what it's trying to do is it's trying to minimize the prediction error of of of its prediction of its neighbors. And so once you're at equilibrium here, you're basically at zero prediction error. Um and all the networks are successfully predicting each other. Um so just at a high level, I'm going to spend a couple slides on these next few topics here. Um first is you know how I actually connect these to an environment through essentially motor coupling. um sensors and motors kind of define that interface between the environment and the model. Um then I'll talk a little bit about embodiment and why this is important. Um and in fact um I think to me the most important aspect of embodiment is the fact uh the existence of time in the system like the time aware awareness of it. you know, for me, I have a hard time looking at traditional deep learning AI models even in robotics where you have a body um being embodied because uh or considering them to have embodiment because they're they're time agnostic. You know, the you you have they have these input output functions and they have no concept of being situated in not just in a place but in a time um where their last action that they took um is relevant to the current one for example. And there are ways to to trick this by feeding, you know, especially in reinforcement learning, this is often what you'll do is you'll feed in your previous uh inference results as inputs to the to the function. But I think of that more of a as a hack. Uh and then I'll talk about um a little bit about synchronization as a learning process itself. Um in case that's not super obvious. Um so once you've built a homeostat, even a large one with a large topology, it's it's still kind of boring. All it's doing is kind of learning itself and learning to get to equilibrium. Um so you know one of the first things you have to do is um so it's it's self-synchronizing you can kind of imagine as a self-s synchronization process. So the next thing you have to do is you know if you want this to do something useful you have to connect it to an environment. Um and so sensors kind of define how the agent is embedded in the environment and uh the the topology itself defines what the adaptive capacity of the homeostatic agent is. You know what it can do and what it can learn. Um, you know, obviously the larger and more complex, uh, the network, the the more expressive and more complex behaviors you'll you'll get out of it. Um, uh, and then the sensor definitions, we've kind of, um, I I call them sensors, but maybe they're inputs. I'm not great at semantics here. Uh, naming things are hard. Um, the sensor the sensors here are actually really important because they also act as uh, tly dynamic priors in terms of getting this to do something useful. uh you basically can define what the homeostatic set points are in the sensors. So the sensors aren't just data. They're not just, you know, pixels coming in, but they have a a um they are a they have goals implicit with them. And an example might be instead of having a temperature sensor that's just saying 68 degrees or 70°, instead it's saying, you know, I am 2 degrees away from optimal or I'm 5 degrees away from optimal. So the idea here is they have kind of implicit goals in them. Um and those can be seen as like input attracting sets um which you can kind of um from externally you can modify those in real time to get more interesting behaviors and to kind of change what the model is doing at any point in time. Um the motor definitions here are basically just coupling points that go outward. So there in inside the network there's a lot of recurrence a lot of birectional connections. The sensors are almost by definition input only. So they're unidirectional and the motors are also unidirectional outward. um and they basically provide an uh affordances to the network for modifying the environment um to you know and it provides the basis for future sensory predictions. You can imagine the network here is basically trying to generate uh motor outputs that help satisfy the input uh goals you know and so uh the preferred states of the sensors become the preferred outcomes of the motors and so you kind of have this synchronization loop that goes outside the network. So the network is doing self-s synchronization and then once you connect it to an environment with sensors and motors that synchronization manifold basically extends outward um you know and it basically is performing a kind of homeostasis through the environment outward through the motors and through the sensors. Um you know the model kind of imagines what it wants and what its preferred state is from the sensors and then generates motor actions to kind of drive it towards that state over time. like I was saying, I think embodiment is um a critical feature of of what we what we're building. Uh, and I think the the the missing piece here is time. So, in these networks here, there's no forward or backwards comp propagation because there's so much recurrence. You can't really define a front or a back or a top or a bottom. You know, signals just propagate in all directions. The closest thing you can get to is distance to the environment. So, how deep into the network you are and how far away you are from a sensor or a motor. um uh you know this is not um like the sandwich model I guess is is what people like to say uh where you take input you compute it and you generate output. Uh instead you know information is flowing into the network from the sensors um and uh actions are kind of you know flowing outward from the network through the motors. Um but once the information hits the network it just starts bouncing around in all directions depending on the network topology. Um you update the network and information flows in all directions. Um and through this kind of when you're reaching equilibrium and you're hitting this state, you know, it's that's when you can consider that the network is successfully predicting the sensory inputs. Um and you know, whether or not the motors are contributing to that uh is u depends on on how on the specifics of the motors and the sensors themselves. Um but the the the expectations of the sensor sensory information here becomes embedded with the motor actions themselves. So the idea here is the network while it's generating sensory predictions towards the front of the network, it's also generating you know um motor connections to the to the motors and these are kind of happening in parallel. So as it's updating its model to better predict the sensory information, it's also updating the motor actions that are being generated. So the the kind of the motor actions and the sensory expectations kind of get intertwined in the network here in and this is why I like to call it the active synchronization process because you know these are kind of being co-created as part of the the model uh the world model that's being direct. Um um this this is you know if those familiar with uh predictive processing or predictive control sorry perceptual control theory um this this will look very kind of familiar in terms of how this actually works. Um um finally I figure I'll talk a little bit about synchronization and think about homeostasis and prediction and where the prediction errors are. If you if you kind of imagine um uh the drive towards homeostasis as kind of a goal, the the distance you are from that equilibrium could be seen as like a prediction error. And so um you're basically trying to drive this prediction error down. And and this becomes more obvious if you actually look at the the chaotic attractor definition in terms of the the way that you're taking a weighted multiply of your input signals. Um and you're trying to get that to be zero. You're trying to you're trying to basically predict that you know what your neighbors are doing. Uh and a successful prediction will result in a local uh prediction error that is near zero. Um, and when when the network h when this happens, you can kind of think of it as a synchronization process. When when you're you can when you can act uh when you can successfully predict your neighbors, you're kind of synchronized with them. You know, you you know what they're going to do before they do it. And so that's how you can have the the sensory expectations meet the sensory inputs through a synchronization process. Um, and synchronization is kind of magical once once you have two systems that have synchronized. Um, or or let's say you have multiple systems synchronizing. You can imagine each node in here is synchronized to its neighbors and so you kind of have this uh synchronization over a distance. Um that's a way for two systems that are not in direct communication to coordinate their behaviors. Uh and so you you know you can kind of get very interesting collective behavior out of these networks uh collective here of the of the nodes um through the synchronization process where the the each individual nodes behavior is going to be con uh consistent with or or or um or uh coordinated with distant nodes in the network. Um and and so this is a very important aspect of of how these functions work because if if you had to if if two distant nodes that are representing different aspects of the environment had to be in direct communication all the time in order to to maintain their their correct modeling of the world um this whole system would become incredibly computationally expensive and impossible and inractable if so um so the synchronization process is very important I think I talked about previously how this is maintained at a critical state and so the smallest prediction error that's coming in through the sensor can have cascading effects through the network that cause very very fast destabilization. U and so the trick here is to synchronize faster than the environment can destabilize it. So if you're updating your network too slowly and the environment is changing very very quickly, you're going to be constantly in a state of chaos. But if you can update your network much much faster than the the the changes that are being experienced in the environment, you can imagine that this network is able to re recynchronize and re uh restabilize um continuously over time faster than the environment can. And this this goes back to kind of Ashby's uh law of requisite variety. Um and the the idea here is that the variety that's being generated is very much time dependent and depends on the rate of change of the environment versus the rate of change of the network. Um but the trick to this is ultra stability is very very efficient and then we can run it really really quickly. Um and in most of our robotic applications for example um these networks are operating usually around uh the update state the update rate of these individual nodes is you for a robotic application usually between 5 kHz and 50 kHz. We do this very very very quickly and depending on the application and and what's being done um that's generally fast enough to to maintain the synchronization manifold between the model and the environment. Um you know it's a it's a prediction error minim minimizing synchronization manifold but if you if you run the network that fast you're able to basically maintain the synchronization manifold over time. Um and that's how we kind of get useful behavior. Um [clears throat] all right I think I've talked enough about synchronization. Um, now I'll get to the fun stuff that I started out talking about the beginning here where the killer application for this kind of learning approach or this adaptive approach um really is robotics. Um, you know, early on we we kind of considered different applications of of this technology for like anomaly detection or even uh analyzing uh you know model drift in AI or deep learning systems for example. Um, but the the challenges in in robotics that AI is facing right now um are are really insurmountable with the amount of data that they need. Uh, and at least in my opinion, I don't I'm not sure they're going to get there um in in in any in in in the next decade. Um, and the they call it a synreal is the other problem. So, if you train a robot in simulation because the simulations are not entirely physically accurate, transferring it to the real world is a big problem. But doing the learning post post uh post deployment in the robot um solves a lot of that. uh a lot of a lot of those challenges not by it doesn't solve it. It more sidesteps it because you're doing the the learning in the real world as opposed to in simulation. Um so this is just a quick look at some of the industrial applications we've been doing um with this technique. Um these models are entirely using homeostats to do the robotic control. Um the these are some of the more recent ones that we've been working on. On the left here is an insertion process for automotive manufacturing where we're kind of inserting small rubber parts into um in this case a door latch assembly. Uh we've also been looking at connector insertions for cables and cable manipulation. Uh and on the right here is an older u uh model that we built for turning valves and bolts. Uh this was um I actually I'll talk about each of these in the coming slides here. This is the one on the right here. We take to conferences. It's a small little robot. Uh, and we let people kind of move it around as and and let the robot figure out how to turn the the bolts as people are kind of playing around with it. Um, so this is one of our first models. Um, we built this several years ago and this was originally uh designed for doing uh inspections on a moving um on a a moving vehicle. And so the vehicle could be bouncing around and they needed to kind of, you know, point the robot at at something that's being inspected as it's being bounced around in unpredictable ways. Um uh this was originally designed for you know full uh solar farms um but can be used and we use this as kind of a base model for all sorts of stuff if you just need a robot that that points at something. Um next uh we started working on um more manipulation use cases here. This was also originally done for the energy industry and they wanted to see if we could um automate the turning of valves. Uh you know you never know what size or shape a valve is going to be in a facility. Uh and sometimes there are injuries if these are high pressure gas valves for example they have you know people get injured and killed very frequently. So you know that's a great great thing to for a robot to do is something that's dangerous and so we would turn this into a demo that we bring around to conferences sometimes and we would let people like move around these bolts while the robot tries to tries to turn them and it's kind of fun. Uh people like like to play around with the robot and torture it a little bit. Um and there also not very many interactive robotic demos at conferences. Um this is this is another video of this conference demo and I included it because it has this video here on the right. So that's our actual um homeostatic network debugger, but you can actually see the network um updating in real time here as people are um are are are jerking around with the robot here um trying to torturing the poor robot while it's trying to turn a bolt. Um, and people like to watch the, you know, it's an internal debugger and so it's, we built it as a tool for us, but um, it's people like looking at it while they're actually playing with the robot. um, these are, this is some, um, some more recent work. Um, and I'll show what the final result of this ended up looking like. This was for, um, a project with in automotive manufacturing where they wanted to basically insert, you know, tight fitting rubber parts. Um, and the problem is you can't really do this with vision inputs alone because, um, they're tight fitting and, you know, humans are really good at this. You can kind of feel your way through it and kind of wiggle it in really easily. Uh, and so that's essentially what we're doing. There's a a force torque sensor in the wrist. Uh, and in this in these videos here, there's we're actually not using vision to guide the process at all. So, we basically tell it uh, and we're actually telling it the wrong location. And we're telling it the wrong location usually between two and 5 millimeters of error in like a random direction. and it's using the force uh sensor in the wrist to kind of feel its way through the insertion process. Uh and the one in the bottom left there, you can see where it kind of has to, you know, it catches an edge and then it figures out that they can insert it and you it kind of pushes it in. Um but we're basically using the same model for all these parts. So, um one of the big goals of this project was to show that we could build one model for insertion and have it work across a large variety of parts. um with traditional AI in a lot of cases you'll have to build and train um with separate data each of these models each of these would be a separate model for each part and that's a problem in manufacturing because these parts change every year or two and and what they don't want to have to do is generate a huge data set every year or two when the parts change. Uh and there's also about 500 of these parts. So they'd have to build a lot of um part specific models to do this. Um and this is um just a video showing what this actually ended up looking like when we finished it out. And so um this you can see this in this case here we've integrated the gripper we are using vision to figure out where the insertion housing is but it's then it's using the force torque sensor in the wrist to kind of drive the rest of this process and um in in case I hadn't mentioned it. So essentially we're using the homeostatic networks to kind of do the control here. So as it's doing this wiggling in, it's basically looking at the forces in the wrist and um we've basically given it the goal of trying to to minimize the you know lateral forces and things like that and maintain that the the insertion um the force vector is perpendicular to the direction of insertion for example and it just kind of you know as it's trying to do the insertion the network is constantly updating its understanding of the you know what it's doing and modifying its insertion uh modifying its behavior to perform the insertion task here. I'll give it one last second here. There's one more part. This is showing all four parts being inserted back to back uh using the same model. So, we're not switching models here. Like I said before, um I'll just let this finish real quick. Hooray. All right. We inserted all the parts. Um so, and then this is some more more work that we've done recently along similar lines. Um this time inserting um I'm not sure why. Wait, let me see. There we go. I just hit hit the button twice. Um doing insertions of various cables. So we're using a single model here to insert cables. Uh in this case here like a 15 pin you know parallel cable and then an Ethernet cable. Uh and the idea here is this should generalize to different kinds of cable insertions. Um again this is um using force to kind of drive the insertion process here. So even if the vision system is not entirely accurate and it's giving a location to insert that's incorrect, you know, a couple millimeters off here and there, it's able to basically figure out how to how to actually insert the the cable in the right uh in the right spot. uh and we're continuing this work right now um in more interesting cable manipulation cases here. So the insertion is one part of this. The next step is um being able to manipulate the cable around, you know, untangle a little bit. You know, cables are very very challenging to work with. Um this is why we're working on it is because this is very hard to automate with traditional robotics. Anything can happen with cables. They can get caught. They can get tangled. They can be in places you can't reach. So you may have to pull another part of it to get it into in a place you can reach because there are obstacles and things like that. Um, so that's uh that's kind of where we are with what with with applications of the stuff. Um, and super excited uh about the next year and what we're going to be doing uh in the future. Um, oops. Uh, that's the end of my talk. Thanks. Um, thanks for having me. This has been great. >> Thank you. >> And feel free to reach out if anybody has questions or anything. Happy to hear from you. >> Very cool. Okay, I'll just ask uh any of the questions that are written in the live chat. All right. >> Oh, yeah. I should have been paying attention to that. I'm sorry. >> Oh, no, no worries. I I'll I'll I'll read it. Okay. So, Ready Player Sid wrote, "Are we headed into a decade of teley operation teley robotics? What is the next decade of robotics going to look like with respect to embodied AI?" >> I mean, uh a great question. Uh I think we are I I think just in the last year I feel like I've seen at least 50 to 100 startups that are doing um uh teleoperation or or selling te teleoperated data or selling systems to do better teleoperation. One of the reasons why we went after uh soft body insertions for for example is because uh tactile data and force data is very very hard to get. um you can't really get accurate data in simulation. Uh and and it's if you need to train a deep reinforcement learning model, you really need accurate data for that. And you know there's obviously doesn't exist out there.

That's why a lot of vision models don't the visual language models don't have any kind of force data in there. Uh and so and so for example, I've seen just like 10 startups in the last year that are providing tactile data collection for teleoperating. So so they're still doing using teleoperation, but they're not adding force sensors and tactile sensors to the to the robot so they can collect that data during teleoperation. But I think I think if the money keeps going in that direction, that's what you're going to see. And I think that's what you have been seeing in a lot of cases here. There are a lot of startups that are just um hiring people to do a task over and over and over again and then they'll train a model to do that task. And often what you'll see is you'll you'll get a successful model that can fold a napkin or fold a towel or I'm those are the ones that are coming to mind. Um but whether or not that that model can then be translated to a different shaped towel or or a shirt um is questionable um mostly because of the science. And so you'll have to tolerate you know tell operate to collect data to for that that you know for the that shirt instead. Um but you know there's so much money involved that's the direction people seem to be going like people don't seem to have a problem with spending $100 million paying people to teleoperate robots day and night. I've even seen a startup where um what they're they're not even selling AI. what they're doing is they're they're basically you rent a robot for like $10 an hour to like operate your shop or I don't know um you know let's say they have a corner store have a robot operating and then it's literally they're just paying somebody in in another country you know $2 an hour to operate the robot and that's literally their business model um so I I think you are going to see a lot of this uh and there's going to be a lot of smoke and mirrors um I think you know I'm a little bit biased I obviously think that there are better approaches and I think um I think eventually the hype will die down and people will realize that we'll need more more efficient and better world models. Uh, and that's that's what I like to think that we're building even though it's it's kind of a an inactive world model and that the world model is comprised of sensory inputs and motor controls kind of intertwined with each other. So, it's not the same kind of world model that most people kind of envision in their minds. Um, but I do think it is it is one that is um relevant to a robotic, you know, task load. Anyway, I kind of got off topic there, but I like to think that there are going to be better approaches kind of like what we're doing here um that will take over at some point here when the when the economics pan out. I I think with self-driving cars, for example, the the value of having a self-driving car, the ROI there is is high enough that it can justify spending a decade of collecting data. Um but for things like inserting rubber parts for automotive manufacturing, that ROI doesn't exist. like you you can't spend a decade collecting insertion teleoperative examples uh and have it be worthwhile in the end. So um I think once you want robots to go after kind of lower value problems like that and it's still highly valuable but just not as much as a car that can drive itself um you're going to need different techniques. That's my opinion. Cool. Okay, I'll read another question in the chat and people can write if they have more. Question two from Sid. Why do Deep Mind Robotics and other companies such as Figure take other approaches when it's so obvious adaptive models are better in so many ways? >> Uh that's a good question. That's a good question. Um I think it's because it's it's a clearer straight line to something that they can imagine working and it's something it's that it's an approach that they're familiar with. Um back propagation is like perceptrons have been around forever but back propagation has been around for 30 40 years which um doesn't sound like I mean in in in the history of man it's not very long but like in in in the AI world that's that's a long time in which case what I mean by that I guess is you have like multiple generations of people trained in in that paradigm and people understanding how to that paradigm. So it's very comfortable and people just can just you know handwave around oh if we get enough data we'll get there. Um I think the reasons why other like approaches with other implementations of active inference for example that are that are that are adaptive um don't have the scalability that they are looking for and um yet and and uh and so I think it's easy to dismiss. Um I think that I think the tide is turning though. I think people are becoming much more aware of these other approaches. Our approach using homeostasis and ashb's techniques is very unique. Like I don't I as far as I'm aware I don't think anybody else is doing this. Um and so you know the reason why nobody else is doing it is because they don't know it exists and they don't know how to do it. And if I'm going to be honest um building a uh getting that chaotic attractor that homeostat that ultra stability definition is very very hard. Like I'm a I'm an engineer by background and so that's I've built my career around taking theory and turning it into real engineering. Um and it's challenging. It's hard. I want to say there's probably been about a half dozen attempts in the last 80 years to build a useful homeostat. And I having looked at them, I disagree with the implementations. I think there are problems with each of them. But this I mean it's all subjective at this point. Nobody there's no uh definitive homeostat that um you really just have to kind of look at the properties and and argue that, you know, it has this property or it doesn't have that property. So I like to think we have something unique here um uh and special, but uh and that that would explain why nobody else has it. Okay. And Sid with the hat trick. What kinds of adversarial cyberphysical attacks are possible on thought forge robots demonstrated in the video that use adaptive models? [sighs] >> You know, I [snorts] haven't really thought about that. Um, and I would like to think that um, you know, yeah, it's it's something I haven't really thought about too much. Um the models themselves are robust to like sensory noise like you know um this kind of inherent in how the algorithms work and so if you have noisy sensors and things like that but if you can like straight lie to the sensors and you tell it that the world exists the world as it exists is different you know um that would I mean that that would certainly work. Um, and I guess you know cyber attacks and cyber security there's there's lots of different ways of of looking at that you know and if you if you get access to the code it depends on what you want to do. Um, but it would be pretty easy for example or not easy it would be possible for example like let's say we have an insertion model that or that goes and picks up a part and does an insertion. If you were somehow able to trick the the perception system into thinking that a person was the part to be inserted, uh it would very happily go and try to grab that person and insert them. Uh obviously, this would be very dangerous for the person. So, um there's no like, you know, if if you can alter the perception of reality for any of these models, then you can get them to do whatever you want. And that's kind of how we're getting to do something useful in the first place. If I'm going to be honest, the the way that we've defined the sensory inputs where we're kind of implicitly putting todnamic prior, we're putting our goals in those sensory inputs specifically so that we can control them to do something useful because otherwise a homeostat will just kind of sit there. If you disconnect it from an environment, it it will reach equilibrium and do nothing. Um, and so the way that we're actually getting these things to do something interesting in environment is by manipulating its perceptual experience, which you know I I I guess the moral and ethical consequences of that is probably for a larger talk. [laughter] >> Yeah. Very interesting. and and uh connects with this tension of the expectations for like a sensory prediction to be veritical but also actionable because just veridically predicting being a spectator. It's passive inference. But then that is the question of how much to to have that kind of biased counterfactual homeostasis that takes you into the step that you need to go towards. >> Yeah. And and maybe there are ways to protect against that by having um higher order the dynamic prior where like goal instead of inserting inserting a a part is to construct a part or construct a car. And so maybe then it can evaluate the effectiveness of its subtask of like, oh, is this thing I'm inserting actually going to get me closer to building a car? You know, that might be one way out of it. We're we're not there yet um with our technology, but uh I think from like a theoretical perspective, I think that's that's one way to approach it. >> Cool. And then sort of in in closing, any last thoughts? Of course. And just what do you see for robotics or active inference? What are some of the constraints or limiting steps right now and where do you see things updating or changing in 2026? >> Oh man, that's a hard prediction. I mean, robotics is super exciting right now. There's tons of stuff going on. Um, I'm super focused on like practical applications, building the business, trying to make sure that this is an economically viable approach. Um, and so that's that's where most of my energy intends to get spent. I'm super excited by all the traction that that active in inference as a field has kind of made in the last year just in terms of um public awareness and and I and I think if the AI bubble bursts I mean I don't know if it's ever going to burst or maybe just deflate a little bit. Uh I definitely think there's a hunger out there for new approaches. I think right now people are starting to see that even even LLM with their huge capabilities and what they can do is super useful, super interesting. There are limits. You know, at some point here, language will not get you everywhere and will not is not what's going to get you to, you know, truly, you know, intelligent software, if you want to you want to put that way. Um, I am much more lower. I'm not trying to create Rosie the robot for your home. I'm trying if if I if I have a a lofty goal, I'd much rather make ants, you know? Um, because they are are they they're they're capable. They can build stuff. They are loyal to their queen. and like you don't have to worry about um does the queen of of an ant colony worry about the ethical consequences for its workers? Um maybe, maybe not. But but I feel like that's a much more lofty goal than trying to create sentient, you know, AI and AGI, which is a lot of what you hear on, you know, on from Silicon Valley these days. And to me, you know, I I I I don't think we need that. I think we need what we want is, you know, machines that can take away the dangerous uh dreary work that nobody wants to do. You know, that's that's what I'm looking forward to. >> Very nice. Thank you, Matt. So, thank you for joining the symposium again and for helping on the scientific advisory board as well. >> Thanks. I'm looking forward to next year. >> All right. Peace. Bye. Cheers. All right. Next is going to be a 33minut recorded talk by Maria Louisa Yanuka. >> Hello everybody. Um, thanks for having me in this wonderful symposium. My name is Maria Louisa Yanaku, but you can call me Maria. I am a PhD candidate in philosophy at the University of S. Paulo and the University of Porto. Uh well my talk will be I think a bit different from from the rest because um I come from philosophy and I will try to discuss the amigos the status of attention in uh predictive models of cognition. So um the content uh will be divided into like a brief description of precision modulation and salience followed by a description of attention across disciplines and then I will finish with this discussion that I've been uh labeling as having an ambiguous status and you know as I am a philosopher again this is a philosophical analysis and so my focus will not be uh on any specific models or any specific formalisms but on the general framework of both hierarchal predictive coding or predictive processing and active inference which I believe you all know very well. Okay. So uh attention in these two frameworks uh is usually related to two things precision modulation and salience. Precision modulation or you know precision waiting or precision optimization is uh formally the expected inverse variance of prediction errors. In other words, uh in the predictive brain, the precision modulation happens within second order predictions that control the confidence of the first order predictions. So if the brain is highly confident in a prediction, the modulation will reduce the gain on error signals choosing to trust its prior beliefs over the unreliable data. However, if the context is uncertain or noisy, uh an arrow will be given a significant weight, you know, prompting a a major update to its internal model. And it is believed uh to be implemented via lateral or self-inhibition in the error units with the biological substrates being uh like inhibitory synapses, neurom modulatory adjustments and also fast gamma synchrony. And attending here uh would be the modulation of uh the synaptic gain for the two top down predictions and bottom up uh prediction errors based on the precision they are estimated to have and this is pretty much how the endogenous and exogenous attention would work. Salance on the other hand has to do with action selection. So salence has been uh characterized uh as intrinsic motivation and denotes actions promising maximum information gain uh that is uh highly unpredictable and yet interpretable observations after which sampling is suppressed in a manner akin to inhibition [clears throat] of return. And in neurobiology this would be tied to action planning uh specifically uh in vision we would say uh the action of basog ganglia uh where the salience ranks potential circadian cicade uh evaluations. Um so attention then comprises actions selected to elicit information that cannot be referred passively and salience indexes the epistemic value of such actions as quantified by uh expected free energy. And of course uh the predictive attention has been around for a while across you know various disciplines. And here's just an example of three tasks that were simulated using predictive models in visual neuroscience. But you know uh there are tons of different other examples in the literature. So this first uh on your left is the famous Pausnner queuing task which revolutionized uh the neuroscience of attention and it was successfully simulated by Feldman and Freeston in 2010. And this task basically uh makes participants fixate uh centrally while cues indicate where a target might appear. If the cues are valid, it will speed up the reaction times when the target appears at the ceed location. If the queue is invalid, slowest responses will happen uh when the target appears as well. And it, you know, uh people say that it reveals how attention can be uh covertly shifted without eye movements. And the example in the middle is an active inference model of selective attention developed by Mertza and colleagues in 2019. Uh they focused on contextual epistemic foraging that is how agents seek out task relevant information while ignoring irrelevant stimuli. And for that they simulated typical attentional behaviors like this one in Yabu's task uh where you basically uh have a picture and you and you do and you have to look to this picture and then uh your sakads will be um recorded. And so on the left uh side of the figure that there is the human sacads and on the right side there there are the simulated ones. And they framed attention uh not just as the uh reactive process but as a proactive part of planning that will be tightly linked to the psychotic eye movements and also believe updating. And the third example on the right is more focused on the movements of the eye itself and also the neurom modulation that happens. uh and this paper was done by part of Fristston and they used an active inference model to simulate how different uh neuromodulator specifically the colonic and the gabiagic modulators affect uh alkalom motor behavior during a delayed sakad task and this task is about a cue that indicates a target location followed by a delayed period requiring fixation and then uh sacads to the remembered location. And so by manipulating precision parameters associated with each neurom modulator, they uh were able to model how pharmacological interventions could alter both the decision to initiate a sacad and the trajectory of the eye movements. Pretty cool. And so uh predictive models have also been used in computational psychiatry that is uh a new and progressively important area that aims to you know understand the psychiatric disorders using both real data and simulations and models and the majority of these studies work with predictive coding schemes instead of active inference. So they kind of focus on the precision modulation parts instead of salience and I'm not going uh in details about these papers that I put uh on my slide but you know I I left them here just for you in case you want to uh check uh later and read if you want but the idea that I wanted to say here is this many if not all the psychiatric disorders in the predictive context happen because of the malfunctioning of precision modulation. And even though the malfunctioning is not the only or the main reason as everything you know about our psyche happens because of multiple things, it is a current and a core reason. And this means that if we understand attention as precision modulation then all psychiatric disorders could be in fact attentional disorders at least in So can we you know really say that attention is precision modulation? Well, attention like uh any other phenomena of the mind has been studied since forever uh across cultures and fields. Uh one definition that was kind of foundational to uh the science of attention was the one made by William James. That is uh the taking possession by the mind in a cleared and vivid form of one out of what seems several simultaneously possible objects or trains of thought. Focalization, concentration of consciousness are of its essence and it implies withdraw from some things in order to deal effectively with others and is a condition which has a real opposite in the confused days in scatterbrain state. This is like the classical definition. But more recently uh neuroscientists Sabine Castner and Kanobi coined a more scientific or more operationalizationable uh definition that is the prioritization of processing information that is relevant to current task goals. And a couple of years ago, philosopher Wayne Wu made an even shorter description and operational operationalizable definition of attention. That is simply attention being the selection of a task to guide behavior. But of course, uh as the literature is huge, there are many different effects, mechanisms and interpretations of what attention might be. So if you take a look in your right you see uh this impressive taxonomy made by Cus and you can see that selection fraction is just one sub kind of the many other sub kinds and kinds of attention that coexists. Nevertheless, uh I'll stick with uh Woo's idea or or these definitions of selection uh because they shaped the visual neuroscience and of course for the sake of time and scope of this analysis. Okay. So um there are three main neuronet networks that people commonly attribute to attention. If you look at this pictures I borrowed from Ross uh you can see in detail uh the activations. [clears throat] The first network is the alerting system which keeps the person awake and not asleep and this neuronet network is composed of the talamus luscelius frontal and peral cortex and has the noradrenaline as the modulator of the network. The second is the orienting network which drives attention overly and covertly and it is composed of the pvener uh superior caliculus superior parietal lobe and paral temporal junction and also frontal eye fields having a saticoline as the modulator. The third is the conflict network which you know monitors conflicts problem solves problem solving or uh you know basically decision- making and is composed of the vententral lateral prefer cortex bas ganglia and the anterior singulate cortex and it has dopamine as its modulator and additionally I included here a figure from paren that illustrates the network of eye movement which you know is fundamental in attention studies in visual neuroscience and this network is composed of the superior calicles uh baso ganglia frontal cortex cerebellum and also the vestibular system and it also has dopamine as the modulator very well so from the [clears throat] 80s to the '9s there was you know a significant progress in understanding the effects of attention um and with that many theories and models were developed. Here I brought uh three theories that are well consolidated in the science of attention at least that I am aware of. And the first one is feature integration theory that was developed by Annie Traceman and Gary Galad in 1980. And this theory is about attention being a kind of glue that binds the perceptive features together in a single percept. And one core item in this uh in this theory is the cognitive maps which have been extensively used in modeling neuronet networks. Uh the second theory is the primoto theory of attention that were developed by Laya Craig Hiro and Jako Richati in 1987. And this theory is very important for spatial attention because it kind of proved that attention shares uh the same activations as eye movements and also action representations in the brain. So they basically described the dynamics of over and covered attention. And the third uh theory is the B bias competition theory which was proposed by Robert Desimon and John Dukan in 1995. And this theory is highly influential because it basically describes uh the dynamics of top down and bottom up attention uh specifically how visual content would compete for cognitive resources. So yeah, pretty important. And when it comes to personal level of attention or clinical attentional effects, there are many uh mainly three disciplines that kind of investigated psychology, psychiatry and philosophy. Of course that you know before technology and tools, neuroscience was psychology. uh today's cognitive psychology and you you you you find attentional literature before the 19s mainly in psych psychology textbooks. So anyway, the personal level effects of attention are basically to focus right when you kind of pay attention to something uh and how well you perform it the you know this will depend u in many things like mood, interests, environmental distractions and so on. And we also have the internal agency because not only do we pay attention but we choose what we pay attention to. So even though uh our attention can be grabbed sometimes like uh you're trying to pay attention in my presentation then your phone rings even though um it will distract you for a moment. you kind of you can choose between giving the attention to your phone and answering the person or whatever or getting back to the presentation. So uh you choose basically uh you you your in personal level you choose and when it comes to attentional disorders when these things do not work well or properly uh science and you know psych psychology psychiatry uh worked a lot with lesions or developmental problems uh that make people kind of neglect one side of the visual field and this was of um sometime later uh labeled as he he special he spatial uh neglect and of course we have the attentional disorders ADHD that is pretty common nowadays and people are being diagnosed even uh when adults uh yeah uh it's like it's important to highlight here that uh not only a a DHD but all psychiatric disorders have a common symptom that is the decrease in attentional task performance. And so of course when we talk about lesions and development uh it's kind of a different thing but when we talk about psychiatric disorders uh this attention performance is a common and current symptom and lastly real quick I would like to present to you some of the philosophical theories of attention so the first here is way selection for action. He proposes that attention is best understood through the role uh in guiding behavior. So in his view, attention would be an active process of selecting information for uh uh that is relevant for action. So for instance, if an agent is preparing to grasp a cup, the visual and the mortal systems will prioritize or select cup related features. And this selection according to him is what constitutes attention. Sebastian Walt's priority structure depicts attention as the structuring of consciousness itself. So it it will be a way of prioritizing mental content across perception uh thought and emotion. And for him, attention is not tied to a specific modality or task, but it reflects how the mind allocates its limited resources. And so this prioritization shapes what becomes salient, accessible, and integrated into our stream of experience. Christopher Mo's cognitive unison theory links attention to uh something related like uh something like an orchestra. So attention would not be a single process but the harmonious functioning of multiple cognitive systems and for because of that attention for him would be an adverbal adverbial phenomena. uh that is not something that the brain does in one place but how v various process operates together. Now uh some theories of consciousness have attention at their core u as it happens with at least these two here. So Michaels Graiano's attention schema theory will argue that the brain builds a simplified model of its own attention. That would be the attention schema that is similar to how it tracks the body. And this internal model helps the brain monitor and control attention. And it's this model that gives rise to the feeling of awareness. So consciousness then would be uh the brain's interpretation of its attentional state. Lastly, uh Jesse brings attended intermediate representation theory. Uh that is to say that consciousness happens when attention targets intermediate level perceptual representations. Those reach enough to be meaningful but still tied to sensory input. So all the attendant representations at this level would become conscious making then attention a gatekeeper of awareness. Okay. So uh now that I've talked a bit about attention, let's see why I think uh its status in predictive models of cognition is a bit ambiguous. So first of all uh what are the differences in attention addressed in hierarchical predictive coding and active inference models? You know I'm aware that this question may be more because I'm not entirely familiar with the formalities and you know I don't know how to reproduce these models. I don't model anything but you know by reading the papers I could not spot real differences in the way attention is treated and I would be happy to learn. I mean if you know if you know to answer how to answer this question please email me. I mean formally uh precision modulation and salience are different things in the models of course but the way attention is related to them is what is not clear to me. uh you know for instance uh precision modulation happens every time in all scales but does it mean that we are attentive at all the time in all scales I don't know I don't think so or even if we are maybe it means that attention would be more like the fio spot of our experience I don't know and even like if it is the case then what's the difference between this and conscious perception And I don't think that attention disorders mess with perceptional in this manner. Of course, there are some lesions in the brain that could uh that could be related with, you know, holes in in a vision or black shadows, I don't know, things that messes with visual perception. But I don't think that uh attention disorders would mess with perception in this specific manner, you know. And what about salience? Is salience the same as the dynamics of over and covert attention? It kind of makes sense to put it that way, but I don't see the relation of this dynamics and what salience means in the predictive models, specifically the action choosing thing. I don't know. But anyway, secondly, uh can these models address uh the various if not all uh types of attention beyond spatial attention? You know, uh this idea of precision modulation and salience uh seem to work well if related to bias competition and the promoter theory of attention. But these two are about spatial visual attention. And so I I don't know. I was just wondering if we could uh if it will be ever possible to address the other kinds of attention. For instance, there is this work by um by Ransom and colleagues that argue that uh the effective biased attention cannot be simulated by predictive models. So I don't know it's just it could be just a matter of times uh until all the kinds of attention will be addressed but I do not. Anyway, uh my third question is uh might also because uh of my naeness towards the formalism and again I'll be happy to learn more but what is the original contribution of these models to attention literature or to attention neuroscience? The way I see it uh there have been no new predictions made by the models only reinterpretations or simulations of non-scientific schemes and more uh many of these schemes are also simulated by other computational models like reinforcement learning normalization and so on. So what's the novelty here? You know, and lastly, where is the external world? You know, exorionous attention is famously known for its dependence on the external world or the external stimuli. But it looks like here uh it's all about stop down modulation even to externally triggered attention. I know I know that prediction errors come externally but uh if whether they are going to be salient or not will depend on this modulation and not on say the intensity of the stimulus or or something like that. Uh and honestly it kind of makes me feel a bit down because you know I'm studying active inference precisely because of the narrative of the organism environment coupling and and stuff and as [clears throat] I understand it now uh it looks like the external world is really a minor detail in the predictive biom machinery so I don't know I would be wrong but okay uh if we hard to accept that attention is precision modulation. Maybe it means that attention is something more fundamental than a simple cognitive feature, right? I mean it can be what Moles described as an adverb in a cognition, meaning that it is a mode that things happen. But I I think it still wouldn't explain the integral functioning of of things as sometimes we do not perform an attentive behavior at all. But it might be an interesting start. And also uh if we understand attention as a more fundamental thing maybe there won't be as many types of attention as we currently have because once we reinterpret these types they will kind of disappear into uh the model updating scheme I guess and another important implication of this take is specifically for psychiatry or computational psychiatry. So it is known that bad performance in attention tasks is common to psychiatric disorders. But what if they are at their core different expressions of attention disorders themselves? You know for instance uh people say that too much gain in prediction in predictions leads to psychosis whereas too much gain in errors leads to autistic behavior. So could that be the case? I mean is everything attention attentional? Um you know this idea is not entirely new. Okay there are some works uh in the literature suggesting that psychiatric disorders are attentional disorders and so this might be a fruitful spot to investigate further. And of course, um, would we be closer to a unifying theory of attention if we continue this reinterpretations of models and theories of attention into the predictive account? I don't know. It looks like at least reinterpreting biased competition and promoter theory within uh active friends framework, I think it worked well. So, uh, we could keep doing this and see what happens. So, yeah. Well, my last slide is what I think it kind of should be addressed by the community of active inference. Um, first of all, what are the advantages of saying that attention is precision modulation and salience? you know, could we say that uh what I said a couple of slides ago is an advantage or is it more confusing like if we have a theory that explains everything then this theory explains nothing right so I don't know what what are the advantages the real advantages of saying that attention is precision modulation [clears throat] and how to acknowledge that sometimes modul ating weight has nothing to do with attention because it may be the case right I mean we don't need to of course if we say that attention is more fundamental than other cognitive features then we could uh say that at least uh a primitive or a minimal uh attential uh function is working all the time but then I don't know I mean is it the case really. Um well, and besides uh the kinds of attention that could be dissolved into uh the model of dating scheme, can the predictive attention say anything about the personal level of attention? That is the act of paying attention or uh the internal agency or you know anything like that that requires personal level interaction with the word. And lastly, this is more like a request that or a suggestion than anything, but uh so I understand that active inference and predictive coding frameworks are still young in the literature and people are applying them to everything, which is cool. But specifically for philosophers, it feels like that philosophers got away from the literature, you know, uh well scientists have always been skeptical and still are, but philosophers were once super attracted by uh the predictive idea and then I think they are kind of getting away from this. uh if we see philosophers talking about it, they either have a very specific idea that may even not corresponds to the formalisms at all of the frameworks or they are critic they are critics of these frameworks also usually having a very specific and maybe even unrealistic idea of the framework. So my request as a philosopher is still to trying to deepen and learn the models and trying to work on their implications to real life is that the community uh should stop scaring philosophers away and I don't know we should try to uh make a second bridge for philosophers again to be interest in the in the active inference models and see what's new uh and to continue working on the implic applications of these frameworks to real life and so to help modelers to not be so abstract and lost in their maths. I don't know, it's just something that I feel like saying here. Anyway, that's it. Thank you for your attention. If you need to contact me or you can scan this QR code and see my website. Thank you. Okay, we are back with Samuel Montanes at all agentic finance. So Samuel, looking forward to your presentation and anyone in the live chat can write any questions. Thanks. Go for it. >> Thanks Dan. Hi everyone. Uh my name is Samuel Montinez. I am a an artificial intelligence PhD student based in Mexico City and I'm here with my co-authors John Clippinger and Matthew Moroni. We're basically uh a working group and a research group uh working mainly on the intersection of finance, economics, uh neuroscience and in this case bridging active inference for financial applications. So that's pretty much what we're going to share today with you. So what we have for today is identic finance active inference applied to dynamic portfolio management. So any questions please put in put them in the chat and also Matt here has some practical industry oriented questions for that that should be interesting that we're going to do do in the end as well as the public's questions. So let's start. First, I like I'd like to share with you the finding a very a very important finding in economics that pretty much launched the behavioral finance revolution in economics and it's called a the excess volatility puzzle. So, in standard finance and economics, we we think about the prices of securities of financial assets pretty much depending on their fundamentals. When we say about fundamentals, it's their economic fundamentals. So for example, for stocks or other type of securities, we usually value them as taking the present value of the expected dividends discounted by some risk rate. So that's pretty much the standard formulation that you will learn in any introductory finance course. But that's also the the most [clears throat] the the standard way that most people do it do valuation both in industry in corporations and that's pretty much what you would be taught in an MBA. So what's behind that formulation in in terms of the theory of what we are assuming about the economic agent is that we are assuming a theory that economists called rational expectations. We assume that economic agents are rational uh rational agents in the sense of having complete information, having uh transitive and static preferences uh and a bunch of the other assumptions that makes makes the model easier makes just the model easier to work for and in general most financial models have this assumption behind it. So in this case, if we assume that if we assume that we have this rational expectations model and we say that we expect that a certain security would be correctly valued depending on the expected value of their dividends discounted back to the present. We're going to check if that's true. We're going to use the standard and 500 the the market index that represents the 500 largest companies in the US. And if we assume a rational expectations models of the dividends I just shared with you, we would have from the 1980s to the 220 to 225 pretty much the red line that we're seeing here. That would be the expected correct price under the rational expectations model. That would be what what we should be looking at or something very close to that. But what we are I mean what we are looking at really is a blue line. That's the the actual real back test on the S&P 500. And pretty much Robert Schieler, one of the founding fathers of behavioral finance at Yale, pretty much uh measured the the difference in volatility between the rational expectations model and the real the real market price. and he found an astonishing finding which was that real S&P or market returns were 25x higher than what would be predicted by a rational expectations model and later on uh subsequent studies tried to find the answer of these difference in in volatility and they pretty much analyze everything different difference for example in costs of trading differences in taxes because of course the index is just like uh well it's just that it's just an index so it has no transaction costs stuff like that. So they accounted for all of those things and none of those things could explain a 20x difference in volatility. So basically this led us to found to I mean this led the the economics uh profession to start importing uh methodologies and knowledge from other fields and in this case psychology to try to answer why is that markets are behaving this way that prices are diverging of what would be expected under a rational model and essentially what Schiller claimed and what what is like the tenant of behavior behavioral finance, behavioral economics is that market are driven by belief dynamics and not only by by their economic or cash flow fundamentals. So given that we're speaking now about beliefs, we need to import things or knowledge or techniques for from psychology and later on from neuroscience. And pretty much what we're proposing here our framing uh is the let's say the integration of different fields coming together. On the one hand, we have Robert Lucas as one of the I mean probably like the last mainstream economist who is a strong proponent of the rational expectations model especially in microeconomics. We can also mention others like Eugene FMA with efficient market hypothesis. We can al also mention um Paul Samuelson at MIT but all of those are the mainstream economists who have uh pretty much established that rational expectations is the way to do economics and it's still the the ruling school in economics. Now later on with uh with this this this uh work from Schiller here uh famously Canam and Antverki two psychologists who were exploring decision-m under certainty uh published their famous prospect theory. You probably uh seen it or heard it or read it in in their famous book uh thinking fast and slow. And they basically found that that individuals consistently perform bad economic decisions and and came to several decision errors uh especially many errors concerning calculations, probabilities and money in general. And later on uh with this gentleman, Richard Thaylor here uh a recent Nobel laureate in economics uh into 2017 uh actually received a Nobel laureate in economics because he was able to integrate those findings in econ in psychology and import them into economics and that that gave rise or formalized the field that we know today as behavioral economics. However, as with uh as with uh rational expectations, economists usually think about behavior. Uh for example, we can remember from Paul Samuelson's uh revealed preferences see behavior and we infer preferences from that observed behavior. And that standard lens to look at economics was pretty much prevalent also in the behavioral economics school. So it didn't change much the experimental methods or the techniques that they were trying to use or that they were using to try to to understand economic behavior. So later on uh especially from neuroscientists like Paul Glimemer at NYU and Colleen Came at Alcaltech there was a neuroeconomics revolution which was basically saying why aren't we using uh the latest neuroscientific findings to trying to inform our understanding of economic decision-m. Uh so basically these two gentlemen are probably the the leading um acadea ac academics or intellectuals following this trend but interestingly I mean they have made ways in economics but but but it is not nearly the most popular field in economics. Um but basically what we are claiming here is that we can come together into a more general theory that accounts for a principlebased uh scientific theory which is the active inference theory or framework and integrate all I mean both of these schools of thought into a single school of thought which we're going to call cognitive economics where we're basically going to unify the behavioral finance behavioral economics and even some some aspects of rational expectations as well with neuroeconomics findings that come naturally in the active inference framing and that's pretty much what we are proposing using active inference as a general framework or a general umbrella to make these fields come together into what we call now cognitive economics. So essentially we are taking I mean more I mean is this is not limitative but we are specifically focusing on bridging these aspects from each field from neuroscience bounded rationality uh also the neural process theory in active inference and also other aspects of decision theory in neuroscience. uh from economics mainly behavioral economics behavioral finance finance but we can also reconcile some aspects of rational expectations as well because it has been tremendously useful in economics from finance asset pricing theory and from active inference in we're taking pretty much everything uh to integrate this in into the framework and actually cognitive economics is not per se uh let's say our own creation uh other major uh economists or mainstream economists like uh Gabay at Harvard or uh Bordaloo in Italy and Schlager at Harvard have have been actually pursuing this this line of thought in the last 5 to 10 years uh let's say and for example Gabay is studying the the effects of attention and limited processing capacity that naturally from for almost anybody in the active inference community uh I mean we are well acquainted with land hours principle uh precisely having a thermmo thermodynamic constraints in decision- making. So that comes obvious to people in in the active inference community but it's actually sort sort of novel in economics. Um also for example Bordalo in uh from Bonia's life at Harvard have been pursuing what they call diagnostic expectations which is basically integrating findings from cognitive psychology some from neuroscience and bridging it or importing them to economics and they actually propose laws of motion for understanding belief dynamics and naturally uh does the I mean that naturally [clears throat] resembles basin mechanics in active inference And finally loss aversion which is uh the famous finding from Canaman in behavioral finance can also uh be tracked like one of the let's say pioneering works uh on trying to understand decision making from a narrow grounded theory. So basically active inference what allows us to bridge all of these field from in in one part from a normative standpoint on the other part part from a descriptive standpoint because it gives us much richer tools to understand economic decision- making and the other is a mechanistic model to understand how those belief dynamics are are are forming and are being updated. And for this we are proposing a change in paradigm in finance and in fin financial valuation. So we're we're we're proposing an agentic finance type of model. So instead of using static models as for example taking the present value of the expected dividends which is basically a standard discounted cash flow type of model and many other financial models. We're saying instead of doing that let's use agentsbased modeling. Let's assume that you we that our economic agents are actual agents and that they perform inference as planning because in a sense all type of economic decision making has a reference to goals for most I mean almost almost mandatorily has to do with economic financial goals and it has to do with planning. So when we integrate those two key aspects that makes the the agentic framing uh very natural very very natural and we are proposing that this agentic finance uh framework of of financial evaluation would allows us first that the agents can infer distributions on their uncertainty and actually update those those inferences about the distributions that they are observing where we're speaking about distributions of asset prices, returns, etc. Because in traditional financial models, most of them assume, for example, a fixed distribution with known parameters, most often the normal distribution. But what if the distribution is not normal? What if we don't know the parameters? And that's what we often observe in financial markets where we see changing distributions. We also see distributions with fatter tails that will than what would be predicted by a normal distribution. So that's very very common. And finally, traditional financial models completely ignore the agent, completely ignore what the agent believes, but uh pretty much always assumes an a type of etogenity in preferences. So all agents are the same. All individuals made the same type of economic decisions. They are fixed. Whereas in agentic finance, we can always tune our graphical model to the specific type of agent with a specific type of investment goals of risk preferences of prior distributions, financial knowledge, etc. So it allows us to really personalize valuations uh with I mean depending on the type of priors that you have whereas in financial in traditional finance that's fixed. So basically we are proponenting uh using active inference to frame financial valuation as planning through inference essentially financial planning in this case. Now the Pond DP architecture that I just showed you in this graph in this visual graph is essentially the standard way to frame uh or to apply active inference. So the the partially observable mark of decision process is partial. I mean you might be acquainted with a mark of decision process in standard reinforcement learning for example but it's partial because the the agent only partially knows the the exter well the the external states in this case it has partial information and it has to infer uh those external states. Now a key aspect that we are are framing here is that we can get into great detail in what financial aspects we want to take into our valuation model. Here here is just an example of how you you could frame it but it's really not limitative to uh public assets. It can also be used in private assets. Uh it can be much more detailed in the actions that you can take or the many observation variables that you can define. This is just for visual clarity and that that we can understand it. So first we have the state spaces and we're going to define two state spaces or two factors. So one is going to be the market regimes. So when we speak about the market regime is what we usually assume naturally. The market is going up, it's going down, it's flat or we're in a crisis like in a financial recession. So those are our four market regimes and those are the ones that the agent the trader wants to infer. And then we have portfolio positions. So portfolio positions means what I'm going to do or I mean where do I stand in my portfolio in the sense of going long, going short, selling or I mean or expecting a decrease in price or just holding I I I am holding I am holding my my position flat. And when we observe what we observe and and what the agent is observing from the external world to try to learn are two things in our model. One is the return observations for the asset that we are going to analyze. It can be a basket of of assets, an ETF, it can be whatever we want. And second, we have the P&L observations. So when we say P&L is the profit and loss of our own portfolio of the portfolio of the agent and we're going to divide that portfolio into quartiles and we're going to divide it into quartiles so the agent can analyze if if I am on the top quartile or in the bottom quartile. So we can have some sort of ranking of order of preferences. And finally we have the action space. So the agent is going to be able to act on its sensory uh inputs to try to change and trying to change their positions and get closer to its expected preferences. So those trading actions are going to be the three standard trading actions hold, buy, sell. But we added a fourth one which is reduce because in some cases you might not want you might not want to sell your entire position. you but you want you might want to want you might want to decrease your exposure. So that's like a fourth uh like detail that we added. Now if you want to think about how would this uh framework would evolve how would it work while I mean while being active basically we we say that the state spaces are the hidden states. So we have two hidden states. One again the market regime and naturally as investors we would want to to learn if the market is going up, if it's going down or if it's just like staying flat as it is. And the interesting thing is that we have no control over that.

We have no control over market returns and nobody really knows what the market is going to do. For example, uh economists all over the world, but especially in the United States, have been forecasting a financial recession or an economic recession in the US for I mean almost since 2020. So we have been five years waiting for a recession that never comes in the US. Uh and also for example, Jamie Diamond, the CEO of Goldman of JP Morgan uh recently said that he expects that perhaps in 2026 we're going to get to the recession uh this time. So if not even the best economies, the world financial leaders can guess in which market regime we are that's pretty much the the best example to see how they are completely unpredictable uncontrollable. But what we can control is our portfolio position because we can decide to go long, go short, I mean just hold our position as it is. So we want to change that. And then we have our the observations that the agent is going to learn from. And we divide the observations in this financial application into exogenous observation and endogenous observations. And why? Because again the external I mean in in the external states we have the market and the mark and there's the financial securities and they generate returns. So we can observe those returns. We can observe them especially for public securities like equities, public equities. Uh let's say for example for an Amazon stock we can observe its price at all points in time but we cannot we cannot have any influence on that price but and then we have the P&L of our own portfolio of our portfolio and we can observe that so that's indulgence to the agent and that distinction is going to be very relevant to what's coming next. Also, I mean, a key difference between a standard financial model and perhaps not that different from an econometric model, but much cleaner in its implementation, at least for an agentic implementation, is that we can factor these different types of information that the agent receives and clearly classify it. So that's key at least for financial information and return attribution which is a key aspect uh of applying different artificial intelligence techniques for financial markets and also we can add action dependent transitions which is that we can act upon our portfolio to alter the P&L but we cannot act on the market. we can make that modality distinction which we often cannot make in a standard model where we just put all of the variables into a black box. And finally in the C matrix which is the preference matrix that would be a quasi version of a reward of a reward in reinforcement learning for example. uh we can also embed into that uh uh risk aversion parameter which is also standard in finance and we are using the 2.25 25 standard uh I mean risk aversion factor uh which was pioneered by Kennean Atverki which basically says that humans compute equal losses to gains in I mean in twice the magnitude. So that's what we are also putting into the model. So we can actually embed this type of neuroscientific or decision theory findings into the model to have theory based parameters. I just mentioned that we are that that the C matrix basically is the matrix where we can put the the traders in this case the economic agents preferences. Now how it is done in the standard active difference is that we work with log preferences of of well we we work with log pre with preferences of defined as log of pre prior probabilities basically. So that's the standard way of framing it as we can see in the upper box. Now instead of using that type of implementation we're changing it a little bit and we are defining our C matrix as it is done in risk sensitive control or uh KL control where we just define preferences in terms of utilities and one question would be but why would you use why would you define the symmetry like this and not use the standard active inference implementation and our reasoning to do this is mainly because of knowledge transfer and let's say industrial applications. So if we want to convince or persuade real investors, financial institutions to use this model, it's much easier to explain to them how preferences are encoded in just expected utility. So if the C matrix is defined as expected utility, we can say the model is maximizing expected utility subject to a specific risk preference. So we can actually frame it just like that. Whereas if we frame it as the log of prior distributions, we will have to say something like our our market returns are expected uh or are modeled as a uniform distribution, let's say. And that's much muddier, let's say, than just saying it out loud as in as any economist would or financeier would say it. And this is not new or or anything uh new. uh this is the standard way to do uh let's say maximum maximum entropy reinforcement learning or KL control as mentioned and actually KL control has been used in financial management and portfolio management before. So that might be a method that also industry uh I mean that I mean general industry uh practitioners are acquainted with. So how are we defining this? Essentially remember that we had market returns and the P&L. We just said that market returns are uncontrollable. I mean the market is just there. We cannot control it. So we're going to define that C matrix as zero because so that means that we are neutral to market returns because we cannot influence them. Whereas the P&L is controllable because we can act upon selling buying etc. So it's position dependent. That's what we mean. So he's not going to be here. And essentially an an additional an additional benefit of framing the C matrix or the or the preferences as utilities is that uh scaling them by the risk aversion scaling the losses by the risk by the 225 kaman and the risk risk aversion is transparent. So it can be done in the active inference framing as well but in this case is super transparent as you can see for example from the utility vector uh the risk the the the risk scaling parameter is only applied to losses because we say that gains are just I mean factor by one so for the 6 and the

1.0 in losses we can scale them as you can see here by 2.25 25 and then that would yield the the the reward preferences for the P&L the updated reward preferences. So you as you can see is pretty much transparent. Now how does this links to standard financial theory? So when we take an introductory corporate finance or investment management course the first pricing model that we usually learn is called the CAPM the capital asset pricing model. So the D the CAPM right here is essentially a regression model a one factor regression model that essentially says all expected returns on a security let's say on an equity on a stock depend on just one factor which is the market risk premier. So essentially the the the model says you are strictly and only being compensated by the amount of market risk that you are taking. So we so so we can decompose the CAPM into two type of essential risks in in financial investments. One is called systematic risk. So that systematic risk is pretty much what I just told you the market risk. How much market risk are you taking? And that's the only risk according to this financial theory that you get compensated for. Whereas the other term is the ideic risk. So that idiosyncratic risk we can refer to it as the firm specific risk. So for example, if I'm investing in a restaurant, I am exposed to specific risk that the restaurant has like having a fire or then employees don't show up or a fraud or whatever. But but that's not systematic risk. When we speak about systematic risk, we refer to an as infl uh as as general market risks like inflation risks uh like interest rate risk. So co as well those are systematic risk that we cannot control in any way. So we cannot diversify it away in any way in standard portfolio management theory. Now what are the disadvantages of this model? Number one it's static meaning that the betas are the betas in the model which has is is often called a market beta is not dynamic. So in the standard framing is not dynamic. Now there are more recent proposals where you have a dynamic beta. But interestingly for I think that most of the viewers if you go today to Harvard Business School and pursue an MBA and you get taught a corporate finance course you're going to get you're I mean you're going to get taught this model not a dynamic beta model. This is the model that you're going to be taught and most investment banks and consultants still use this standard model and not more sophisticated models that do exist in the literature. So that's very important because perhaps there's uh updated models but those are not the models that most people use in industry. Um and f and and additionally it's a passive model in the sense that there's no agency because again uh economic agents are just static in a sense. So it's just I mean is in a sense just saying you can buy an index fund and just stay there for I mean as long as you can is in a sense is saying that because you cannot really uh adopt m adopt much more uh return unless you're willing to take firm specific risk idiosyncratic risk but that's up to you and in theory this theor says you shouldn't because you're not being compensated for that. So, so it's it's not even logical. Now, in our active active inference framing, I just explained that we cannot control market returns. We can learn from them, but we cannot control them. So, those are neutral preferences in the sense that we cannot act upon them. But we can act upon our P&L. And our P&L is a function of two things of the weight that we want to have exposure to in a specific asset times the market return on that asset. So that's what we can act upon. That's the controllable modality that we can uh act upon in our model. So how's different from the CAPM? Uh it pretty much uh changes the whole framing because number one the W the weight is adaptive. It's not fixed. It's not a static. So that's one main difference. Second, it learns because even though we cannot act upon I mean act directly upon markets, we can learn from that market information to change our weights in the position. So we can learn from that information and we can act upon our portfolio position to modify our P&L. So as you can see it completely changes the way to see finance as just measuring temperature to an agentic vision of So how does this links to standard framing in economics? So as I was saying even though the rational expectations model comes across sometimes as like very antiquated let's say for many people because they say how do economists like believe or I mean it's not that they believe but how do they take those axiums which are very extreme into their models and they just think that uh the models are going to be very successful as those actions are extremely radical let's say but Robert Lucas is probably the I mean uh I mean the one of the the the most famous living economists uh he won the Nobel laureate because of his famous Lucas critique and basically what he claimed was the following in econometric models uh those econometric models again implied a certain stability of financial agents. So policy makers just assume that again individual ones in their societies were fully rational respond to information and the most important aspect is that preferences were stable that people don't change don't change on their preferences so preference are stable and essentially what Robert Lucas said says is that parameters of econometric models are not structured so what they what that means is that they are agenteped dependent. So agents, I mean humans, we respond to information. So we respond to actions that the government takes. So when the government implements a certain policy, they should not expect that people are just going to be fixed and acting as they always did. I mean people are going to react and probably or in some cases likely are going to modify their behavior. And as surprising and that as that may look that wasn't the accounted for in standard microeconomic models uh in the 80s9s and again you can see that here's another another place where we start thinking about beliefs in economics. Now how does this critique is relevant for our model? Because if we if we frame our active inference agent as a standard active inference implementation where we say we have different modalities where one observation is going to be let's say the different senses. One is going to be visual information the the other is going to be auditory information. Naturally that information is coming into the agent. we can wait its importance but it's all coming into the agent and and you can put on glasses you can put on phone I mean headphones or whatever so you can act upon those observations but in this case if we if we were to conflate that you can act you upon the P&L and to the markets as well that would cause a problem into the model because we just explained why they are not different they they are actually different so What's our approach? We want to break the potential is issue in this type of implementation of policy endogenity. So we are proposing an application where we have selective basian learning instead of full basian learning as it as it would be done in standard active inference. So for example we have in the standard active inference model uh we just mentioned that we have two modalities. So we would have two learning equations as as they are posed here and in those in that in that case of standard application we would be just using the two modalities entirely to update our beliefs. But as I just mentioned we don't want to use the two modalities fully. We want to learn from the market but we don't want to learn from our portfolio. And as and why do we don't why do we why we don't want to learn from our portfolio? That's like like a a weird question but imagine this. I am looking at my portfolio and my portfolio performed well during the last six months and given that it performed well I am going to update my internal model to uh act and start trading with that information from my portfolio. So my portfolio if I am a reg if I am not Vanguard or Black Rockck is not going to be nearly representative at all of the general financial market. So if I am learning for from that information, I'm going to contaminate my internal model to actually predict to try to predict what's going to be the next market regime. Because again, if Samuel owns 0.00001% of Tesla, it has literally no my position has literally no effect upon Tesla or the robotics market or the electric cars market at all. So what are so what we are doing here is frozen that part of the learning equations in our model. So that's what we call it select selective basin learning and we are only updating our internal model based on external market information which is what we can learn from and which actually can provide relevant information. Additionally we have adaptive confidence confidence gating. So we're implementing ad hoc confidence gates uh where where where we are saying that when there's a certain level of policy entropy the agent is not going to act. So in standard portfolio management we have different types of risk management tools but there what I mean some standard ways to to frame a trading strategy is to st to standardize I mean is to implement some systematic trading rules. So for example, some of the rules can be you can never invest more than 5% in a specific asset or you can never invest less than 1% in a specific asset or you cannot have an overconentration to a specific sector of 20%. So those are standard riskmanagement trading rules and in the same sense we we're are imposing let's say uh an active inference flavor trading rule which is having a confidence gate that where if the policy entropy goes beyond a certain threshold even if the agent wants to act not going to act and that could be naturally uh naturally adjusted depending on how conservative we want our trading agent to be. Additionally, we are also using an exponential forgetting factor as well and also that can be adapted depending on the type of agent that we want. But it is also recognizing that people don't remember m market history perfectly. Uh I mean not nearly. So it also recognizes that economic agents forget naturally forget. Now if you want to frame this financial agent or this financial market in terms of the mark of blanket like the standard framing in active inference we could say that we have an external world an external market and in this case the spec we're specifically going to focus on trying to predict the market regimes. So that's going to be the external state the market regime that we want to predict again if the market is going to go up it's going to go down what's going to happen with the market but we cannot have direct access into that market and and those are uncontrollable dynamics on the other hand we have in the right hand side the internal model of the agent so that internal model of the agent is going to be the internal model about what's going to be the market regime I'm going to try to predict as a trader what's going to be the next market regime And how this I mean the how this model actually works is between the sensory and active states. So the external market is going to yield as is going to yield sensory states. And again we just mentioned that two sensory states are going to be or the two observations are going to be market returns from the assets and the P&L from my portfolio. we just explained uh I cannot I can learn from I want to learn only from the returns of the of the assets in the market and I'm going to froze the information on my P&L because I don't want to contaminate uh or I don't want to have an an endogenity issue in my learning and finally I can act upon that information and in this case I'm going to act in two ways one is going to be performing trading and the other is going to be looking for information the epistemic uh component into the active inference framework. So as you can see this type of framing I think more naturally accounts for standard trading and economic decision making in financial markets than for example the the model we we showed. So essentially how we can look at it now I mean when it's actually updating in the in the action perception cycle in active inference is depending on the expected free energy that can be decomposed on a systemic value the intrinsic value and the extrinsic value which is going to be the pragmatic component the expected value in economic terms essentially what we I mean what one of the key advantages I mean well known to to everyone in the active inference community from the active inference framing. At least in this case against economic type of uh approaches is that it allows us to embed in a single functional in the optimization process both the risk or the epistemic component or the intrinsic value and the the expected value or the extrinsic value or the expected utility into a single optimization process. So is balancing these two components or another way of framing it is balancing the exploration exploitation dilemma automatically into the model. So how this I mean how is this different from standard models in economics in finance? Well in economics in finance we almost never ever use epistemic value whatsoever or intrinsic value we only care about expected utility expect extrinsic value. Now in other type of approaches for example in reinforcement learning you can add a not hawk parameter to add an exploratory term for example in e-gritty methods but that's a n hawk uh addition to the model in this case an advantage an advantage is that it's embedded into the optimization model of the model in the optimization model itself. Now for just to provide an example, we're going to use the ARC Innovation ETF. So for those who who don't know it, uh ARK is an investment management firm, one of the most popular today in the in the US. It's owned by a famous investor called Katy Wood and she became very popular at least in financial markets and uh let's say investment community because she has been a a strong proponent of thematic investment and especially focus on what we could say we could define as transformative technology uh highly growth stocks and yeah like transformative technologies like uh ranging fromies to Bital health, um, clean technology, uh, many like many of those mega trends and her asset management firm issues, is issues and manages several exchange traded funds. And we are portraying here the last 10 years of performance of her most popular fund, the ARC Innovation ETF. That pretty much is an investment fund that gives exposure to investors to the major trends again in innovation. Now, as you can see, the fund performed extraordinarily well since 2015 up to 2021. So, it went up 700%. In five to six years, 700%. So, at that time, Katy Wood was appearing in TV all of the time. She was claimed to be the next Warren Buffett. Uh, and her fund draws, I mean, draw uh investments from for for more than $20 billion. So it became extremely popular. Now as of now I mean after that peak it went down 80% in it low in its lowest point and even after almost six years people who were most of the investors who entered into the peak have not recovered their even their initial capital. Now what I mean why we like it is because this type of popular trendy momentumchasing type of investment is precisely the type of asset that allows us to measure regime shift because almost visually we can see that that would be a bull a bull uh a bare uh regime shift and then more or less a flat reg shift. So we are using arc as a as an example and how we can train our I mean we're going to train our active inference agent taking the almost like a standard machine learning type of training methodology where we where we're using 70% of the in of the historical information and we're performing a back test on the last three years uh from 2020 uh 2 to 205 to see how would our agent would perform. Now what's the advantage because one may say why not use just deep learning and LLM and the advantage is that that we don't require retraining to just keep following the performance of the ATF because by definition this type of active inference implementation is adaptive it's adaptive learning is full online learn based in learning so that's a a completely a completely different way to do uh machine learning and I would claim it's a more much more efficient way and also I think for financial applications this adaptivity makes you much more uh prepared to adapt to changing market circumstances especially with these type of volatile assets which are one of many in the financial markets. Now just a brief theoretical uh bridge as well with standard financial and economic theory. Uh using the Helmholms decomposition uh which is the a standard part of the active inference and free energy principle we can we know from Helmho that we can decompose any vector field into two components or two parameters. One is the gradient we can say almost a deterministic component and the other is the selenoidal flow which is the yeah the cycle that that is never changing uh into the model. Now how can we build a link between this theory behind active inference and financial economics? Uh one model which is one of the most important models in valuation is the black shells meritton option pricing formula. So this model was was created in the 70s and very quickly became the standard to price uh financial insurance financial options in in markets in general. But originally the model was created to value corporate corporate corporate liabilities actually. Uh so so so another extension of the black shields model is what we call the real options black shields merit model which can be actually applied to value private projects, private equity projects and mo I mostly innovative projects where there's a contingent aspect into that type of investment where where there's like huge changes for example like all of the innovation trends embedded in a fund like ARC Now what I mean the the black shells marathon option pricing formula defines that stock prices move as a geometric brownian motion where you have a drift the first term here defined by mu and a stochastic component where you have the volatility here and the vener process which does the stoastic part. But what is the probability evolution of this model? And the probability evolution pretty much will yield this equation here which basically just left you with the gradient meaning that by equilibrium economics uh that that dissipative aspect uh I mean comes to an equilibrium. So there's no selenoidal flow. I mean put into economic terms or into physical terms in this case because again the model is an equilibrium model. So it will always reach equilibrium. So if it always reaches equilibrium there will be no exploration no epistemic value into the model. So what we are claiming here is that this agentic active inference framing would actually is a non-equilibrium dynamics applied to financial markets and it can be applied to other type of markets where we know that even though at certain very quick point in time points in time the financial market or one specific market might reach a certain equilibrium that equilibrium would rapidly dissipate and start moving towards another equilibrium. And the reason why again is because standard equilibrium economics assumed that agents were s sort of statics sort of robots. But when you have changing belief dynamics, it's always going to be non-equilibrium dynamics. And in this sense we say that the synergetics theory also behind a lot of the active inference can naturally be applied in the self-organization and market regimes and interestingly in the econopysics literature which is like a what's a what we could say like a niche group uh within the economics field has basically been studying this being I mean applying synergetics and applying other type of phase transition physics uh literature to try to to try to understand market regime transition and also the emergence of prices and different market phenomena from the microscopic and we and we define microscopic as for example the acting of an individual like some in his own portfolio and how that accumulates between millions of investors and that gives rise to the the broad uh macroeconomic behavior. behavior that we see in markets and eventually even in world financial markets. So we can see this emergence of economic phenomena coming from essentially microeconomic foundations and naturally that framing as well can be embedded into trying to understand uh the theory of market regimes from an active influence perspective as well. So what we are just saying here is that this framing of gradient flows, cenoidal flows, synergetics gives you a n gives you a very natural technology to see these type of microeconomic phenomena and microeconomic questions and address them with this type of technology and in an agentic fashion with the with the with the applied active inference agent. So what are the advantages? I have discussed some against both for example LLMs and in this case a closer application would be deep reinforcement learning and also against traditional finance. So against deep reinforcement learning uh number one is the sample efficiency. Instead of requiring sillions of data uh of data points to train the the LLM in active inference uh you have an adaptive agent that is training from real data or coming data. So it's much more efficient. Now in deep reinforcement learning we can add ad hoc exploration parameters whereas in active inference it's inbuilt within the optimization function. DRL the mo or in general deep learning the most important aspect is the explanability and transparency aspect because by definition is a black box. uh and in investment management and also from coming regulation in future years is going to be extremely important uh when applying AI methods to financial decision making that you can actually explain how the agent got to a certain recommendation. is going to be key for regulation purposes. And the active inference framework with a graphical model allows you to clearly understand how and why the agent recommended a certain decision or why not very clearly very transparently. And finally, you have implicit uncertainty in deep reinforcement learning. Whereas again in active inference uh the entropic term is explicit into the model and you can also add confidence gates uh regulated by entropy policy entropy like like we did in our implementation. Now again traditional finance uh I mean not to joke but it's it's sort of a lowhanging fruit because again most of fi standard finance has stayed in models that were built in the 60s and 70s and they haven't been updated much now naturally academics have updated them but in industry practice uh I mean besides some small quant shops uh in very few places most financial departments in most corporations are just using models that were built literally in the 70s and 80s which are static uh which have I mean have no no way to adapt themselves to changing circumstances have no account for uh information gathering or epistemic affordance uh none of that pretty much so as I mentioned epistemic value is completely ignored in traditional finance uh most standard applied finance often use findings from behavioral economics. I mean more more often is now included in in some aspects in more in marketing but not that much in trading. Um and in terms of reging adaptation uh at least in the applied sense is almost completely uh ignored in in applications uh of real investments. And we can see that again standard discounted cash flow models the blackshot models uh which are real models that most people learn and most people use are is completely I mean completely ignored all of these points and right I mean fine I mean we're about to conclude but just before the standard metric to measure the performance of portfolios in finance is what we call the sharpie ratio. So the Sharpie ratio is essentially this measuring the expected excess return of my portfolio against the risk-free rate. So that's what we call the excess return because the the risk-free rate would be like that the the baseline of money of or investments divided by the standard deviation which in finance we call the volatility of that portfolio. So in plain English it's basically measuring a ratio for risk adjusted returns. So what we care in investments is not the absolute return is the risk adjusted return because if I have a very large return but taking huge amounts of risk that's not interesting because probably if I follow on on that strategy I'm going to lose money eventually. So what I want is a high return with low risk or control risk let's say. That's why I care what I care for. Now, what we're proposing here is that given that our agent is adaptively learning, has online learning, we actually want to measure that information gathering efficiency. So, we're proposing including an additional metric to investments called called the entropic sharpie ratio. So we are doing the same thing that we do in the sharpie ratio taking the excess return of our portfolio against a a risk-free rate but we are dividing it not by the portfolio volatility but by the policy entropy. So that's what we are dividing it for and is essentially measuring the excess return per bit of informationational work performed. So is is measuring the ratio of how information gathering efficient or how efficient the exploration component is happening within our active inference agent. So for example, how can could could we read it? Uh if we have an expected sharpie ratio above 50%, we can say that the model is well calibrated is being very efficient on using the information it is gathering to trade and to generate profits. If it's low, we can say that that information seeking aspect is low is not being efficient. Now, why that why this is important is because increasingly investment management firms are are are are implementing uh LLM, natural process, natural language processing strategies to gather lots and millions of bits of data of market information to trade. And it would be useful to measure how efficient energy efficient and costefficient that information gathering is being for generating profits into the portfolio. And if you can actually show that your model is much more efficient into that that can be also accounted for in the economic let's say valuation of how how effective the agent is. And almost last last slide the explanability as I just mentioned a key aspect that is coming in re in in I mean very very very few years is going to be AI regulation and especially in fields like finance in fields like healthcare given I mean I mean just having recommendation agents uh that that you cannot actually see how they are coming into that recommendations it's output that's probably not going to be accept acceptable for regulators. uh because if people lose money in if if people get bad medical recommendations somebody needs to be accountable for and and and if that if if deep learning or LLMs cannot be actually interpreted in that sense in how they came into the output then this type of machine learning we believe is going to be uh the standard way to do financial machine learning because first I mean first and foremost you can decompose the entire model to see exactly what caused the model to to come to a specific uh decision to a specific recommendation. So we believe that that's a key advantage against other standard deep learning methods. So to conclude what we reviewed today was essentially that active inference is not per se a different theory. is a much more general theory that can actually account for most of what has been done in economics uh as of now but done right let's say or done or done more much more robustly. Uh it also integrates other aspects that have been missing into economics for example adaptive learning or online learning. Uh and it's framing again valuation or asset pricing as planning plus control. And also we can show that standard financial models like discounted cash flows emerge as special cases where you only focus on the pragmatic aspect of expected free energy whereas this model is much more more robust. It's also including the epistemic value component which again we just mentioned is becoming increasingly relevant in the world where uh AI agents are going to be all over the place trying to gathering data and to distinguish from fake data as well. That's also going to be a key uh a key value that we want to have in agents performing any sort of task. And finally uh more specifically to the regime uh to the regime to the market regime theory. uh this is a type of agent that will allow you to understand more microscopic market phenomena or more micro finance type of of of phenomena but at the same time being embedded into a single individual agent. So it allows you to let's say reconcile uh some like pending uh phenomena in microeconomic theory w with micro microeconomics and finance in this case. So we believe that our core contribution is that active inference provides a principal framework for portfolio management under uncertaintity. And as we just showed with demonstrated risk management advantages because the I mean if you if you define that the agent is risk averse as naturally people are risk averse it will usually engage in much better risk management than standard traditional methods or also reinforcement learning methods where it's very much focused on the reward. So thank you very much. Uh, so we're going to go with some prepared questions from Matt and then we go we're going to address some questions from the audience as well or we're running out of time. Oh, >> awesome. >> Awesome. >> Uh, thanks Daniel and Sam. Uh, excellent stuff. I want to take the perspective of someone who would be a practitioner in the industry and there's about riskreward management. Uh there's a lot to be said about portfolio construction and the separation of trading and compliance in most firms. Well, I really want to focus in on this contribution of the entropic Sharpie ratio because to me this fundamentally reframes how we think about data economics. An asset manager might spend tens of millions of dollars on Bloomberg terminals and satellite imagery without any idea of whether each new data source actually improves their returns. >> Right? This actually suggests that we can measure the value of each information source by its policy entropy. So I wanted just to make it super practical. Could you outline how you would do real time scoring of different information signals like tweets or data releases or earnings reports that could help readjust a portfolio based on the quality of those different information feeds over time? >> Yeah, sure. Basically we would say that I mean pretty much the higher the the ratio is that means that whatever information you're feeding into the model is being much more efficient in terms of the signal to noise ratio. So it's been much more efficient on signaling the correct information or the valuable information to I mean to maximize its model evidence or minimize its surprise let's say whereas if you're fitting whatever information other information if the policy entropy is low that means that is is doing the opposite effect is being uh is not helping the agent much on maximizing its model evidence or minimizing its surprise. So that would be like the the the clear like the clear implementation and what's valuable as you just said is that later on firms could could literally quantify it in monetary terms on I mean on waiting how much money they're spending on uh on feeding data into a specific strategy and how much that data uh is providing in returns. So that also gives market opportunities for people willing to create methodologies, data sets, etc. to make that that process much more energy efficient and data efficient as well. >> That's really important. >> I have a quick question sort of a a big p big picture question. I if you if you really look at what you're saying, you're you're providing a really new paradigm for identifying value and and it it fuses economics and finance and it it actually combines monetary policy and fiscal policy in some form. And when you look at the current economic system where it is look using an arbitrage of different opportunities and risks and spread bidden as risk spreads. This actually is a replacement mechanis mechanism viewing the market as as as a markoff blanket that aligns different agents around specific outcomes. So this becomes is is is far bigger than sort of me being simple technical trades. It's a very different way of envisioning the market and and yes you and I think that is going to have really significant implications at multiple scales. Would you agree with that or do you is that overreach? Um >> no totally totally as you as you just mentioned um right I mean we are literally refraraming how we would conceptualize macroeconomic uh structures on or or market structures uh with I mean within mark of blankets and how the agents as active agents not as passive like just uh static agents in standard economics are actually uh I mean taken I mean are accounted for into that theory. So I as I mentioned in the beginning it's a way of framing in general finance and in this case I said pricing but in general framing economics into an agentbased theory. >> Which I think is the right way because in in because it's how people behave in their economic decisions. So I think it's the natural way to frame the theory. What? >> Yeah, just as we're out of time, I'll just for for um completeness and just for your all's future work, active lumology wrote, I hope they discuss how they propose integrating this model into underwriting. Bert wrote, "What is epistemic value when actions do not influence the market?" And then also active bloomology wrote about the parameters of defined efficiency and financial modeling. So just just a few notes for possibly a future symposium, but thank you all for the awesome work. >> Thanks a lot. And for those who send questions, you can send them to me through this email and we'll we'll get back to you. Thanks a lot for your time. >> Farewell. Bye. >> Bye-bye. Hello, welcome back. We are with Denise Hold for from coordination to cognition rethinking agent communication protocols for active inference in the spatial web. So Denise, thank you for joining again. Anyone in the live chat can ask any questions and thank you for the >> Thank you so much Daniel. Thank you for having me. Okay, I'll just jump right in. So, today I'm going to be talking about current agent communication protocols that we're using with current AI versus what's coming with the spatial web and active inference. So, with the rise of intelligent agents, agentic AI comes essential need for multi-agent coordination. So, as a step beyond reactive chat bots or standalone systems, LLM agents are now being designed to collaborate across tasks, domains, and organizations. But how do we get these agents, these LLMs, to talk to each other, share context, access tools, perform tasks, and make decisions in a way that is secure, scalable, and interoperable. The response has been a new class of communication protocols. You have MCP which is model context protocol. It's from anthropic and this gives agents access to essential data sources. ACP agent communication protocol from IBM and Cisco enables structured collaboration between agents. A2A agentto agent protocol from Google standardizes agent tasks across vendor systems and then there's which is an agent networking protocol and this is open source and it supports decentralized agent discovery and secure identity management over the web. Now why do agent communication protocols matter? Each of these approaches has unique strengths and limitations. They're all new and they're beginning to be implemented, piloted, and standardized today. Uh they just came on scene this this last spring. And they they help developers solve urgent problems enabling the first wave of autonomous systems. So they're not just valuable, they're an essential part uh at this moment in time. And it's agent communication for the worldwide web. But a larger shift is underway. Parallel to these efforts, the spatial web is rising. We have HSTP, hyperspace transaction protocol and HSML, hyperspace modeling language. And this forms a radically unified infrastructure to contextualize and connect everything from people and places to devices and data. It enables a globally consistent framework for describing, discovering and interacting with entities across physical and digital realms. And this framework was ratified by the ILE E, one of the largest core standards bodies in the world on May 29th of this year as the third official foundation foundational internet protocol layering on top of HTTP and HTML. And this protocol doesn't just accommodate agents, it redefineses the ter the terrain that they operate in. So this raises a pressing question. As the spatial web's universal domain graph begins to take shape, will today's agent communication protocols evolve to align with it, or will they eventually become redundant? These [snorts] new agent communication protocols reflect a legacy of centralized computing and static data models. Active inference on the other hand requires something more. Realtime action perception loops, causal modeling and dynamic adaptation to unfolding environments. The current agent protocols are solving today's problems, but they may not be structurally suited for the agents of tomorrow. So agent protocols at a glance bridging the gaps in today's intelligence systems. So let's go one by one just so we can understand what these protocols are doing currently. MCP by Anthropic. The purpose it provides a standardized way for LLMs or agent frameworks to retrieve structured context like files, APIs, databases, system tools and it acts as a content feed that simplifies how a model pulls in relevant data. The strengths its simple standardized integration of tools and knowledge for LLMs like GPT or Claude. It enables straightforward integration reducing development time and cost and it enables developers to plug in live information to otherwise static models. But the limitations MCP agents they don't re reason about the world. They pull in data. They process it and output a result based on prior training and prompted instructions. You're still dealing with LLMs. The agent's understanding is contextually injected. It's not self-modled. Active inference doesn't just fetch context. It continuously infers it, updates it, and acts upon it in real time. The MCP model isn't designed to support that cycle. Now, ACP, agent communication protocol by IBM and Cisco. This enables structured multi-agent collaboration between agents in enterprise environments. Its strengths, it supports stateful threads of communication and allows agents to negotiate and reason together across tasks. Facilitates task coordination and collaborative workflows. Uses uses standard APIs and JSON schemas. helps AIdriven enterprise applications coordinate complex tasks like order fulfillment and it establishes clear workflows and roles for agents. But the limitations are it requires a centralized directory for discovery and coordination and context is carried in messages. It's not modeled internally in the agents. In an active inference architecture, agents don't just cooperate, they co-regulate. ACP enables inter agent chat. Active inference agents exchange beliefs, align internal models, and adapt as they act. Now, A2A by Google, its purpose is cross vendor interoperability for agent tasks between tools like CRM, HR systems, etc. from various vendors like Salesforce, SAP, Oracle and others. The strengths it solves real interoperability pain between tools like CRM or HR systems or productivity agents by reducing operational silos with within your tech infrastructure. It's well supported by major enterprise vendors and it's built on familiar standards like HTTP, JSON, and OOTH. >> [snorts] >> Now the limitations it's optimized for task exchange but not cognitive computing. So there's no real world grounding or embodied understanding. Agents operate like modular services in a cloudnative workflow. Active inference agents perform perception planning and action within a unified framework. A2A enables interoperability but it doesn't enable inference. And then you have and this one is open source. It it enables decentralized discovery and semantic identity for agents and it comes closest to aligning with what's needed for active inference in the spatial web. And the strengths are that it utilizes decentralized identifiers and linked data. uh potential foundation for semantic interoperability, discovery via open registries and search indexed agent descriptions and it empowers secure agent interaction without a single point of failure. But the limitations are that it lacks shared global context and transactional knowledge graph that the spatial web provides. challenges in scalability and real time context synchronization. It [clears throat] connects agents but it doesn't connect them to their environments and it's difficult to scale enterprisewide without global context management. So to sum it up, you've got MCP which is easy context access but there's no internal inference and it's static and non-addaptive. ACP is for task coordination, but it's centralized and limited semantics and it's not flexible. [snorts] A2A is cross vendor integration, but there's no adaptive context modeling and it's not contextaware. And then is decentralized security, but it li it has limited global synchronization and it's complex to scale. So introduction to active inference AI and this is [clears throat] a complete paradigm shift the principles of active inference agents continuously update internal models to reduce uncertainty. It's based on the free energy principle. It employs generative models for decision making and predictions and it moves beyond reactive machine learning towards adaptive predictive behavior. and you minimize uncertainty. These agents adapt in real time. So why does active inference matter to a business active advanced AI approach? It's an advanced AI approach that helps agents predict and adapt dynamically enables proactive responses to emerging business challenges and it significantly reduces uncertainty and improves operational resilience. Now, generative world models and causal reasoning. Active inference agents proactively model environments. They are continually creating generative models of understanding about these environments. They're simulating multiple business scenarios and outcomes. And this allows agents to understand why something happened, not just what happened. And it improves strategic decision-making and risk Now, one type of active inference uh model is renormalizing generative models and these are an advanced form of active inference modeling. They're they're hierarchical and scale-free and they efficiently manage complexity. They enable real-time modeling and adaptive predictions across domains, efficiently handle complex operational scenarios, and significantly reduce data requirements and computational overhead. So while generative models like RGMs reveal how agents learn and predict, the next evolution including multi-agent systems brings physics itself into the inference process where belief and behavior evolve inside a shared field for multi-agent coherence and belief propagation. So when you're talking about this field of belief for multi-agent belief propagation, there is an architecture called field of belief Hamiltonian Monte Carlo. And this is this takes us from models to fields. It moves beyond hierarchical layers to a unified field of beliefs shared across agents. It has Hamiltonian dynamics. It applies energybased equilibrium for efficient inference and real time adaptation. It includes isohedral symmetry. Each agent is seated from the same geometric program enabling instantaneous alignment and a silent consensus among the agents and decentralized coherence. Agents act locally yet remain globally synchronized through the shared field. And this establishes a physics grounded architecture for scalable self-organizing intelligent systems. So given all of that, what kind of communication protocols are needed for active inference? Active inference requires continuous dynamic updating of context. It needs protocols supporting real time semantic synchronization and it pushes beyond task oriented protocols towards holistic environmental modeling. So why businesses need active inference compatible protocols? They need them for tradition because traditional protocols lack the dynamic adaptability that businesses require. Active inference demands protocols capable of managing real time changing conditions and this is crucial for responsive operations for customer experiences and for innovation. So the spatial web building the world agents can think inside of. So when we look at the spatial web infrastructure and we consider the the space between the physical and digital worlds, right? You have connectivity and then you have this spatial management area that includes things like identity and activity and location and it's in this space where trust and privacy and security and interoperability lie. So this is a new infrastructure layer that unifies physical and digital worlds and it provides unified context, semantic, shared meaning, spatial and temporal interoperability and it's a foundational layer that enables complex adaptive agent interactions. So it's the next generation internet that it integrates digit digital and physical business operations with intelligence layering over all complex systems and it allows for the coordination of processes, people, entities and places seamlessly. So what is hyperspace modeling language HSML? It's an ontologybased semantic modeling that provides universal consistent meaning and definitions. It describes entities, relationships, permissions, locations, and states. And it ensures clear consistent communication and interoperability across domains. So when you think of a spatial domain and a web of spaces, you need to know who or what is authorized to access the domains. What content or data is available to view? Who can publish or modify content? Who can transact or interact within it? So this standardizes business semantics. It provides a universal business language for digital and physical entities ensuring consistent meaning across all business units and partnerships. So, hyperspace transaction protocol this protocol is it provides secure contextaware queries and transactions. It supports decentralized multi-dimensional data exchanges. It enables real-time consistency and context coherence across the spatial web. So it it offers secure and contextrich transactions, a secure trusted method to coordinate complex multi-party interactions. It enables enterprises to respond immediately and precisely in complex environments. And it enables decentralized multi-dimensional interactions across the enterprise. And what it results in is this universal domain graph or UDG. A decentralized permissioned and constantly evolving model of everything. Not just information but real time spatially anchored representations of the physical world with global decentralized knowledge graph of all entities and states. It stores spatial, semantic, and temporal data. It's a common source of context for agents and systems. And it enables real-time queries and context updates and synchronization across domains. So context is not fetched, it's embedded. Entities don't just reference data. They're part of the global graph. Every agent, location, object, and event can be described, queried, and updated in a universally consistent way. So agents can locate themselves, interpret relationships, and respond to unfolding changes without relying on private databases or hard-coded integrations. So [snorts] the UDG creates this shared business reality, a common realtime operational view for your entire enterprise ecosystem and it dramatically improves visibility, responsiveness and decisionmaking accuracy. So for active inference, the spatial web offers something profound. Active inference agents operate on the principle of minimizing uncertainty about the world by continuously updating internal generative models of understanding based on incoming data. They don't merely perform tasks. They infer the hidden cost the hidden causes of their observations and act in the world to fulfill expectations while adjusting their beliefs. And this requires a deeply contextual understanding of space, time, causality, and intent. The spatial web delivers what active inference needs. It delivers spatial temporal grounding. Every entity is located in time and space, enabling agents to align their models with real world dynamics. It offers semantic consistency. HSML ensures that room, temperature or flight path mean the same thing across domains. It offers contextual granularity. Agents can navigate the universal domain graph to retrieve just the data they need at the right level of abstraction and scale. And it offers permissioned access and trusted frameworks. And this is crucial for real world applications. agents can access content securely and act within regulated boundaries. So the spatial web provides the shared external memory, sensory input, and environmental structure that active inference agents need to operate effectively at scale. So contrasting current agent protoc protocols with the spatial web, excuse me. this takes us from API calls to adaptive worlds. The spatial web is not just a medium of interaction. It's a model of the world itself, accessible in real time. It isn't static. It evolves as agents act. The feedback loop between agent and environment is central to active inference and it's what makes the spatial web a uniquely compatible architecture for the next era of intelligent systems. Where legacy agents process tasks in isolation, active inference agents embedded in the spatial web can act as contextual participants, constantly sensing, reasoning, and responding to a dynamic multiscale world There are architectural differences with current agent protocols. They're pointto-point versus shared context. Agent protocols like MCP, ACP, etc. They they're pointto-point, isolated context, taskdriven, and centralized. Spatial web is graph-based, unified, global context, inferenced driven, integrated, decentralized, and globally synchronized context. So you have centralized versus integrated. Current protocols are narrow taskbased communication. And the spatial web offers unified realtime operational visibility and context. So the spatial web supports active inference agents by maintaining a persistent structured reality that they can pull from, update and adapt to. Discovery and identity management, isolated registries versus global semantics. Current agent protocols, their localized states, centralized registries or basic decentralized discovery. Spatial web is real time global context management, semantic global entity identification and universal discovery via spatial web identifiers which are kind of like URLs uh but within the universal domain graph. So in the spatial web entities are discoverable by meaning, location, time, function or any combination. And if you look at this graphic that I

I inserted in here, it's one of my [clears throat] slides that I used often uh when explaining this because HSTP and domains in this universal domain graph, these new spatial web domains, they're very different than what we know uh as a worldwide web domain. And if you look at current AI, it's built for the world worldwide web. So a a a worldwide web domain is includes websites and apps and these serve up content and generate data and they enable you to do things with that content and data. Right? So current AI is built for this system. Now a spatial web domain is an entity with a persistent identity through time with rights and credentials defining people, places, things and rules in machine readable context and spatial grounding. So active inference AI is built for this new uh framework. [snorts] So for highlevel causal reasoning of active inference, agents need more than just who to talk to. They need to know what exists, where, when, and why it matters. State management, local versus global agent protocols. These current agent protocols, their local state tasks, messages, context, and ACP and A2A allow for message threads or longived tasks between agents, but the state is local to the conversation. and MCP supports context injection but not persistent world state synchronization. Whereas the spatial web, the universal domain graph becomes the single source of truth for all agents. Every entity event and state change is part of the global context accessible and queryable by permissioned agents at any time. To minimize free energy and make accurate p predictions, active inference agents must align their internal models with an externally observable state. A common source of synchronized reality like the universal domain graph allows agents to coordinate without needing custom logic to reconcile individual perspectives and semantic interoperability. current agent protocols, they're message-based formats without shared meaning. Whereas the spatial web is ontologydriven semantic interoperability via HSML. So clear semantic standards ensure consistent meaning so that a term like location or battery status refers to the same concept across systems. And then there's causal reasoning and adaptation. and you have fixed workflows versus embodied understanding. So today's agent protocols treat agents as isolated services reacting to tasks and triggering responses. Active inference by contrast enables agents to simulate multiple futures, plan under uncertainty, and infer hidden variables based on context and change. The spatial web enhances this by providing real-time structured context that agents can reason about causally, bridging the gap between perception and action across a shared world model. This enables enterprisewide responsive intelligent decisionmaking. So, active inference agents within the spatial web. This ushers in a new era of autonomous intelligence. Agents don't just cooperate, they co-regulate. They learn, infer, and adapt in concert with their environments. In

FOBHNC terms, each agent is a localized excitation within a shared belief field, acting independently yet maintaining global coherence. Agents don't just follow commands. They understand situations, simulate consequences, and act accordingly. And most critically, active inference agents don't need to be tethered to a centralized database or cloud tools. They can operate on the edge, inferring, adapting, and acting autonomously. But to thrive, they need an environment where context is available as structure, not just as data. And that's where the spatial web becomes indispensable. The future of agent protocols in the spatial web era. So navigating the road ahead, the agent protocols emerging today, MCP, ACP, A2A, A&P, they seem to be solving different problems than the spatial web. They manage task execution, data retrieval, peer messaging. And the spatial web aims to model the entire digital and physical universe. But when we examine how these systems interact and where technology is heading, a pattern begins to emerge. These protocols are not competitors to the spatial web. They're early bridges, temporary scaffolding across a digital landscape that's still taking form. They're a necessary layer for now. There's no doubt that today's agent protocols are doing important work. These protocols are gaining traction not just because they're useful, but because there there's no shared substrate yet. Each protocol is a patch for a fractured internet, stitching together siloed APIs, services, and data under a common interface. They're vital in a world where agents are still grounded in machine learning architectures that lack self-modeling, physical grounding, or native contextual awareness. The spatial web from protocol stack to contextual substrate. The spatial web is not simply a next generation protocol stack. It's a paradigm shift in how agents interact not just with each other but with their entire environment. Active inference provides the internal generative model. The spatial web provides the external semantic and spatial context. Together they enable real time scale-free causally grounded intelligence across distributed networks. In this convergence, the universal domain graph becomes the ambient knowledge base for all agents, no matter who built them. HSML becomes the language agents use to interpret their world. HSTP becomes the protocol they use to act upon it. So what happens to these current agent protocols? Well, in the short term, there's collaboration. Today's agent protocols continue to address urgent interoperability needs. They form essential bridges toward broader spatial web adoption. Midterm, we're looking at convergence. Protocols will begin to integrate or evolve to align with the spatial web standards, integration pathways, HSML for agent descriptions, HSTP for transactions. Businesses should start aligning protocol strategies with the spatial web principles and early uh adaptation will ensure competitive advantage and future readiness. Now in the long term there's a risk of protocol inaction. Many protocol functions like discovery, negotiation, task management, they're likely to be natively supported in the spatial web. interoperability challenges will disappear and failure to adapt may lead to technological redundancy. And if you look at you know the evolution of the internet historically uni universal and open standards inevitably replace fragmented systems early web protocols were displaced by HTTP and HTML. So with my company with uh AIX uh global innovations we're working on an ecosystem of adoption that includes education and guided activation and tools and solutions a continuous path from learning to deployment. And if you'd like to learn more uh feel free to scan this QR code. It'll take you to learning lab central. It is a global education hub that uh we have built that is hyperfocused on active inference AI and spatial web technologies. I host monthly learning labs and uh we have courses and certifications available there and it's just a great place to to come together as a community and find other people who are interested in building in this uh field of technology as well. So, I will stop sharing now. >> Thank you, Denise. Awesome. >> Thank you, Daniel. >> Okay, some fun questions in the live chat, so I'm just going to read them off and and we'll go with it. Okay, Bert Burkers wrote, "When and how can people get started building on the spatial web?" >> Yeah, that's a great question. And um I don't have an exact date. I know that the standard was approved uh and ratified by the the ITE in May and uh I know that the consensus is that towards the end of this year I'm assuming end of this year beginning of next year we're we'll start to see some tools that will make it really easy to start building uh registering domains in the universal domain graph. I know the spatial webb foundation.org org is the authority that will be um handling the universal domain graph. And so I'm assuming there's going to be tools where that can instantly start to register. And I've seen tools where um you can put in documents, you can put in information, it'll register the domains, it'll spit out the HSML for you. And I know those tools are on their way. I just don't have a date for those yet. So, I feel like as soon as those things land, then it'll be really easy for people to just start, you know, kind of going to town on this and and exploring and figuring out how how this looks in the wild. >> Awesome. Okay. Scott David wrote, I I'll put it in the chat just in case you want to see it. He wrote, "What are the steps through which the spatial web and contextrich AI interactions can be onboarded to existing organizations, companies, governments, etc. cost savings, risk reducing?" >> Yeah. Uh, and that's similar to the answer you're going to get that I just gave you. [laughter] I don't have all those details right now. I know that they're coming. Um and uh yeah, I mean I I would keep an eye on the spatial web foundation because uh they will be the entity that starts to uh onboard people and give uh information on how to on board and how to start building out in this space. Yeah, I think it was a great great framing and very helpful to get that review of these protocols and I just got the sense of all these different pairwise or hub and spoke type interaction architectures but they're still very generic and they're not necessarily about any real physical or embodied space. So some domain ontology is still required >> to have the kind of content of that agent to agent channel be meaningful beyond just being like a handshake. >> Right. Right. I mean, we're really good at message passing and injecting, you know, data, [laughter] but you're still dealing with the LLMs, you know, and there, you know, it's it's uh there they have hard limitations, right? You know, and when we talk about active inference and we talk about the capabilities of active inference, we really want to enable all of those uh within within a framework that enables this embodied AI, right? that can tie these agents to the physical world um and this kind of digital twinning of everything. >> All right, I'm going to read a question in the chat and then um I'll just read read a few more things that were followed from there. So, Alex Sabine wrote, "There are some quite profound onlogical implications for humans who share this field of belief with agents. How are we managing the ethical discussions around this?" And then Scott David added to that plus one. Ethics is about human ideas about human behavior. So maybe we need new approaches to ethics more akin to an ethics of treatment of other species, ethics of environment interactions with nonhumans. So I know that's a few different pieces, but how do how do you engage with this at at that level? So what's really interesting to me about that is you know when you when you try to think of you know what we've the approach we've tried to take with you know ethical ethical use of AI and you know bias mitigation and different things like that it doesn't really work because you can't there's no amalgamation of that globally if you look at our world you have all kinds of belief systems all kinds of different cultural differences is all kinds of different governing styles. So you have to have a way that preserves all of that. And so this enables that. The spatial web enables people to be able to build agents from wherever they are in their world in the world. And they can imbue them with their their um you know their own beliefs, their own you know uh they can be customized to to the various differences and preserve those differences throughout societies throughout the world. Right? So I think that this is the system that will actually enable us to you know solve these create ethical AI localized right and then because these agents can share beliefs there can be a global consensus you know of certain things the same way we've done with human knowledge right I mean the way human knowledge grows is it depends on the diversity in thought and and ideas but we also come together as a species to have a consensus over, you know, um larger overarching uh governing ideas, right? And so I think we're going to see that play out with these agents within this next evolution of the internet as well. Okay. you have traveled around and given many educational talks and hosted a lot of learning labs and and experiences. So over this last year, what would you say has been surprising or interesting in in those experiences? Like how over this journey have you seen the kinds of questions or interests that people are showing up with in education settings evolving? Well, it's interesting because I I see that people are starting to understand the implications of this and you know, it it's also really funny because um you know, everybody's been really focused on these LLMs and you know, current AI because they're they're amazing. They're powerful tools, right? But I've been out here talking about what's coming with spatial web and with active inference you know a lot of people don't really know what to or you know I I for a while I was a lone voice in the wilderness right you know people would be like I don't really know what to make of it but this is really interesting because it's right here and it's right in front of me but that I think has helped with this kind of adoption cycle and this this understanding because people have been able to develop expectations for what they really want from these tools and then they've also had firsthand experience with what we have currently to understand that they have hard limitations and they're not really going to get those some of those things from these tools. So I feel like over this last year there's been a lot more people waking up to the fact of of really understanding what active inference offers the the potential of this adaptive self-evolving you know uh superefficient intelligence and distributed intelligence. you know, getting away from the uh being tethered to, you know, the mothership of the giant databases and being able to operate at the edge and and everything else. So, I would say that's probably the biggest thing I've noticed over this last year is that people are starting to really, you know, develop more of an understanding, expectations, and having the ability to recognize why active inference would be ideal compared to what they're experiencing currently. >> Yes. Interesting. And then also just sort of on the education wavelength, how do you hold the space and make it relevant when a lot of the models are very technical or people might not be building basian statistical models in their basement as some do but not many. So how how how do you convey that or what have you just found interesting or useful about how to bring technical ideas and try to make it relevant? >> So most of the people who are attending like my learning [clears throat] labs and you know learning lab central a lot of them are not technical people they're business leaders. They're they're wanting to understand why they need to choose this versus something else and how it's going to affect their industry, how it's going to affect their business. And you know, they need this deep understanding about it so that they can make proper choices, you know, moving forward with their business. And so, you have to be able to take these complex ideas and make them understandable for anybody. And so, that's what I really try to do. And it starts with meeting people where they are. You know, you you don't stand over here and talk about something and expect that someone is going to come and just start understanding it eventually. You have to start talking to people where they are, you know, with uh a common ground of understanding with with where what they understand and then you bring them to you. Then you start to guide them, you know, towards you as far as what you're trying to explain to them. So that's really the approach that I take. Um you know and and so that means you know getting rid of a lot of the jargon and just you know talking in plain plain terms you know um but I think a lot of it too there's a lot of moving parts to all of this. So repetition is important. You know, people [clears throat] being exposed to the ideas over and over and maybe exposed from different angles, you know, and that helps to kind of bring this cohesive understanding together. >> Yes. All all make sense. Um what are you going to be most looking forward to? like what will reduce our uncertainty the most in the coming months or in the year ahead. >> There's a lot to look forward to. [laughter] Um you know I one of the things that I'm I'm doing with my company is really um focusing on establishing this ecosystem of adoption. And you know I know that as these as these tools become more and more available the ecosystem of adoption has to be able to scale with the demand. And I mean if you look at the reality of what active inference does what the spatial web does and we're talking about the internet right and we're talking about this layer of intelligence that will be operating throughout this new expansion of the internet. This is going to affect every industry across the globe, every business and and anybody with a website or an app is going to need to evolve their presence to take advantage of these new capabilities. So the demand is going to be high. And so with my company, what I've been trying to really wrap my head around is how do you scale for that demand? How do you scale the adoption for that demand? So those are the those are the problems that I'm I'm attempting to solve. And so it's exciting for me because I I really see the framework uh starting to be established within my company for that. And so I'm just I'm really excited for that. I'm excited for what's coming in the next 6 months, 12 months. >> Very cool. Any last comments or anything else you want to bring up or ask for people to consider? You know, Daniel, I just want to give you a big shout out and thank you. You do so much for the active inference community and you know, I think a lot of it's done behind the scenes and I for one am really grateful for you. Um, you know, and I'm always here to help support the Active Inference Institute in any way that I can. I think that what you guys do is a phenomenal job at putting education out there for active inference. And uh you know I I I have a whole section for you on Learning Labs Central [laughter] so in the resource locker and I'm just I'm I'm just very grateful for you and and all the hard work that you do. >> Thank you Denise. It's it's a team ecosystem effort and it's really exciting times. So thank you. Till next time. >> All right. Thanks so much. Thanks everyone for tuning in. right, we're back for our final live presenter of the day. Viet Dunwin will be presenting on active neurobotics, deep prior preference learning guided by learnable goals. So thank you via for joining again and for the NAC lab for sending their students to present. Take it away. >> So yeah uh thanks Daniel for having me uh today. Um I'm Viet. I'm a PhD candidate at Rochester Institute of Technology. Um I'm currently working with Dr. Robia um in the neuro adapted computing lab. Um and yeah today's I will be discussing um our work um active neurobotics deep uh prior preference learning guided by learnable goals. So first of all I want to talk a little bit about um introductions and motivations. Um yeah so we can see um nowadays there's a lot of like applications of robotics in several um different fields right so for example robotics can be applied in industry industrial manufacturing for example speeding up assembly processes in factories can be um applied in healthcare cani in farming agriculture um and constructions right um this can also be further applied in locomotions for example self-driving car and space explor exploration. Um but there's this um several problems in this where um most robots cannot operate um and learn autonomously. Um and then they actually cannot adapt to new environments as well. Um and generally training a um robot learning algorithms actually cost a lot of amount of resource for data collections um that's a lot of money. And then um the safety measures of robots are often not learned. So it's cannot be flexible enough to work safely around human. So to overcome these barriers right um research has actually investigates different uh ways to tackle this. For example, one ways to uh tackle this is to investigate neurobotics which is a sub field of robot learning. Um basically it's modeled a robot's learning process um very similar to that of a human. Um so a biologically inspired automat learning system often investigates um human perception, memory, attention, sensory motor control um and so on. And so uh this new robotics field also features um robot learning from real world interactions or online learning, right? Um so with online learning u robot can adapt to new environments uh with less data collection cost and and then can be learned to be safe around human. um first of all we need to in order to do like neurobotics we need to investigate um cognitive architecture. Um so the cognitive architecture is um it's very much um studied in the active inference process theory. Basically it's investigates different modules inside the brain that can be used to learn a perception system. Right? So first of all there's like the generative model which basically the procession system itself right. So it's basically trying to um through the sensors um making observations in the environments and then uh learn the um um actual interactions um between the en environments and the agent. Um the second one is the motor which is the action controller. um it's actual uh the action will be taken by the agent um and then executed inside the environments um to return a new and then there's like a memory module um which stores the interactions um between the agent and environments right so that memories kind of memories will be uh stored in here in order for future use of like training or um you know just uh maybe remembering um of knowledge and knowledge uh purifications for example or distillations. Um and then there's prior preference. Um so basically the prior preference distributions trying to predict the desired representation or the desired goal that the agents want to uh achieved in the future. So in so generally in active inference a prior preference distribution is provided in advance. Um but in our work we we are trying to have like a separate uh neuronet network to predicts the prior preference in the future um dynamically right. So uh the last one is um sensors right like for example the robots has like eyes and arms and so on. So before going to the main topic um I want to touch on a little bit about deep activ um and the background around it. So basically uh the interactions between the agent and the environments can be described as a um as the interaction between the generative process and the generative model. So first of all is the generative mode process. Basically it's the real war environment itself right and inside the real world environment there's like a hidden state. So this is like the true probability distributions of the environments and we cannot actually see it. Um [snorts] and uh in uh in in here we can see that there's an arrow between the true state and the observations which means that um the agentive process actually generate the observation that can be observed. observable, right? So the agents can actually see this observations. Um and this um when compared to like this is very similar to a partially observable macation process um that is often formulated in the reinforcement learning So now when we have the genative process or what the genative model which is the actual agents itself, right? So uh the agents also have a state distributions here, right? So um it is actually the agent's best approximations of this generative process. Basically it's we can call it like a dynamic states or a um agent's belief state right or a world model state however you want to call it. So this state distribution will be best estimates of the hidden um states inside the environments. So the hidden states here can be like um the world dynamics for example or the physics or um the logic of the world for example something like that. Um and in active inference process theory um the agents basically continuously trying to make the observation through time um and also continuously modify it belief after making the observation. So um how do we uh how do we further formulate this? Um first of all uh we have the agent state here right it is like the initial belief or representation or the world dynamic states right um it's it's going to be called the prior over state which is P ofS um and then we also have PFO which is the model evidence uh which is the observation distribution um and then in active inference we trying to um have the agents both observing and predicting right so first of all we predict the um observations which is basically the likelihood observations um it is the agent's best estimate of the observation distribution given like the prior or the posterior um information that it have and then finally the P of S given O is posterior function basically it's the agent's best estimate of the state after making the observation so um It's going to be um after making the observations um it's uh the agents actually making or changing the posterior uh belief right. So now we have like all these definitions and of all these functions. Um how do we learn this generative model? So we have to learn um first of all the likelihood functions and then um the posterior approximation right and then the uh and then we have to best estimate the uh model evidence distribution here as well right and then we estimate the posterior um and then the prior right so how do we learn this so inference process theory is proposed that um we trying to um minimize ize the variation of free energy or the marginal free energy in the amount. So in order to do that we first have to uh predict the likelihood of the observation right or the modal evidence. So in debuctive inference generally um this is actually called the decoder and it is often parameterized by your neuronet network. Um next um is the posterior functions right? So at the same time when we're trying to predict the observation we also have to make an observations right and then we make uh we produce the approximate posterior at the current time step. So um in deep learning literature this is often called an encoder and it and it is also often parameterized by a neural network as So uh in here active active inference says that we have to um trying to continuously correcting the belief after making new observations. Right? So we are trying to minimize the cha divergence between the belief states uh after we have make the we have made observations and the belief states um prior to making observations. Right? So we're trying to minimize a surprise between those two kind of distribution. Um that also means that we're trying to minimize the K divergence between the posterior P of S and the prior P of S. So this is also means that um we are trying to correct the belief after making any observations. And then next is um in parallel to uh minimizing the between the posterior and the prior or the surprise between the posterior and the prior we also have to maximize the prediction accuracy of the observation as well. Right? So in this case we are trying to maximizing the similarity between the true observations and the predicted observations and this also called maximizing model evidence or maximizing observation prediction accuracy and we do this by uh minimizing negative lock likelihood um of the predictions. So that is for like one step right and now we want to extend it to multiple step and then we have to integrate actions here as well right so at this at the time step t there is um several different um there's hidden states right that produce the O of T which is the observation um that the agent can see and then the S of T is like the actual initial um it can be prior or posterior uh of the agents and then there's the actions of the agent that the agents like plan to take right so when the agents take the actions um in the environments it is executed and it's produce a new state a new hidden state that we cannot see and produce a new observations right at the same time the agents um can estimate the next state given the previous state ST and the action ST uh it can estimate the the next state ST plus one here um and Then we do the same process right? So um SC plus one is actually the prior of the next stage and um we can do same process by predicting the observations um of the next stage and at the same time we observe uh the observation of the next stage um and then encode it to a an approximate posterior and then we minimize the variational free energy um by minimizing the ka divergence between the posterior and the prior and then maximizing the prediction accuracy by minimizing negative lock likelihood of the prediction um OT hat plus one. So here's a formula for um the variational or marginal free energy. Um you're going to see that in here there's like a complexity term where it's minimizing the car divergence between the Q of S given O um and and P of S given uh ST + 1 minus A T + one um and A T+1 basically the previous state um and previous actions that's cause um the um the current state right leads to the current states in the current prior um And then um this complexity term is like measuring the amount of surprise um that we want to minimize. And then finally we have like the accuracy term where we are trying to maximize the prediction accuracy of the observation. And so we're trying to minimize the negative lo likelihood of the prediction um observation distribution. So now when we have done um the learning for the world model or the generative model we have to learn the policy as well right so we have to know a way to like decide like what kind of actions should be executed in order to optimize or solving the task. Um so there's so many as action that can be taken in one single time step. Um and each of those actions can lead to like different possibilities, right? So the agents can like estimate like each possibilities um based on like previous actions and the current um belief, right? And so on, right? So there's like as the time steps or the horizons spans out um the number of possibilities of the belief states actually goes up a lot, right? So how do we choose the optimal um trajectories of actions that we want to take? Um that's uh brings us to um the expect expected free energy. Right? So we are trying to say that um whatever kind of trajectories um in the future that yields the lowest expected free energy then we choose that as a optimal choose and then we learn the policy from that's exactly like this um the expected free energies um eventually decompose into um an epistemic term and an instrumental terms so in this uh topic I only focus on the instrumental signals. so the theta here is the uh neuronet networks actually parameterized by theta right the pi is planned action right so the instrumental signal is de composing here um it is uh we can look at all these three terms first of all is the um estimation of future observations here and then q of s given pi is basically the approximate posterior in the future like given the policy in the future Right. Um and then there's log of P of O given uh log of P O T. Uh it's basically the prior preference distribution at time t. Um generally in active inference right we provide the agents um with the um prior preference distribution in advance. Um but in yeah but uh I've mentioned before um in our work we will predict this kind of distributions um so dynamically right so that the agents can follow um its own wanted or preferred trajectories and learn the policies that can follow the preferred trajectories. So yeah that brings us to the main parts active uh narrow robotics and goalg guided preference learning. So before going to the main uh main topic is basically I want to um reiterate a lot of problems in current model based uh robot learning nowadays. So popular robot um or model based robot learning algorithms often deploys um the offline policy onto a real-time robot. What does it mean? So it's basically it's learn an online an offline policy and then it's evaluates in the real time. So there's no little to no online learning capabilities. Um second of all is uh they do not leverage the concept of prior preference learning. So no goal predictions. Uh so the agents might not want like might not know what it wants right. Most of the algorithms actually only maps from the vision or the language or both um directly to the actions. Um that means um it is only learning uh it's only treating all these experience as um identically distributed independent and identically distributed right and then next is depend depends largely on expert demonstrations. So if there's no if there's little to no expert administrations then the policy will suffer from suboptimal policy. Another problem here is um nowadays we we can see that a lot of um big or large organizations um doing like all these vision language action models like there's foundation models and fancy um and so on right so all of that is just come come down to um training a very large model um that have a very large language model and large vision model um and then encode that um with transformers and then map it directly to the action tokens. Right? So this is basically um a schematic of um model behavior cloning and imitation learning. Right? So in this case um expert demonstrations are treated as identically and um independent um distributed right um and every single interactions is um is put into like a single data set and then it's trained right and then it's going to be evaluate in the real world or you know in simulations. Um so this is more like a supervised learning setting where um they'll they could um yield a lot of problems. Um the classic problems here is mentioned by also um Sergey Levine um and a lot of other uh popular uh people uh is cascading problem right so cascading errors actually accumulates over time in actual trajectories um because of like the domain data on the domains um that's the expert demonstrations um is trained on uh is used to train on the robots it's different from the actual real world domain, right? So it's kind of similar to a data set distribution shift or domain shift. Um and therefore the robots actually deviates from the demonstrated examples. Um and then this leads to policy collapse um in uh real time or online executions. right. So in here we are trying to develop a uh very intuitive way of overcoming all these problems. Right? So uh behavior cloning and imitation learning um they actually can learn to match the expert performance at best. Right? And the expert policy is potentially suboptimal. Um as you can see in the figure on the um we can see that like behavior cloning in red um for example um it's going to be um solving all these um all these it has like the reward um trend it's very low for example um this is like the suboptimal policy uh it's learning is learned uh through expert demonstrations for prior preference learning, right? Um the agent first learn to predict the expert trajectories as its prior preference and then um throughout several interactions or you know further interactions of the between the agents and the environments um the pre preference learning model can trying to dynamically produce more optimized goal trajectories or more optimized preference prime preference trajectories. And so the prior preference the predicted prior preference trajectories will actually um eventually be better than um the expert demonstrations. Um and then we can see that um the prior preference the dynamic prior preference is independent of potentially suboptimal So that is the intuition of the method. in our work um we also have a um generative world model. Um it is also very similar to what I've talked about in the debug inference background section. Um but in here is there's a little bit difference. We have uh another model. So in this case um the the similar thing is we still built our generative models on um the concept of a war model which is basically backs to um

David Ha. Um and yeah so we still have like um we have like encoder posterior that encodes the current observations and then we have like the transition model um that have like that predicts the next state um prior given the previous state and the previous actions. So this prior will be reserves the prior informations um prior to making the observations of the next stage of the next time stack. Um and at the same time we also have like the uh observation at OT right and then we encode it um to an approximate posterior. So at the same time we have like two kind of state um one is prior to making observation and one is after making um Um and then we also have the decoder reconstruction as well where we have the posterior and then we reconstruct it in order to predict the observation. Um and then we also have the complexity loss which is the K divergence between the posterior and the prior. We also have the accuracy loss which is the uh negative lock likelihood of the predicted observation. Um there's a different here right we have the uh RSS which is basically called recurrent states um yeah recurrent state space model right. So this actually um is very simple and this is introduced in the dreamer series from Dunar Huffner. Um so the RSSM is um here's the inside of the RSS. So we still have like the post encoder posterior encoder posterior in orange here and then we have an additional recurrent neural network. Normally it's a gated recurrent unit. It takes in um the recurrent states and then it takes in the um distributions of states um and then it's predicting um the prior of the next state itself and then it's also predicting the recurrent state HD um or the deterministic state. the prior the next prior um over stage um is often be called stochastic state right and this HD is also um serve as an input into um predicting um the next state posterior in addition to um the next observation OT right so now we have trained um our world generative world model um that is RSSM right we can now learn the prior preference model right so um learning the pri prior preference model um requires um I'd say uh requires the state of the world model itself so in this case we are trying to propose a uh dynamic contrastive learning system where CRPP is contrastive recurrent states prior preference model Um basically we're trying to keep the RSSM the same um or fixed um in the when we are trying to learn the CRSP or the prior preference model. Um so the point or the goal of learning the prior preference model is trying to have that model predicting the optimal trajectories, right? the optim the optimal um trajectories of the state um in the future um such that those state trajectories represent the goal or the preferred um state or however you want to call it. So in order to do that we have to collapse the state of the U prime preference model to the state of the RSSM whenever the trajectory is successful. And in contrast we have to pull away or push away from from the RSSM state when the trajectory is not successful. Right? So in order to do that we have to um introduce a term called row or prior preference rate. So we're literally saying um whenever um the trajectory is successful the row is positive right this positive term will collapse all these two states together right um and then whenever the row um or p preference rate is negative then these two state are pushed away from each other right so u when we keep the state of the RSM fixed or uh stop the gradients of the RSSM um then only the state that get pushed and pulled um is the state of the prior preference model down here. Right? So in the end right um the CRPP or the private preference model will learns to only predicting the optimal state of the RSSM without predicting the um the failing state right um so you can see that in here there's a little bit difference right so the RSSM is not conditioned uh or the RSSM is conditioned on the actions right but the CRPP or the prime preference model is not conditioned on the actions, right? This basically trying to say that give me a an observation, I will predict future trajectories um that is needed or future goal trajectories that is needed to be in or future optimal state that is needed to be that the agent needed to be in. So eventually we will use this goal trajectories to um guide the policy learning. Right? So now there's um here's an example of how we calculate the prior preference rate row. So basically um we are trying to say whatever the immediate step that leads to successfulness of the trajectories then that one will have the most strength um of collapsing two states to each other that the two states of the RSSM and the the state of the uh prime preference model right so we can say that whenever the trajectory is successful for example the car is in like in the flap right here is successful then that's um proper prior preference rate at that point is one and then we decate backward. Similarly here um the push button uh we also decide backward. Um on the other hand um when the agent is in an unsuccessful state or trajectories for example um then the prior preference rate will be minus one um near the unsuccessfulness states right and then we also decay it backward. So here's an example of positive row that is used um in um the successful trajectories. So in here when the positive throw is used these two RSSM state and the CRPP state is collapsed to each other right um and the CRPP uh has um been able to predict all the state of the RSS and that is successful right so here's an example or example representation of the CPP state which is trying to predict the successful And for example in here um there's negative row um when the row is negative or pence rate is negative these two states from the SSM and the CRP is push away from each other so that the CRPP will not predict any failing goal or failing states from the SSM right so there's no complexity objectives here no car divergence between the posterior of the CRPP and the prior of the CRPP So uh throughout time steps or future horizon um which is in the x-axxis right. So we see we have like two environments and benchmark here. First of all is the observation um which is the first uh line first row second is the second row is the prior preference rollout. So we plot it um in order for v visualization. So we can see that throughout all these observations um the car has to go to the left in order to build enough momentum to go toward the right hill in order to reach the flag otherwise it cannot right. So we can see for every single um observation in the mountain car environments [snorts] in the top row we can see that's the prior preference roll out um is trying to predict the goal that the car needs to be in right for example in here the first observation um the car is in the middle or the bottom of the hill um then the prior program is trying to predict the car is trying to going to the left in order to build momentum And then until like the frame number five and six and seven um the prior preference model is trying to predict the goal of the cars fully moving toward the right. Right. And in the next benchmark um and environments it is also happening the same case even though there's a little bit of noise here but we can see that the prior preference rollout wants the agents um to push the button right so it's trying to predict the preference roll out. So we are halfway there. Um now we need to uh integrate uh the language part right. So when we want to compete with like all these vision language action models um very popular for example like RT1 RT2 right um we also need to have like a form some form of like multiodal encoder it is because language instruction or dynamically provided goal um can be considered as a form of uh observation right so in this case we're going to use like pre-train large language which models as frozen feature extractors for example GPT lama T5 and so on right and then there's um vision encoders um basically it's trying to extract real-time observation um and we can just train um it as normal so generally I often saw people do um trainable text encoder as well um but uh generally it should yeah should not uh keep the same embeddings um and you know integrity of the language encoder. So um there's there's a lot to be experimented. So what I'm trying to say um and then finally right we have like the language embedding and the image embedding right and then potentially we can have like several other modalities as well but here's like I demonstrate the two case um most common case right with the language and the vision right then we concatenate them together as embedding like a 1D vector and then it's going to be further also known as uh modality fusion techniques mid fusion um and then this can be served as the observation embedding that can be fed into the posterior um network. So here right you can see that the multimodal encoder is producing all these um for example uh embeddings right this embeddings fit into the posterior network as normal um in our system that we have described right this is the RSSM um and then we can see that um the posterior trying to feed in the recurrent neural network and then producing the deterministic state at the same time the recurrent network also predicting the next state prior distribution um or the stoastic state, right? And then um the deterministic state um also affects the predictions of the next u posterior distribution as So we can see that um this is kind of like a multimodal world system where it can execute actions scanned by the language instruction. So now that we learned um the world model right um we also need to learn the private preference model as well right. So not notice that here is the CRPP without the language um input right now we need to expand um the CRPP modules with uh so that it can be guided by the language as well so that's the private preference model and predicts the uh optimal trajectories that is conditioned on the language right so in here is the actual way that we did it um the CSP is now can um be conditioned on the language age model. The language here is basically can be encoded using all those um models um large language models uh for examples GPT lama right um and then we do the same process as well as uh I described um in the previous slides. So now we need to learn um the policy as well right. So after we have like the generative world model, we have the prior preference model. Um now we need to find a way to like determine the robot's actions, right? So when when we are trying to there's a lot of popular um actions determin determinations models, right? So when we're trying to do that, we have to be very mindful of like like what kind of uh policy learning that is that yields the best performance as well as efficiency. So there's two competing methods. So first of all is cross entropy methods where um trying to say for every single step um we sampled um n Gaussian distributions of actions, right? So usually n is 1,00 and then we're trying to prune all these um distributions using like um the world model and the preference model to estimate the scores uh of all these possible trajectories that is yielded by all these 10,000 distribution of actions. Right? And then after that we um tried to take the top K um maybe um K is um 30 to 50 um normally 30 to 50 and we choose the best um K um performance um trajectories, right? Um and then we take all those actions um to make a new mean and dist mean uh and the standard deviations of the the action distributions and then we initialize um we sampled n Gaussian distributions again right so we trying to iterate that those like five times for example um normally it's two to five times um and optimize actions populations um so generally this process is very flow we have to do it for every single time step. Um that will yields the computational time complexity of like O of N by K. Basically N is like the populations and K is the iteration numbers, right? So imagine we have to forward the world model 5,000 times for every single step that we want that we want to take the actions, right? So it's very slow. So um other methods here um can be uh active critic. It is very classic in a reinforcement learning literature. Um and it is faster because we we only needs to make one single u forward pass in the world model um and then make one single forward pass in the policy model. So in order to learn this model right we trying to have this state dynamics model here and then it's produced the representation and the preference um states throughout all these trajectories roll out right and then we can use the representation and the preference um to compute the instrumental signal and the epistemic signal for example um the representation preference can be used in here is basically um for example instrumental signal um can be computed by the difference or the divergence between um the representations actual representations and the preferred states that the agents want to be in. Right? So if it's far away from each other then the instrumental signal is low. basically it's um the score of like rewards is is low and on the other hand right so if um the action that use the represent the representation that match our um preference or goal or predicted goal right then the score internal score should be like higher right so it's um and then for epistemic signal here um uh basically we're trying to to boost the explorations of the agents, right? We are trying to say that the agents want to take actions that potentially um reduce the uncertainty in the futures, right? So uncertainty can be um in several different ways we can understand it, right? So uncertainty in the agent's knowledge, right? Or un uncertainty in the agents um or uncertainty in the environment itself, right? So um to reduce the uncertainty we have to take action like appropriate actions that explores agent explores the environment enough right and there's also other um epistemic signal um that is investigated in um by car as well um so it is a beautiful literature so yeah we use all those score compute all those scores Um and then finally we accumulate them um to compute the advantage value and return value. Um this will serve as the label for the value models to predicts um the return right. And then finally um we learn the policy learning by you know back prop through the value model or value value network right um this is also called policy gradient. So yeah uh after like learning the world model the private preference model and then the um policy model right now we are able to actually perform online learning in simulations and the real world um there's several different benchmarks to it. So there's first of all the front cuta kitchen um which employs front cuta kitchen robot arm uh there's a lot of task eight task um for example the robots has to um open the microwave or turn the switch or move the kettle and so on right and then there's also meta world um where the saw robot arms features with um over 30 m manipulation tasks and then there's um robbo suit which is a very classic and popular benchmark for testing robots algorithms. Um so it it features so many robots but in here we actually use Panda robot arm with um a total of uh 20 plus manip manipulation tasks and in the future u we also may want to integrate um real world robot learning So here's our preliminary results. Um we have compared um our agents on the top left here. Um we have compared um our agents with a lot of other state-of-the-arts benchmarks or baselines um on different benchmarks, right? So we have like VM v3 um and deactive inference v variational policy gradients, right? And then um some other active inference baselines as well as well as model based enforcement learning baselines. Um in general we have our agents um actually perform um has the best performance um in most tasks um and it usually converge faster um and learn learn to solve the task faster as well as uh can solve complex tasks where other agents cannot. Um for more complex tasks, for example, um a uh where the tasks requires completing multiple subtasks. Um for examp and then it's going to have the language instruction here as well. For example, open the microwave and then move the kettle and then finally open the slider. Um you can see that the robots actually uh do according to the language instruction. yeah and then after that um we also test on um the robot suit as well. For example in this case um open the drawer, pick the cup, place the cup into the drawer and close the drawer. We can see that the robots actually have a little bit of struggles to put the cup into the dryer, but eventually it did it. Just just while you're on this slide, how does the model come to understand the semantics of what a cup is? Is that the vision model connecting it to the embedding or how does that how does it know what a cup is? Um so it doesn't know um by a concept per se. Um but I've say like all these um actions is encoded um based on the visual uh informations that is received. So it's basically the robots has received the informations of both views like from this view and from this view. Um, and it's kind of kind of encode like use the convolution neural networks to like encode the perception informations into some form of a state and then use the states to um to uh decide on actions using the policy network. Um, did I answer your question? >> Yes. So it doesn't necessarily have a like a knowledge graph world model, but it uses the embedding. So then if if the model could represent um >> a chair that would go from the visual input to the embedding space of chair that gets hooked up with the then how does it know how how to if you said put it to the left of there. How would it know that? >> Right? So in this case um when we have like expert demonstration data, right? Um it kind of have like the initial understanding of like of like which pixels um is on which pixel, right? So uh there's like different set of pixels inside the visual image that's kind of activates the representation that's can yield the um policy activations um and trying to solve um trying to like actual do um solving the task. But if you actually move the cup to the left a little bit, it will still struggle, right? Because um the distribution of pixels now is different, right? So it's I think it's going to be totally dependent on like the distribution of pixels as very um susceptible to like the um image quality and stuff. Um if it is like different from the image quality of the expert insertion that it then it will struggle. So to to answer your question um there is no knowledge graph or um I'd say causal model um that's the robot learn. So there is no causal model and no concept models. Um it's just the representation or like the embedding that is learned from the distribution of pixels um throughout the encoder CNN. Um and that by that we called um all those representation the approximate posterior and um the prior itself. Yeah. >> Yeah. That's very interesting how even without the formal causal model you could still in language say something like move the cup to a place that will look better. But then that implicitly entails a causal model of how actions would lead to something looking better or looking different. >> So there is a sort of way in which the causality is at least actionable even if it's not represented in that declarative way. And then yeah I would say there's also failure case as well right so um this is um because like there's like the you know pixel distributions kind of different um from trajectories to trajectories right so um for example with the same problems right here um and the distributions of like action of the robots is different as well then the robot will struggle to uh to interact with the environment and yields the optimal trajectories. So yeah it's because for example the robot like move the kettle um but now the kettle is in different position as the learned one right. So it's going to be um struggle to like having all those informations encoded into the policy and the policy making you know appropriate actions to solve the task. So yeah um that is one challenge. Um there's also other challenges as well. For example, um you can see that's like the um in the figure on the right here, right? Um we can see like this is an example of like the inability to predict um different blocks in the language table bench benchmark. So this uh the figure on the left is the true actual observation that the agents saw. But in here like the agents actually predicts the observations like has no blocks at all. Right? So you can see that like the subtle changes in pixels um cause the agents to struggle to predict especially in the envir environment stats you have like so many different blocks that is like in subto pixel right there small pixels inside the environments and then there's like the all the trained um trained interactions um and observations as the uh subtopixels varies right have all these blocks like varies in positions in size and in colors, right? So, this actually caused the agents to uh predict all these frames with disappeared block. Um and then you know there's also other several problems as well. For example, like decoder learning is computationally expensive and then research work has investigates um encoder only conference model. For example, in the very popular paper from Neripss um it's uh contrastive active inference is also that um paper that investigates and and coder free um active in right um and then um experimental design finally. Yeah. Right. It's it's also challenging um because you know experimental design for real world online robot learning something that's not usually possible because we have to design the reward function right we have to design the reset functions for example for every single episodes right we want to reset the state of the environment we cannot like use the human to like manually like reset the state per today because the world the real world is kind of like a you know single life process where the it's like one episode one trajectory is like long forever it doesn't reset right um and um for example in here for example pick and place the block the robots can play around with the block inside one episode and but when can it put the blocks push the blocks away um then technically the environment needs to be reset. But um there is kind of to build a system that trying to pick and place the block and put it back to the initial position so that the robot can learn. It's either um chicken and egg problem or um maybe needs we need like a larger system to be in very tedious to set up for example. Um and if we can use like human to like manually pick and place the block um it will cost um a lot of um effort and time and money for data collections and stuff. So that is also very challenging to design a real world online robot learning experiments. So that um concludes my uh presentation um about my our topic. Thank you very much for having me. Um if you have any question feel free to feel free to mention. Thank you. >> Awesome. Very comprehensive presentation. Um I'll just see if anyone in the live chat wrote a question. Um meanwhile though what stage are you at in your PhD and what what are you looking to kind of get towards with this phase? >> Um yeah so I'm currently a fourth year PhD now. Um and I am very active in u developing different robotic learning algorithms using biologically inspired um algorithms um and active inference. Right. So I focus on building all these um generative world model um and making use of all these um concepts that active inference literature proposed um yeah and all these ma mathematical concepts well in active inference. So I think it's very interesting to um further extend this to like real world robotics humanoid. Um it's very interesting to like um say a humanoid um can you know automatically or autonomously explore the environments um in order to do safe and rescue um operations and stuff for example, right? or it can do like autonomous explorations of like deep sea mission for example or we can send like humanoid robot on like Mars for example. It's um going to be a long way but um it's inspiring me so much about that. Um >> Cool. Okay. from the live chat. Jason Fox wrote, "What number of expert demonstrations did each task require to train a good prior preference model?" >> So in previous research of R, um we used five trajectories. Um generally in you know RT1 RT2 they use like several thousands um in order to do like a single task. Um but um in in our task we use five trajectories. Yeah. >> Yeah. That was an interesting part with the behavior cloning but then there was a second plateau of performance with the secondary prior shaping. So what is the what is the lift or the improvement from the expert trajectory? What is that secondary improvement? And then do you think that that could be replayed back to a human expert to kind of update their improvement or is it learning a technique that doesn't really for for some reason it's it's more it's more optimal but not for a human or something. So, so yeah, um when when we say it's um it's going to be uh learning the suboptimal policies and then be better um than the sub-optimal policies. Um we are trying to like we're trying to proof that um the like our algorithms um kind of use less um expert demonstrations than like normal behavior cloning. um and um and eventually right the prior preference model um predicts the expert demonstrations um accurately right and then eventually through the process of self-imitation learning right so when when we're trying to um collect all these interactions from um between the agents and the environments um eventually the agents als also explores the environments right um when the agents is currently having like suboptimal policies equal to the expert administrations it also explore environment um using the epistemic terms right and then it can trying to it can try to explore the more optimal policy and then now when we have like further interactions um between the agents and the environments that is more optimal than the suboptimal policy. Um we save that into the memories, right? And then the prior preference model will um look at those memories and then see that oh this one is a better interactions. This one is a more optimal than than the um suboptimal uh trajectories from the experts. then the prior preference model will try to learn that and try to predict the more optimal one instead of now predicting the suboptimal uh explor trajectories. Right? So with that when the prime model um predicting the more optimal trajectories now it can guide and influence the the policy learning right. So that is the reason why um the agents can outperform um the experts uh in case of um in case here we leverage like self-imutation learning and um prior preference learning. Um in case um to answer your question about can we go back and improve the expert um I would say yes um that would there's several different ways to achieve this right so um we can treat the prior preference model as another expert for example um and then we have the expert model right we have the expert model and then now the role has changed, right? The uh the main agents, the active inference agents now is an expert and then it's trying to interact inside the environment and then produce all these sort of interactions between the agent and the environments and then um all these interactions saved into the memories and then the memories used to trains the experts um using normal supervised learning. Um so that would increase the amount of um increase the the uh optimality of the expert itself. So yeah. So did I answer the questions? >> Yes. Cool. Anything else you want to mention or point towards? >> I think um for now it's good. Yeah. >> Yeah. That was an awesome presentation. Really cool to see how going from that basic state estimation. This project is really taking it all the way through the vision, language, action models and um cool to see. So thank you again Viet for joining. See you next Okay, next I will play a few pre-recorded videos. We're over with the live presentations for day two and those will return tomorrow. I'm going to play several pre-recorded videos and we'll see how far we get on this stream. First video is going to be a 57 minute by JF Cotcher, research fellow at the institute. So here it goes. My name is Jean Franis Kutier and I'm a research fellow at the active inference institute where I carry out a research project in artificial agency. The title of my talk is the phenomenal-logical robot. So my research project asks the question, what does it take at a minimum for an autonomous robot to learn to survive in a world it knows initially nothing about? So in the hope of finding answers, I'm programming a mortal robot to have felt experiences to make sense of them and to grow the agency it needs to survive. The robot in question is a rover made of uh Lego EV3 parts that is equipped with a variety of sensors and motors. So what this project is not first of all well this is not an input output simulation of human level cognition which is what LLMs do. It is also not meant to be a general model of cognition. It is at best an attempt at implementing one form of basil cognition for a simple robot. This is not a robot solving free energy principle equations in the same way that bacteria do not do math. And this is not a robot as an extension of its programmer. Very importantly, the robot's experiences are not going to be grounded by its observer. It has to make its own sense of its own reality. So what is this? What is this project? Well, I want to achieve basil cognition, not simulate the more complex human cognition. I am building an artificial agent not proposing a general model of cognition. Though this project is principled free energy principle is applied here as an analytical framework not in implementation. It will be used but a And finally experiences are to be ext intrinsically meaningful to the robot and not necessarily to me its programmer. So this project is a learning by building exploration which is at in the intersection of phenomenology, active inference and agency. Phenomenology seeks to determine the essential properties and structures of experience itself and to provide an objective account of subjective experience. to what is experience and not is this experience correct in terms of some objective reality. Active inference. Well, according to active inference, an enduring self actively infers the causes of its sensory input and shapes its niche to minimize prediction errors. and agency. Well, agency is the ability of a self to act purposefully to achieve its own goals in an uncertain world it experiences. So my project is to program a robot so it learns to survive by experiencing, engaging, and making sense of its world. And the robot is to do so all on its own unassisted by its programmer or some observer. Okay. So first let's look at the principles guiding this project. Well to have agency a self must be mortal. We take that position. I take that position. Agency implies a self that is distinct from its environment. I think that's pretty straightforward. Though the self is not is not disconnected from its environment. So distinct but connected. A self is not a thing but processes that are organized to endure. In other words, a self is a dynamic organization of processes that constrain and sustain one another. The organization is not fixed but can adapt to avoid dissipation. What matters is the lack of discontinuity. A self's overarching goal is to survive and survive in a dissipative environment. It depends on the processes that produce the self must adapt their organization to preserve continuity of being. The self strives to remain self, but not necessarily to stay the same. Mortality is critical because it introduces value. What helps the self persist is good for it. What threatens to terminate it is bad for it. Value is subjective. Uh here we have we see a small bacterium that's breaching a larger one to feed on its insides. Well, clearly what's good for one is bad for the other. So for our purposes, let's define an agent as a mortal self choosing how it survives. So to survive, an agent must make enough sense of itself in its environment to predict well enough the consequences of its actions or inaction. To endure a self must have felt experiences. That's very important. So what are experiences and where do they come from? First of all, well the agent's environment impinges impinges on its senses and so do the agents own actions. The agent must constantly pro process streams of multimodal sensations. The agent integrates these sensory streams in spacetime in order to create its experiences. Actively create its So for example, when I bite into an apple, I experience its taste, crunchiness, color, weight at once in unified moments of It's worth noting that an agent need not be aware it is having experiences in order to have them. But that's the word in it's a different topics. It's a I think if we go deeper into this we start entering the the domain of of consciousness. Let's not do that. Not yet. So the agent keeps track of risks to survival by constantly measuring its well-being. It is mortal and measures of well-being indicate how its survival is at risk. Let's uh posit that the well-being of an agent is a function of three things. Its fullness, how energetic it is, its integrity, how healthy it is. And its engagement, how involved it is. And let's assume that the functions calculating well-being are predefined and unchanging. They are given to the uh agent um at birth so to speak. So the agent will correlate its experiences with fluctuations in its well-being, detected fluctuations in its well-being. And this is how it will have felt experiences. An agent can thus have good and bad So sensemaking, an agent must make sense of its world if it is to survive in it. It cannot just act blindly. An agent makes sense of its experiences by modeling what causes its sensation. It builds a causal theory. The causal theory explains what causes the sensations. The causal theory is used to make predictions. It is a generative model. It generates predicts incoming Now, actual sensations that contradict the predicted sensations become prediction errors. This happens when the causal theory is inaccurate or incomplete or when the world has changed since the causal theory was formulated. So this is this is basic predictive coding predictive processing uh principles. So what an agent senses will be the uncontested predictions it makes plus the prediction errors it's it receives. So the the the the perceived sensations are a combination of predictions and What the agent experiences then is limited by to what it can predict. But what it can predict changes over time. Let's not forget that the causal theory uh that generates these predictions is uh going to be changing and evolving So we remember that an experience is the integration of streams of sensation and so an experience will be surprising when it incorporates sensations that are prediction errors. So the more prediction errors are integrated into an experience, the more surprising the experience will be. So at first an agent will hold a trivial theory of causal theory, one that simply predicts that nothing changes. So it predicts unchanging sensations. But clearly expecting a world that never changes leads quickly to many prediction errors. The agent updates its causal theory when too many experiences are surprising. As we've discussed, the agent uses its causal theory to predict incoming sensations, but also to predict what actions favor good experiences over bad one. So the causal theory is involved in planning. So not all experiences are worthy of being acted on. So there we need to look at the concepts of attention and goals. The agent pays attention to the experiences that matter most to its survival. Those it feels most intensely. That will be primarily the ones that feel worse, but also the ones that feel best. The agent will assign itself the goals of impacting experiences that it pays attention to. Experiences it doesn't pay attention to will be neglected. The confidence an agent has in individual experiences will modulate how it responds to them. So an agent seeks to gain confidence in its experiences before it attempts to persist good ones or attempts to terminate bad ones. and agitous confidence in experiences that are not surprising which is experiences that are not made up of many uh prediction errors or if they are surprising it will have confidence in in experiences that are enduring. So even if it is surprising the longer an experience persists the more confident uh the the agent will be about it. An agent will prioritize uh existential goals. It will prioritize goals its goals so that it they they maximize so it it maximizes uh its well-being. But when survival is not at stake, the agent can choose to act just for the sake of new experiences. It will pursue exploration goals. We say that it bables. Okay. So having reviewed at at a very very very top level the principles guiding this project, let's now look at its implementation. Uh implementing a robot that has agency does present significant challenges. Well, first somehow the robot must be made mortal. So it has good versus bad experiences somehow. Well, the robot must experience its world. It must make sense of its experiences and act to promote good one and do that entirely on its own. Which means that the the robot's experiences, its sense making and its goals must be entirely its own, not the programmers. Now what role does active inference play in uh the implementation of this project? Well, we have to to to uh recognize that active inference does not tell us how to program a mortal agent, especially one as we'll see that has an expanded That's fine because that's not its purpose uh or intent. What active inference does is formalize what is required informationally for itself to persist in a dissipative world. So active inference is provides us the requirements on the implementation of our of our uh autonomous agent. If I am to be successful, the observable states of the robot will conform a posterior to the requirements set forth by the free energy principle. Now let's talk about building a robot's mind. So how am I programming a mortal experiential and agentic robot? Well, my implementation draws inspiration mainly from cognitive biology, biosemiotics and philosophy. I find inspiration in concepts from cognitive biology such as the organism being mutually sustaining constraining processes uh concept of multiscale competency uh stress as cognitive glue problem solving in morphospace and uh cognitive light cones. Those of you who follow the work of Michael Lean will recognize these as uh some of the key uh concepts underlying his his uh his work and the work of his lab. I also take inspiration quite a lot from um with concepts such as aboutness signal processes and sense making and my two principal sources of in of inspiration are Dr. Terrence Deacon and uh Dr. Jesper Hoffmier. I also lean pretty heavy on uh ideas that uh are coming out from from philosophy. the application of K can't K can't K can't K can't K can't K can't K can't K can't K can't K can't K's uh metaphys physics its synthetic unity of our perception plays a central role in my implementation and we'll go into a little bit more details um and more specifically I I take very very uh keen inspiration from the work of Richard Evans and his team in the development of the app perception engine again we'll talk about this a little bit later for you programmers watching watching this. Uh the software is built using um the elixir and prologue programming languages and a an extension to prologue called uh uh constraint handling rules CHR. So elixir is a processoriented and functional programming language. Prologue is a logic programming language and CHR is a constraintsolving extension to prologue. These are my programming tools. Uh, the hardware used in this project is a combination of the robot's devices plus a high-end server. On the robot, we find um a set of Lego AV3 sensors and motors. um a Raspberry Pi 3 for for uh computing aspects and a Dexter Industries uh brick pie board that interfaces the EV3 sensors and motors with the Raspberry Pi. So, they can be controlled uh through the Raspberry Pi. And the Raspberry Pi, of course, is is connected through Wi-Fi uh to the the network, making all of these devices remotely On the back end, I have a 128 core server that uh I actually do need to do the compute inensive uh processing of various cognitive uh functions. Well, that's a that's a necessary uh piece of hardware because the the the computing power of the Raspberry Pi is is far too far insufficient for what I'm trying to do. Okay, let's talk about mortal code. Can mortality be programmed? Can a robot be made mortal? Well, obviously the robot's physical structure is immortal. However, the processes that animate it can come in and out of existence. And if these processes implement, let's call it the mind of the robot, well, the robot's mind can be So as it as it operates, as it explores its world, the robot will grow a mortal mind. This mortal mind is what animates its immortal body so that it can survivably explore and exploit the world it's in. the ro the goal is for the robot to learn to maintain sufficient well-being or else its mind uh will die. I uh built a virtual grid world to provide the a virtual robot with an environment that can both nourish, provide food and damage the robot through obstacles and and and poison patches. The reason why I need a virtual world and virtual robots is just to accelerate development. If if I were to do all the testing debugging in uh with uh physical hardware and in the physical world, this would be uh largely unfeasible. But however, at some point, a a physical environment will be built that recreates the conditions of this virtual world, but add all the unpredictability, messiness, and chaos of an analog world, which we can't do without. to really fully test this implementation. All right. So, we said that the uh the robot will grow its mind as it explore exploit its world uh in order to capture what it learns. So, let's talk about growing the robot's mind more specifically. So, what are the architecture and dynamics of this robot's mind? Well, first of all, I truly believe that all intelligence is collective intelligence, which means that the robot is not a monolithic agent. Its mind contains multitudes. So, the robot's mind is implemented as a dynamic collective of cognition actors. So you you should think of a cognition actor as a process that's part of a community of mind processes. It is a rich interaction between cognition actors that instantiates uh the agents society of mind its collective intelligence. Each cognition actor is implemented as a process that makes observations which is a combination of predictions and prediction errors. Unifies these observations into experiences, makes sense of these experiences, assigns value to them. Are they good or bad experiences? And then acts to have better experiences. Each cognition actor in a collective of cognition actors has its own oovel belvelt uh you can you can think of it as the self-centered world of of of of a uh of a cognitive process in this case the cognition actors. So each cognition actor has its own belvelt that it observes makes sense of and acts on. A function actor zoom belvelt is made up of other cognition actors except for the more low-level cognition actors who connect directly to sensors andectors. So this collective of connection cognition actors that grows during the the life of the robot is organized by their own belts and organized into abstraction layers. The bottom layer is populated apriori so before the robot is turned on by static cognition actors. These static cognition actors interface directly with the robot sensors. I call them sensor cognition actors. And with motors, I call themector cognition actors. Their onum belts is the robot's immediate immediate external world. It's the distances perceived. It's a distance perceived, the the the ambient light perceived. It's the the contact perceived. So, but the static function actors connect a robot's minds with its body's sensors and motors. So, that's the bottom layer. All other layers come in and out of existence during the lifetime of the robot's mind. They capture, as I said, what the robot learns from striving to survive. So if the robot's mind is a collective, how does the collective learn to survive? Well, clearly the apriori bottom layer of composed of the static cognition actors is not enough by itself for the robot to find nourishment and avoid hazards. Additional layers providing abstractions are needed to create the big picture experiences and affordances that are needed to for competency and survival. Pure low-level perception and action are simply not enough. So, as we said, the first row is prepopulated by the static cognition actors, but all of the layers are populated by what I call dynamic cognition actors. As the robot experiences its world and learns how to survive in it, we have to imagine that the society of mind grows, shrinks, reshapes itself by adding and removing dynamic CA cognition actors. And and for example, a dynamic function actors with insufficient well-being can trigger the birth with sufficient well-being, I'm sorry, can trigger the birth of another to feel a gap in coverage. So obviously we have a bunch of static cognition actors um that are not in the umbelt of of a higher level one.

So coverage is incomplete. And so we fill in this layer and a layer must fill up before a higher level is started. There there's a a logic [snorts] to the growth of uh this um hierarchy of cognition actors. So as dynamic cognition actors are added into new layers, well clearly they form an abstraction hierarchy. But it's not only growth. A dynamic cognition actor will remove itself if it persistently serves no purpose. If if a a dynamic cognition actor um does not participate in another belvelt or if if it um if it's never called upon to to to share its experiences or to execute its affordances. If it's useless, it will eventually remove itself and this will this will allow maybe new and different uh cognition actors to take its place and maybe be more successful. So this adding and removing of cognition actors and layers of cognition actors in fact explores the space of possible organizations of the mind. So let's say that this particular collective illustrated here is struggling to maintain the robot robots's uh overall well-being. Well-being is going down. This is not working. We're where death is impending. So we are searching for a competent collective but this is not it. So the top cognition actor might remove itself and it can because no other cognition actor depends on it. This allows for other un under underperforming coction actors to also remove themselves and so on reducing the energy costs of the entire collective and thus the loss of fullness by the collective. So we've essentially staunch stop the bleeding of of of of fullness of well-being and we're now in a position to grow again. So the collective having now having a lower energy draw can afford to regrow itself and this time differently. This is there's a lot of randomness in in in how things grow. So the odds are that as it regrows itself it will be different from the collective that uh existed before and was found to be insufficient. So the robot's mind searches for an organization of cognition actors that keeps it nourished and safe. The robot's mind is doing problem solving in morphospace. So what about experiencing? How's that implemented? Well, a cognition actor assembles unified experiences from its observations. uh sensor cognition actors. So that these would be the static cognition actors in the bottom layers bottom layer have very simple experiences from observing distance, color, luminance, touch etc. Their observations of sensor readings are their experiences. So while a static function actor observes and experiences the readings of its associated sensor, a dynamic function actor observes the experiences of its umbelt function actor. So when a a dynamic function actor is born, it is born with an oomvelt a subset of the function actors from the layer below and that's a static uh omvelt. And this is what the uh conction actor will be observing. It will be observing the experiences of its uh its omvelt. But in order to observe it, it needs to predict. So the uh cognition actor in an envelate its experiential domain to the parents cognition actors so they they so that these can make meaningful predictions. An experiential domain describes the kinds of experiences a cognition actor has. So if I'm a K shhatter, I'm part of an envel. My first task is to communicate to my parent, these are the kinds of experiences I have. So you can make predictions about the specific experiences I'm having. And that's what the parent function actor will do. It will observe its uh its velvet, the function actors in its velvet by predicting their experiences and then receiving or not prediction errors and the observations of the parent cognition actor are the summation of predictions and prediction errors but predictions of experiences of its envel. So there's a there's a a a a sense where um the experiences of your children become your observation. But the parent cognition actor then unifies its observations into its own experiences. So multiple observations of its child function actor experiences will be combined into single um experiences. So multiple an observation multiple observations of child experiences become a parents So for example having observed in its own belt that uh distance keep shrinking a dynamic function actor would experience the more abstract uh concept of approaching. So the experience of a cognition actor becomes the raw material that are [snorts] unified into the experiences of more abstract cognition actors and so forth all the way up from static cognition actors. So clearly higher up dynamic function actors have more abstract experiences than the lower down function actors. For example, the the the experience of I stopped approaching is more ab abstract than the experience of approaching which is more abstract than the experience of distance. It's worth noting that experiences are constructed and named by the dynamic cognition actors and they're not constructed nor named by the programmer. In fact, they need not make sense to me as an external So, we talked about felt experiences. Where does the robot get its feelings from? the feelings it attaches to experiences so that the experience are either good or bad or indifferent. Well, everything a cognition actor does increases or decreases its feeling of well-being. So well-being is determined as we saw by fullness, integrity, and engagement measures. any internal or external work by a cognition actor reduces fullness in predefined ways. So for example, spending spending time over a food patch um will be captured by a cognition actor as increasing fullness or hitting an obstacle will uh be perceived by a cognition actor and it will reduce integrity. uh part uh cognition actor participating in a nonvelt will increase its engagement. So throughout its life each cog actor comprising the collective that makes up the mind of the uh the robot will have fluctuations in its own well-being. And this fluctuating well-being determines whether uh a co a cognition actor's concurrent experiences feel good or bad. It might be good to note that well-being measures are not entirely separate. For example, activating motors is work that causes loss of fullness. It's an energy cost, but also a loss of integrity. We're going to be simulating wear and tear while also increasing engagement. We are moving into our environment. So we are engaging. We lost integrity is regained over time. Think of healing for example. But that drains fullness and on and on. Now if a a cognition actor senses [snorts] touch when senses a collision it will have a loss of its [snorts] integrity uh well-being measure. It will experience a loss but the it's important to note that the loss or gain in well-being of a an individual cognition actor will diffuse throughout the collective. We can think of losing well-being as stress. So losing well-being cause a stress and stress is contagious. uh stress experienced by a a a cognition actor in in in another in another's envelip will diffuse to the parent cognition actor which will then diffuse downward to the children and upward and eventually through the connections the umbel collection connections of cognition actors stress experienced by one will diffuse and be experienced by everyone one. And because a cognition actor cannot distinguish between a self-induced change to its well-being and a change caused by the diffusion of well-being from other cognition actors, well, one cognition actor's problem eventually through diffusion becomes every cognition actor's problem. when a cognition actor feels a a a a reduction in well-being, it feels stress and stress alters its behavior. A stressed cognition actor is more likely to try something different from what has worked in the past. Obviously um things need something different needs to be tried. But a comfortable cognition actor, one that does not feel stressed, is more likely to settle into habits. Why change what has worked before? Now, let's take a little bit of a dive into acting. So, a dynamic cognition actor, again, it's part of a collective of dynamic coction actors, acts for different reasons. One is to confirm its experiences then to terminate its worst one or and or persist uh the better ones. So to confirm, persist or terminate an experience, a dynamic cognition actor must first identify the observed belvelt experiences that it unified to its own experience that it now seeks to impact. is a unification of observed experiences by cognition actors in its own belt. So if I want to change as a parent cognition actors if I want to change an experience I have I have to understand its composition in terms of the observed experiences of myvelt cognition actors. Now the conction actor plans will plan how to alter these experiences so as to impact its own experiences as intended. the dynamic contractor. Each one has a causal theory and we'll see more in details how it acquires it that that explains the observations it makes in its own belt and then it will then thus it will use its disausal theory to anticipate what would happen if experiences in it belt were to change this way or that way. Would this achieve its goal of goal of persisting or terminating uh one of its wor best or worst experience? So knowing how its own experiences are made up of umbelt experiences having a causal theory that explains how the observe umbelt experiences change over time then it can formulate a plan to change these experiences in order to alter its own experiences. So question number one as we said which observed and val experiences need to change to impact my experience says a cognition actor and then cognition actor would say okay these are the changes that need to happen to the experiences in my own belt that I've observed and unified into my own experiences that I want to change and it will then formulate a plan as a set of directives and what's a directive it's simply a a a a desire to modify impact an experience in the in in its own belt. So it's it's basically a goal a plan is a set of goals that are sent to the umbvelt for the velv. So the the parent dyn dynamic cognition actor sends goals to its omvelt goals to uh persist or terminate experiences that are being had in the envelt exactly how to change these experiences. The valvel cognition actors will interpret these directives they receive and then make their own plans because their experiences are also made of observations of their own umbels. So this this this uh way of at a high level saying I want you to accomplish this and then that the lower level say okay fine and I'm going to decide on my own how to accomplish it and then ask the lower level to do the same. This implements a multiscale competency. Again, for each goal it takes on, a cognition actor interprets it as directives, plan directives that it then sends to its own belt for them to interpret and so on. So plan action is goes all the way down from the top to the bottom all the way down toector cognition actors where uh a directive might be simply to spin a wheel. Now, a dynamic function actor keeps track of directives that sent and the goal they are meant to achieve. Directives that appear to have achieved their goal are remembered as affordances for later reuse. Cognition actors at the very bottom have atomic preset affordances such as spin or reverse spin. Whereas the conction actors higher up, the dynamic function actors discover abstract affordances such as go forward or even go up the luminosity gradient. So as you go up the extraction layer, the affordances become more abstract, more high level. Affordances are discovered and named by the dynamic function actors, not by the programmer. That's that's that's important. In fact, then they don't even need to make sense to an ex- external observer such as myself, the programmer. Now, let's look at how cognition actors build the causal theories they use to predict incoming uh observations and plan their actions. So a cognition actor observes its envel and and the more abstract the cognition actor the longer its time frame. So we could say that a more abstract cognition actor has a bigger cognitive light cone if you wish and observation made within a time frame are considered synchronous. So even though they may be made in sequentially so within a time frame of say let's say uh half a second 2 seconds 5 seconds they are considered to have to be concurrent to to be happening at once in the real world. So a dynamic function actor constructs it causal theory to make sense of what it observes within each individual time frame. And and this causal theory this construction of a causal theory is a form of induction. That is the production of general rules from specific examples. Examples being the observations. A causal theory here is a set of in if then rules in the program. It's a set of if then rules that are constructed and live about three things, three kinds of things. What observations ought to occur at the same time in the same time frame? What observations cannot coexist in the same time frame? They're mutually exclusive. And what kind of observations what observations would follow from previous observations? For example, we we might have uh this this causal theory that says if the distance is getting shorter, then nothing is being touched. If if if you uh are observing that distance is getting shorter, you should also be observing that nothing is being touched and if you don't, you can infer that nothing is being touched. Another uh constraint, another rule, which is a constraint here that says the distance cannot be getting both smaller and larger at once. So these you cannot have these two observations coexisting. And a third rule which is more of a causality rule proper is if [snorts] both wheels spin forward then the distance will get smaller. So this rule predicts what happens in the next time frame. So these are examples of this is an example of a causal theory. Uh an interesting side note is a causal theory can posit hidden that is latent facts that are needed to make sense of actual observations. So um a causal theory is can be partly an act of imagination and this this positing of hidden facts is a form of abduction that is the inference of likely causes from effects given general rules for those who enjoy logic. So the the discovered rules that make up the causal theory are expressed in the implementation as a logic program. The logic program embodies a causal theory and it is constructed without human intervention. Learning is achieved entirely from experiencing. There's no prior instructions or or feeding of examples before the robot is turned on. And this this this type of learning happens repeatedly over the lifetime of a cognition actor and it happens for all cognition actors making up the uh the collective the mind. So the logic program serves as a generative model. You can you can feed it the current observations and it will produce what it the predictions of the next observations. Each dynamic function actor constructs and runs its own logic program. Though at first it's it's they run the same trivial program. So the whole thing boots bootstrap with every coction actor has a simple program that says it is as it is and nothing ever changes. And so the dynamic function actor will run its logic program to predict the next observations but also to do what if reasoning before acting. Okay. So when predictions made from a causal theory result in too many prediction errors that's because the causal theory is inadequate then the dynamic function actor will replace it with a better one if one can be found. So how are these logic programs that embody causal theories generated? Well by an app perception engine. The app perception engine is a a a computer process that inputs a conction actor's observations. It outputs a causal theory that explains them. So you feed in a history of observations, turn the crank, out comes a little program, logic program that if you uh feed it the latest observations will produce the likely next set of observation in the next time frame, the So when they need a causal theory and they will need it pretty quickly especially in the beginning dynamic function actors call upon the app perception engine as a as a service when they need to replace their current causal theories. Again they do so because they experience too many surprises when using their current causal theories or they fail consistently to predict the consequences of acting upon their own belts. Now finding a logic program from remembered observations, one that makes accurate predictions about future observations is a difficult search problem. It's a hard nut to crack because the space of all candidate logic programs where the the hay stack if you wish is ridiculously large. So to find a good enough program in what I call useful time while it's still useful to have that logic program the search needs to not look over the entire hstack. It needs to be biased so it productively ignores much of the search space. That's where the work of uh Emanuel Kant comes in is synthetic unity of app perception. It's a part of its methysics focuses the search on causal theories that are coherent and possess explanatory power. So Kant's theory says these are the these are the properties of causal theories that are worthy. They are coherent. They have explanatory power and you can ignore any uh causal theory that does not have these properties. So the uh the app perception engine is the is a program that applies Kant's synthetic unity of app perception to find the logic programs that encode coherent and useful causal theory. So the the it's this the original engine was built by Richard Evans and all at the Google Deep Mind and I simply quote unquote re-implemented and optimized it and it was a subject of my presentation last year and there's a um there's a paper on uh that you can read to find more of the details. Okay. So all in all, this project aims to express and connect all of these concepts into working code. And this code will animate a mortal and adaptive robot that is capable of agency. That is the task. So where's that code? Well, it's it's the the open source project uh that I'm working on that contains this code uh and and design notes is called uh the karma system in uh all the code all the notes are completely open sourced and uh here you have the URL of the GitHub repo which you can uh go see and peruse. There are a number of subreos under the karma system. There's uh one with a prologue actor framework. Um the actor framework is kind of the runtime uh platform for the cognition actors. There's a implementation of a practical app perception engine. There's um code that implements agency. So code that implements the actual cognition actors, a bunch of utilities, there's code to um uh implement the uh the um the logic on the body of the robot itself. So the the access to the sensors and and and and um motors. There's an implementation of a virtual simulation environment, Karma World. And there are other uh repos uh to come uh that will uh provide an experimental platform in to so I can run experiments collect information collect data and submit this data to to analysis. So these are the main uh subsystems of the karma system and they communicate uh over a network. So that's how uh the the um sensors and and and and motors can be accessed on the robot and uh and by uh the cognition actors running on the backend servers so that it forms a unified uh system. Bit of a progress report here. Well, a lot of things have been accomplished done. I have uh a detailed conceptual and implementation design that's documented in the karma system uh repo. Uh the interface of the robots sensor andectors is implemented and and working. I have a virtual grid world that's also functional. The app reception engine is is running and and and seems to be doing well. uh I have a runtime platform for cognition actors and um I have implemented the uh static cognition actors for s that that interface to with sensors andectors. So that's that's code that works tested. What's in progress right now is the implementation of the dynamic cognition actors and that's on in the karma agency uh repo. What's uh haven't been hasn't haven't started yet and still to do is a physical recreation of the grid world implementing the observer and analyst subsystems. So observer to collect the data from running the robot analyst to subject them to analysis more specifically um to validate that u the um the requirements uh from the free energy principle are uh are realized in the data that we uh captured from running the robots. Then it find and of course uh run doing the experimental runs and and collecting the data and doing the free energy principle data analysis that's to do. Now a parting thought now should I succeed at programming a robot that has felt experiences makes sense of its world and strives to survive. Could there be something it is like to be this robot? I for one don't know. But if you answer definitely not, then I do have follow-up All right. So, here's how I can be reached. Email. And uh I'm on the uh Simco Robotics channel in the uh active inference institutees discord and u you have here the uh the URL to the the the top of the GitHub repo. So I'd like to conclude by thanking the active inference institute for giving this project a supportive home. Thank you. Next up is going to be Harshel Shaw and Satyaki Maitra who will be presenting on their drone work. This is a 40inut pre-recorded video. Okay, here we go. Hi everyone and thank you for joining us. My name is Hershel Sha. >> My name is Shiki Maitra and today we'll be presenting our project on combining hierarchical active inference with affordance theory for scalable poly selection and autonomous drone navigation. We recently presented at the international workshop in active inference and a lot of people were interested in how we got started as we're sophomores in high school. Um we're from California and attend Mission Tenosia High School and we've been working in active inference for around a year now as they combine your interests. Uh after learning the fundamentals we went into autonomous draw navigation around April uh and we found that it had a lot of potential with active inference for many of the issues and challenges that we face. Uh for me uh my personal connection to active inference was mathematics. I was really interested in how statistics and probability was connected with neuroscience and I wanted to connect this theory uh of mathematical modeling with actual systems which again drew us to this project. Um, I outside of research, I also enjoy math competitions, uh, piano reading and video games, certain types of video games that like strategy games that require some sort of exploration or problem solving. And, uh, here's my info at the bottom. Hi, I'm Hershel. So, I originally came to active inference from the side of machine learning, deep learning, that whole space. Um, what really captivated me about active inference is that I think it's the path to developing truly intelligent AI, which is what I really want to do. And I think it's also really applicable to solving critical problems that machine learning in that space just really can't do. Um, I'm interested in the applications in many fields um, beyond just our autonomous navigation. I'm passionate about medicine, biology, neuroscience. Other than research, I love competitive debate. Um, reading. I love reading books all the time and playing tennis for my high school team. And I just love learning a lot of new things. I have a website where I also write medium articles and I put all of that there. I track my reading journey. So, we'll begin with motivation and background. And then we'll go into the four main corners of our project. Uh, then implementation, how we did it, and then how we evaluated how well the model did. and then just the last conclusions and whatnot. So first let's start with motivation and background. So the first challenge that motivated this project is defining action spaces. When you consider action spaces, you have to look at continuous spaces and discrete spaces. Each space has its own advantages with continuous spaces. You have to worry about combinational efficiency and suboptimal solutions. But defining discrete spaces that are super flexible is really tough, especially when you consider that you have infinitely possible discrete actions, but you need to define a space somehow. Our next challenge is abstractly representing the environment. So for planning, an agent needs to understand what its environment actually is. For humans, we can just say there's a tree or that's a computer, but an agent can't just do that. So how do we turn these raw sensor data that we can give an agent into meaningful ideas? There are some issues with deep learning like it doesn't have an internal world model uh and it learns correlations. And while deep learning scales by avoiding the structure, active inference actually builds up a structure from the ground up. And we want to combine deep learning scalability with active inference structured understanding for a good environment representation. The last challenge that motivated this project is considering computational efficiency in robotics. Offloading computation to GPUs or cloud servers is often not possible when you consider issues like latency and bandwidth limits. This requires us to consider edge processing or ondevice processing. Both deep learning and active inference are computationally in intensive for various reasons. And our goal with this project is to maintain the active inference framework's interpretability and adapt adaptability while maintaining real time inference and control on edge devices. To address these challenges, we combined four main concepts. First, hierarchive inference. Uh as from the diagram on the left, the brain works hierarchically with higher level prediction and planning and lower level perception modules. Uh each layer updates beliefs to reduce prediction error. Um hierarchical active inference models this top down bottom up like blue by minimizing uncertainty at each level to guide adaptive behavior. uh information from lower levels is compressed into higher levels and you still retain these key concepts which is the main thing which will address a lot of our challenges and it also allows in practical use computation can be distributed among multiple levels. The next concept is affordance theory. Affordance theory was originally described in the context of ecological psychology as abstract ideas offered by the environment that can restrict or possible actions. However, this is very adjacent and can be directly used with the actor inference framework to enhance policy selection while contextualizing the environment. We can take the concept of affordances which can replace traditional states with abstract ideas and reducing the need for compute with for inference and this serves as a complimentary role as behavior emerges from interactions between the agent and the environment which bridges the abstract cognition with real world context. Another approach we took is the idea of scalable policy selection. Uh to scale policies, you need to have some sort of structure that can allow you to uh maintain low computation while still scaling up the amount of policies you consider. Uh in our case, we use a latent suitability state in an affordance map. What that essentially means is we have an affordance called suitability which encodes uh numerical or categorical evaluations of some region or path that the drone is about to take. So as you can see in the diagram below um low suitability indicates high obstacle counts and it's not very desirable but high suitability means you have lower obstacle counts and you're closer to the target location and the agent wants to be there. Uh this contextualization allows you to scale up policy selection because you can filter out those low suitably actions. You don't need to consider them because they're just not suitable. Um and this reduced pol policy set can be efficiently evaluated through EF without the need for some advanced hardware or offsourcing compute. The last area we need to consider is autonomous joint navigation. So notoriously autonomous joint navigation has a lot of different issues that we need to consider. So just quickly going over some of them. The first is that sensor data is often noisy which makes perception and localization very tough challenges. The next is that environmental factors like low light glare or clutter can additionally reduce our sensing's reliability. After that, you have to consider that real-time adaptation to dynamic and uncertain environments is very difficult. If you have a drone autonomously navigating in an unknown environment, it needs to be able to move and think very fast to adapt to brand new situations and issues that it's never come across before. After that, we have to think about that the fact there's limited interpretability and robustness in existing navigation algorithms. We already talked about some of the issues with deep learning, but in the field of autonomous drone navigation, these issues are more and more important. Then high computational load restricts deployment on low power edge hardware. Because in autonomous drone navigation, we're navigating environments where issues like latency are amplified. Almost everything has to be on edge hardware. But this isn't always possible with the most frontier models. Um, finally, there's often difficulty infusing multimodal data from lighter, camera, and IMU consistently. We need to have all these different data sources to enable all the different elements of our navigation but to fuse them and provide them to the model is a challenge. So now let's go into the actual architectural structure. So we have a three- tiered architecture. We have high level inference and planning and these lower levels. All of this builds up and sets up the high level planning to be as efficient and streamlined as possible. We have our mid-level perception encoding which builds our world model. It builds it from observations and its suitability and then creates a world model and beliefs about the world model uh beliefs about the environment. Finally, we have low-level control which right now is very simple. It just executes movement and records change in position to maintain egocentric Let's go over these areas in a bit more in depth. So first is perception where we take environmental data and process it into a point cloud. So you can see that we have the air simulation and we collect data from the LAR and camera which are two primary methods of perception. We turn these into structured observations specifically target distance as in elevation and obstacle distance obstacle density and we use these to update our models state base with distance as inim with elevation and suitability here. Um then we take our affordances which calculate expected suitability scores for the region around us based on the same perception data we got from Microsoft airsim. Then we move to high level planning where we sample optimal policies from a filtered action space that's created based on our affordances and this high level planning works to minimize expected free energy with its pragmatic and epismic component. Um finally we do our final execution setting it to a very simple low-level module for physical execution in Microsoft air. So now let's talk about a factor graph of our architecture. Our architecture is currently very simple. So there's not much special about our uh factor graph. It's a typical message passing structure between T and T+1. Uh but there's a very key difference. Um this S4 or suitability actually feeds directly into the observation and it dynamically weights that preference to then move into EF. We'll definitely talk about this more in detail once we get to the planning calculations, but this is a very key difference and it highlights a key problem with our architecture because it directly feeds in. It's actually completely deterministic and there is no probability involved. It just okay, I look at my information, I see this amount of suitability. That's my observation. We need to make this probabilistic so that the drone doesn't have 100% trust in its sensors because sensors are noisy as you mentioned earlier. Um yeah, so that's definitely a feature improvement that we need to do. Now let's go more in depth onto the perception and road model area. So the first thing to consider is the multimodal multimodal data acquisition. As you mentioned earlier, our two primary methods are from the lighter and depth camera. So a ladder gives us distance data. The effective range we chose is from 0.1 to 50 m which is kind of arbitrary but we found this based on testing in Microsoft sim. This is the effective range that we need for our purposes. Then is a depth camera which is only from 0.1 to 20 m and this gives us very accurate depth data. Um this is an image you can see of what this some kind of sensor um sensor data would look like. Um for us what we decided to do was use a DB scan algorithm to cluster all of the fuse points we get and we have an epsilon of 1.0 mters which basically means that the max cluster size is 1.0 m. So in each area of 1.0 m you're going to cluster into one point and min samples too meaning for one cluster you need at least two points. Um this process then abstractly represents the raw no points into a discrete set of obstacles and this inherently performs dinoising. For example, if there was a random point somewhere in an individual point, then the DB scan algorithm would ignore that because it's below min samples too. Um then going more into this for pre-processing and fusion, we take our raw sensor data and filter it for some additional noise reduction and remove out of range points within the effective ranges we talked about. And there's an optional subsampling in case computational efficiency means that we can't handle large amounts of data. Um for fusion we again just take both LAR data and camera data and transform it into a single unified coordinate frame which creates one large point cloud which is more similar to like the image you see here. Then we do our data packaging and output for the technical part of it. So, so far everything's been done on Python. And we're calculating the more information about the target like angle, elevation, and distance. This is mostly supplied by information from the airsim functions, which is kind of like the replacement for the IMU since it's a simulation. Although obstacle data is also packaged here, we take the obstacle positions, which we got from DB scan, and the obstacle density, which is a very simple calculation that represents how dense the environment is. And this information is then finally packaged into a JSON file. And we use a ZMQ system to communicate between the Python and Julia areas of our project. So all of this is in Python. And ZMQ then sends it to Julia which is going to go over our planning module and it's going to process it. All right. So let's go into this actual suitability calculation. So our suitability looks like this. We have a weight. We have safety factor, another weight, and then density factor. The weights are just normalized to add to one. And then we have these weights here. These are hyperparameters. Uh there's a lot of room for improvement, which we'll talk about later. But here's how the safety factor looks. It's an exponential where essentially um when d the obstacle distance is very close to zero, uh you immediately know it's unsafe because there's an obstacle right in front of me or right next to me. Um, and our safety factor crashes as the it's an exponential decay function. Um, then as D approaches larger and larger numbers, it gets safer and safer and safer because um the exponential increases and then we know, okay, I'm not close to an obstacle. Let's just ignore and let's keep going. Uh, and then we also have this penalty which is like if there's an obstacle within 1.2 2 m. Uh we replace safety or we multiply safety by an exponential 1 /d which basically says okay there's this obstacle that's some amount of close to me. Let's quantify that and then go from there. So this is kind of a penalty for very unsafe environments. When there's a lot of obstacles it comes into play a lot. Then let's go to the density factor. It's based on optical density. Uh both of these are what considered observations 04 and 05 as we saw earlier. Um so we take the density factor which is very similarly structured. It's based on obstacle density and then there's a cutoff for both but for this one the cutoff is more noticeable. Um this is like a just a density threshold that if it gets below this it's come like it just the density factor becomes uh one immediately because we know it's very safe. Um and then steepness is just quantifies how sensitive it is to changes in density. So as density gets low um as you can probably guess the factor gets close to one so that okay I know it's quite safe and then as density gets very high the factor gets lower to lower suitability. There's also another density penalty for high densities. uh it's a bunch of stuff but essentially um it takes the maximum of a hyperparameter and then a calculation with optical density and checks okay how much should I scale the factor to penal or uh penalize this factor so that suitability begins to lower that eventually ends up with a normalized score from 0 to one which represents this overall navigability and safety now you can clearly see that there's a lot of rule-based elements that have require high parameter sweeps and that is a very very glaring problem with architecture that has a fairly arbitrary fix which we'll definitely talk about later. Um let's go over the overall planning process. So we start with waypoint generation. Once we have our internal suitability we're able to do this where we generate 25 to 75 candidate waypoints based on obstacle density. So again this is another example of the agent taking environmental context and using it to inform its planning. We do a dual strategy where the sampling 70% about it is exploring the nearby space and there's some target bias sampling an initial 30% that's pushing the agent more towards the goal. Um we have an adaptive radius which is where the action length is between 1 to 10 m which is again based on internal suitability. So we use a very simple calculation where the minimum is one and then you raise suitability to power of three and multiply by 10 minus one. So this basically means as suitability approaches zero then the action length would be approaching one and vice versa suitability is going really well towards one then our action length is going to be 10. There's an additional check that's performed after this to ensure that we have at least enough to reach the target. when you're in some area that's nearby the target. So you can see from this whole process we have flexible actions. If the agent's in a very open space then there's absolutely no need for be taking a small action like one will be taking something more larger like seven or eight and vice versa. If you're very dense space we can take actions of when and yeah so let's go more into how this process works. um when we're get the set of way points we calculate the expected suitability. This is a very similar calculation as Siki talked about with the intra suitability. It has a few additional modifications and penalties. Um so one of them is that there's more conservative parameters used. We pretty much use trigger cut offs like for distance and steepness which makes this a safer check. We then add some additional cluster and octant penalties. So we check for any dense obstacle clusters in any of the eight octants around the way points. And if any octants is dense which has like five more than five obstacles then its local density score is penalized. And if it and if it's extremely dense like even more and has more than 10 obstacles its local obstacle distance is also penalized. We also have a few other penalties like corridor penalty where if the path from the current position is checked for narrow gaps and if the path is too narrow then suitability is severely penalized. This means that we're not just considering where the agent's going to end up but the process of how the agent's going to get there. Um finally we have this target proximity bonus where again if the agent is close to target vapes moving more directly towards it receive an additional suitability boost. >> Yeah. And something to add is these local density score and local optical density are the same things as the factors that we talked about before. So we basically penalize or increase these for expected suitability based on these factors. Finally we filter. So after all the calculations the system discards all these way points with a score less than 0.8. Um and then you calculate transition for other states uh with a suitability of 0.95. Um and then we calculate EF. So we evaluate EF for this action set by encoding uh encoding paths uh with EF based on uh their location and all these factors. Uh so our equation looks like this. We take in the current state so distance, angle and suitability. Um and then the action which is a 3D vector and then the agent's current beliefs as probability distributions. So we have the goal seeking behavior, the uncertainty reduction term, and the VFE term, which essentially we'll talk about later, but essentially uses VF uh to penalize or increase EF. All right, so let's break down each part of EF and each term. So first the pragmatic term uh it's just a pragmatic weight and then preference score distance bonus and this magnitude term. So preference score is just your preferences for distance angle stability as a weighted average which we'll talk about the next slide. Um the distance bonus which is uh just your expected distance minus the current target distance and then a factor of two. Uh and finally action magnitude. So larger actions would incur greater like scaling of this because your pragmatic value should increase or rather be minimized as you get closer to the target. So basically in essence a higher pragmatic value would lead to more efficient cold actions while the adaptive weighing does ensure some caveats in very dense environments. Now let's go over the epistemic value. So just keep in mind in general in any active inference framework the epistemic value is in charge of driving the agent to explore and reduce uncertainty in its beliefs. So for example if the agents like for this state I have a very low certainty in it what action can incentivize and help me find more information about the state. That's what the epistmic values encoding. So let's go over specifically in our architecture what we have. So the equation is very simple. We have this epistemic weight which is by default 0.2. It's a hyperparameter and we do dynamically increase this in our system and high density areas which promotes more exploration and looking around for safe paths. Um we then multiply by an additional factor of 0.5. This is arbitrarily chosen by us. um after that being multiplied by the total untrippy or which is uncoding the uncertainty of the agent's beliefs. This is computed as the equation h of q is equal to h of the q of the location plus h of q of our angle plus h of q suitability. So in essence it's just calculating the total entropy of our states as per the active inference framework. Um this is a more detailed equation as for the belief distribution for how the calculation works where you have the summation and you take each part of the belief distribution and add this term epsilon to it and take the log of that it's very standard calculation and finally in this going back to the equation we multiply by the action magnitude. Um again if you remember our actions vary from length one to length 10 and this term this reflects that increased epistemic value for the larger actions. Um you might think this is a bit strange but logically the motivation is if an agent is moving further then it's able to uncover more information about the environment and thus it's has a greater epistemic value. Um just note because of the way things are set up with our architecture, we are multiplying this by a negative and our total EF ends up being negative. This is just how things are supposed to be. And overall once we get this epistemic value, we see that a higher epistemic value is going to incentivize the agent to reduce uncertainty about its environment, thus balancing exploration and glo goal pursuit. Our final part of BF is this BF penalty. This is a very non-traditional and not super necessary term but it is still there. So we decided to talk about it. Uh it's a small regularization term that links our planning to the inference because we don't have very explicit perceptual inference. Um we do this term. So lambda is just a small fixed hyperparameter uh which ensures the penalty is about 10% of EF. So it very doesn't really affect EF that much. Um and then VFE is just your variation for energy calculated during inference. Uh it measures how well current beliefs explain the observation essentially just our traditional EF uh VF and we have complexity or your entropy term and then you have belief accuracy. So just your expected uh probability of observation over states i.e i.e. your likelihood uh and then your prior belief just a simple belief updating system which is how BFU is struct uh also this all leads into control which is just our final steps. These are simple airum functions. They take an abund all these values. Most of these values are completely irrelevant. Uh but you have xyz and velocity. You move it move from waypoint to waypoint. Uh and then rotate to yaw. Just make sure sensors stay straight. So you rotate to a given yaw based on which direction you are facing when your action is moving. Well, that was a lot. But we finished covering our architecture and now it's time to look at the actual experiment we conducted in Microsoft airsim going over our testing results. So this is the way the experiment set up. Again our core simulator is Microsoft airsim which is built on Unreal Engine. It's a very useful thing. It's a highfidelity open source platform which is built for rapid and safe testing which was able to drive all the development in this project so far. It prod it provides very realistic drone dynamics and simulated sensor data from lighter and camera. So you can see we have this environment. It's very realistic where the lighter and camera sensors are able to pick up very detailed images different things. Uh this is a picture of our core environment. It's called airsim neighborhood and it's a complex suburban environment with houses trees and other obstacles. So this was the very balanced um environment to choose for a drone. It's not too sparse where the entire thing is opened, but it's not super dense where our simple architecture can't exactly just so it can at least have a chance. There's areas where there's open road. This which is where the drone can adapt and take longer actions. There's also areas where sometimes targets are located near houses or trees and the drone needs to take smaller actions. So this allows us to extensively test our architecture. We have verified targets that are made sure that it's at least possible and we pseudo randomly generated them in the range of 20 to 80 80 meters for the purpose of this experiment. Um we fully ran 97 episodes and 96 of them passed with very low collision amounts which we thought was pretty good. Um let's go more in depth into the data we collected and the results we got. Uh so uh for EF minimization as you can see over all the episodes uh the expected free energy reduces as the agent gets closer to target uh as intended. It shows the agent minimizing uncertainty as those both the pragmatic and epistemic values lower as it understands okay I know where I'm going. I know what I have to do. Um the next is exploration and exploitation balance. So we concluded that the agent demonstrates an effective balance of exploration and exploitation. Um let's look at the graphs to analyze this. So this is our main graph. It compares the pragmatic value and epistemic value. Red is pragmatic, blue is epistemic um in comparison to the distance. So the graph goes as distance to target decreases. And we have on the y-axis the log normalized uh component. We see that throughout the entire uh run that the epistemic value is fairly consistent and it's a pragmatic value that's mainly changing. So at the left side the agent is prioritizing more exploration because initially it has more uncertainty about the environment and this means that the do the dominating value is the epistemic value. Um at some point around here the pragmatic value starts to play a much greater role. So as the agent gets closer to the target, the pragmatic value starts taking over and the agent prioritizes exploitation which means it's acting in more goal oriented behavior. So it has less uncertaintity and it's working towards m towards reaching a target as fast as it can. These are some more detailed graphs that show individual elements of the pragmatic and epistemic component and provide more information on the exploration and exploitation balance. >> Yeah, if you're interested, a lot of this is up on our website that we mentioned earlier. Uh so for computational efficiency which was a big thing for one of our challenges we measured it uh for the time it took for each planning step as you can see EF specifically which was the main thing because VF wasn't super impactful here uh was very short only taking 6.1 uh 6.01 01 mills on average with RX infer. Um, and this demonstrated suitability based waypoint filtering that actually reduced the computational cost. The total processing took 117 milliseconds on average mostly due to airsome calls and some other miscellaneous processing. Uh, but all of this computation was run on a standard CPU. Uh, there was no offsourcing or anything. So, this is pretty promising for the practical implementation of actor inference, but there's definitely room for improvement. uh we we talked with a lot of the arcs and fur people. They said this can definitely be improved. So we hope to get that these computation down as we move to a real draw. There's just the general results. Yeah. And the last thing to look at was after this main experiment that was mainly used to evaluate our model, we did run it on a more challenging environment. Um the reason we're not looking at the main data from it is that the success rate was only 50%. um 17 after out of 34 episodes that we ran with the condition that there was a six-minute timeout per episode. So, there was a couple of issues for why the drone didn't have that great performance. But, as we're going to get to later, and as you've briefly mentioned throughout the entire presentation, this isn't due to fundamental issues with our architecture, but more so as to the implementation. And of course, our future work is going to address the main fundamental issues that do actually play an impact. So going more over them, we saw that the drone often got stuck and had a higher collision amounts in this environment due to two main things. The first is myopic planning. So the agent essentially lacks a global strategy. Um it's planning more reactively than we wanted to, which is over considering immediate safety and proximity to the goal, which leads to poor longer term positioning. So, for example, if a drone would get stuck in an area, it would be taking actions that maybe would help it a little bit, but don't strategically move it out of that bad area and out into the open. This also plays more into learning instability where the agent is unable to fully learn from the mistakes it made and it would abandon a mostly good strategy after a single failure. Yeah. So in this challenging environment called Jeangja in airsim it starts within all these trees which makes it immensely challenging but we have confidence that our architecture should be able to do this if these issues are circumvented. Uh an example of this myopic planning would be like it would get itself stuck in like one of these little buildings and it just would not know how to get out because it doesn't have that long-term uh vision to think okay I can go up and around. Instead, it just thinks I have to get to my goal. I have to avoid an obstacle. That's it. It doesn't have the ability to look forward. Thus, we look at some conclusions. Despite those issues, we did see that combining affordance theory and active hierarchal active inference does enable efficiency of context aware navigation as we illustrated with our chressing those challenges in these complex 3D environments. So, we were able to address all three of these challenges in different manners. First for the first challenge which was uh policy selection we were able to use affordances to create discrete highle planning with a flexible action space. We were also able to lower computation because we only have to consider certain paths because of the suitability filter. The second challenge which was contextualizing the environment uh we were able to use just avoidances which fundamentally contextualize the environment. uh and our suitability state was able to contextualize specifically the safety and navigability of the environment. Again, this also decreases computation for the obvious reasons that it just doesn't have to consider unsafe areas. It only considers certain parts. Um and challenge three was we were able to address computational efficiency and see that uh the model is efficient and capable of running on a standard CPU. And uh with our preliminary tests on our real hardware, we've seen that it's likely being able to run on edge device. In our case, a Raspberry Pi 4, but there are obviously more powerful devices that could run this even more efficiently. Now, let's talk about what we're going to do in the future because this is a very much incomplete project and there's a lot more work to do. So, we currently have a built drone. Uh we want to do it fully GNSS 109 and with the open source RGup pilot um firmware. So we have a Raspberry Pi which a little bit hard to see but it's in this big black box over here. [snorts] Uh we have a Liar A2M12 which is currently not attached. Uh we'll put a depth camera on it. Intel specifically Intel Real Sense D455. And this entire thing was built upon the Hawk Works F450 drum kit which came with a Pix 2.4.8 which is up here. It allows us to make very streamlined architecture, but there's a lot of issues with the GNSS denied uh and like frames and whatnot because obviously when you move to the real world GNSS without

GPS, there can be a lot of issues. Uh our goal is to test full architecture in the real world and collect sufficient data like we did with our airsim neighborhood environment and we hope to see some success with this and with active inference as a proof of concept. Um as you mentioned throughout the presentation there were some miscellaneous issues that we're working on addressing right now. So we talked a lot about how we have rule-based methods that are essentially like band-aids on issues we notice in the sim but the more fundamentally sound way to fix all of this is of course the Beijian framework Beijian methods um one example is that for the point cloud when we get the sensor data instead of everything being deterministic which then proliferate through our architecture with even suitability we want to have a probabilistic representation of a point cloud representing probabilities of how likely different areas are to have a point. This is going to inherently fix some of the issues and simplify a lot of our calculations. This ties more into our second point which is fine-tuning the suitability weighing in and calculations. We have a lot of fixed hyperparameters and unnecessary calculations. Proisting methods are going to reduce a lot of this but even the way we get some of the hyperparameters can definitely be improved on in more sophisticated methods. After that is action fine-tuning. So the flexible action approach that we took using civility to internally parameterize where our action is is already pretty good. But we think we can take a more sophisticated approach where we once we sample our first action, we can then make more precise adjustments to where the action is using secondary and um additional expected free energy steps which is then going to make our drone's movement more and more precise while remaining still in a fundamentally discrete action space. Um finally due to some issues with airsim calls we have this whole stop and move cycle rone takes an action stops to get its data which is unnecessarily long and then move on. We're planning on addition on on addressing this with things like a sensor cache and a couple of other codebased adjustments and hopefully eventually the drone will be able to move continuously where it's still playing discreetly but you don't you're not going to see the drone stopping in between which is obviously unfavorable in the real world. Yeah. And just one example on the probabilistic side. So for a probabilistic representative in a point cloud for example this would forgo the need of the corridor penalty for instance because if you have a probabilistic represented point cloud there's literally a cloud which would present pre prevent the drone from choosing an action that would be very close to the point cloud because it thinks okay there's some probability that I could be running to an obstacle here. Just an example. And the last area is this is directly addressing some of the challenges we talked about in our challenging environment learning and cognitive maps. So we want to be modeling our transitions with the approach of cognitive maps. This means that we're essentially looking at expected and current suitability of regions rather than paths. We have a more true application of affordance theory. And this is also going to enable us to learn the tendencies of the environment. So, for example, in the Jang Shi environment we talked about earlier in a very dense area, you'll know that the drone is going to be tending to get stuck there. And thus, in general, those kinds of areas are unfavorable. And the drone is going to have the strategy of wanting to get out of those areas entirely, not just move to slightly safer areas. And there's a lot more concepts from cognitive maps that can help address even some of the more miscellaneous issues of our architecture. And this is going to move us towards a more true hierarchal architecture where the drone and the agent is thinking in layers and really considering strategy with better environmental um contextualization. Thank you uh for listening to our presentation and if you want more information on us and the research as it evolves, please visit our website. It'll be consistently updated as the research progresses. right. Next up is Bradley Alysia. Open source sustainability progress and connections to active inference. This is a 53 minute recorded video. Here it goes. Hello, my name is Bradley Alisa and I'm going to talk to you today about our project open source sustainability. I'm going to talk about our progress over the past few years and connections to active inference. And this features a number of people's work uh including myself, Jesse Parent, Morgan Huff and Gox Scholars which will be mentioned in the talk. We're from the orthogonal research and education lab and we have a preprint on this work at researchgate. So what is open source sustainability? So open source sustainability is a way to understand how we can manage working teams and software infrastructure over long periods of time. There's been a history of trying to make software development better through different types of project management techniques. uh we've had things like agile and devops come around to improve that process. This is hel held true for both industry and for academia and we're trying to look at how to understand the key components of this longetivity. So we started our project in 2022 and we've taken a number of approaches and we've developed these over time. Uh we started in 2022 with an active inference agent-based modeling approach and we developed that um in 2022 and then we've been maintaining that since. In 2023 we worked with multi- aent reinforcement learning and a combination of reinforcement learning and agent-based modeling paradigm. In 2024, we worked with a large language model which we are currently maintaining. And then up till this year 24 and 25, we worked on multi-agent reinforcement learning and reinforcement learning more generally. So we have three basic approaches. One of which is active inference. And in the talk here, I'm going to highlight some a lot of work on reinforcement learning. a lot of work on active inference and some on large language models. So the Linux foundation has published this report called the open source sustainability ecosystem and so they define open source sustainability as the tools conditions and platforms under which open source can thrive. So they talk about tools which is licensing workflows and containerization. They also focus on conditions. So the conditions under which open- source sustainability thrives is where we have healthy teams with trust and coordination. And then finally they talk about platforms which are automation techniques that enable working teams. They also pose some big questions and so these are questions that people often encounter in practice and there are no clear answers to these questions. A lot of projects struggle with this and no one's ever really developed a good methodology to ensure that you do a good job because a lot of teams are very diverse. They have, you know, you have different types of projects. You have different aims and goals. So it's hard to really find a single set of answers. The first of these is how to push production builds in a distributed decentralized manner. So a lot of teams in open source are working in a distributed decentralized manner and pushing to production is notoriously hard. It's difficult to coordinate people's efforts. It's also hard to do put a lot of workflows into practice. And so this is something that depending on the team is easy or hard. It's relatively easy or hard in in it's relatively easy or extremely hard. The second one is foss governance and that is a a social contract uh that Debian kind of proposed uh about 30 years ago where they proposed a way to keep your organization relevant and grounded in the type of uh ethics that you want to you want to see in the world. So a lot of open source software projects will drift from where they began and they will decay in different ways and we want to have a governance structure that maintains projects over the long period of time. The third point is how to maintain collaborations and software systems. So we want to maintain collaborations, we want to have uh healthy relationships in our teams but we also want to be able to maintain software systems over long periods of time. And uh last year we had an interview with Stefan Bowman who talked about containerization as a strategy for that where you can build a container and run different types of programs in there distributed to different teams and it makes it easy for them to implement different types and in this case it was uh neuroiming analysis. Then the fourth question is how to coordinate efforts and get things done. Very simple. Now, we're going to be uh hitting on some of these big questions as we go along. But first, I want to talk about So, there are have been attempts at this in the past. There have been platforms that people have tried to build or, you know, different types of models that people have tried to build and it's it's kind of hard to do because as I said, uh open source projects are very heterogeneous. Some projects focus on, you know, they use one set of tools. Other projects use other sets of tools and they have different aims and goals with their software. Some are driven by large companies, some are driven by small concerns and others are driven by academic groups. And so they're going to have all different sorts of practices, all different sorts of goals. And their teams are going to be managed quite differently. In any case, uh the first implementation of this sort of way to measure open source sustainability was source cred and source cred analyze discourse, discord and GitHub activity. So they pick different tools that teams might use and they find a way to analyze that activity. So you're dealing with producing code, you're dealing with discussions amongst colleagues, and you're dealing with um other types of communications within say an open source community. And so this was proposed by Miazono on the protocol research labs blog in 2020. Another implementation is git compare where it focuses more directly on GitHub and only GitHub. So you can you know take all aspects of a project or all aspects of an open source community into account or you can just focus on something like GitHub and depending on how your measurements how accurate your measurements are, how robust your measurements are, you know, one strategy may be better than the other. So it might be better just to focus on GitHub if you don't really have a good methodology for analyzing discussions on discord or discourse. So, git compare builds a model of general development activity for GitHub repos based on five criteria. So, we're talking about GitHub repositories where a project is storing their code. They're uh implementing their developmental pipeline and they're you know working on building software and they want to basically know how popular this is with the open source community and they also want to know how recent things are. So the and also they want to know if you can protect your assets properly. And you know sometimes licenses aren't just about protecting assets. Sometimes they're about mediating conflict in the team. So they base their model on five criterion. There's a creation date, a license, forks, stars, and watchers. And so stars and watchers are sort of popularity mechanisms. Uh social social media type measurements. A fork is basically when I copy a project and I make a copy for myself so I can modify it which suggests that people find your project generally useful. The license of course is the legal framework or the governance framework and then creation date is going to be where it started. So you can look at the creation date and if there haven't been updates since then or you know maybe in the last two years you know that the project is not sustainable. Um so these are all things that they measure in git comparison is from the git compare medium in

2018. And then another approach that was interesting uh that's been done is a governance surface. So a lot of times we develop these surfaces to describe all the different states of a system that we can describe and then we explore that surface to find you know the transitions between states or sort of the trajectory of the project as a whole. And so this is a governance surface that was developed using an analysis of discord with chat bots. So they analyzed discord with chatbots. They developed this governance surface and they described kind of you know the discussion activity in that team or in that community and this was published by Renie and qualitative inquiry. So the final implementation I want to highlight here is a tool called GMO lab and it's put out by the chaos software community. And so this is a tool that takes a number of places where you can track data in open source communities such as GitHub, Slack, uh other places where teams meet up together and development happens. And so we can take these data, we can retrieve them, we can store them, we can do other types of things uh and enrich them with, you know, all sorts of other analytics and charts and reports. So Grimar Lab analyzes data in the service of being a management tool. This diagram here shows like kind of how we can take data from an open source community and we can look at how well it's performing and then we can look at those charts and graphs and improve our community places where it's failing. We can shore things up. We can optimize things. Now, our goal is a little bit different and in the sense that we sort of base everything on agent-based models and so we're not really interested in being a management tool. We're interested in scenario building. So, agent-based models are famous in the social sciences for enabling scenario building or asking what if questions. So you know we may not observe something directly in a community but we can model a community as a simulation and then say what if some intervention happens in the community? What if something good happens in the community? What if something bad happens in the community? What happens if you scale up the community? What does that look like? And so it gives us a sort of a scenario building tool. And so our goal is scenario building and insight not performance optimization. Now, one of the interesting things about say using tools like Git or GitHub and many other uh project management tools, but we'll focus on Git for now, is that they introduce on individuals a certain amount of cognitive load. So, cognitive load is the idea that you have too many things going on, maybe too many steps in your tasks that you need to perform or too many things to think about at one time. and it impacts your performance. It slows you down. It confuses you. It, you know, it doesn't lead to optimal outcomes. And so, this is a blog post on Git and cognitive load. And this is a cartoon that kind of talks about how we use tools sometimes that are very arcane and very hard to master. And so, if you can't master them, you can't really realize your goals. And so this is one of the main sticking points of sustainability is that people can't go from idea to implementation very easily. And so you know there are a lot of ways we can look at and measure cognitive load. One of those is looking at the complexity of programming environments. So we might say for example if we look at the complexity of programming environments we might want to measure how long m you know if we have long multiple long processes in our workflows that might introduce cognitive load. If we require people to learn a multitude of tools to get their work done that might also in induce cognitive load and then if we force people to memorize many similar or cryptic commands we can also introduce cognitive load. So this uh cartoon basically uh spells out the problem with cognitive load in a lot of open source projects. And unfortunately open source projects uh there isn't a focus on mitigating a lot of these things. These are very common things and problems in open source projects. So this is just one example of the things we can measure and the kinds of constructs we can build looking at open source sustainability. There are a lot of things that are involved in this. So, you know, if we think about like individuals in the team and they're all sort of having to process a lot of information to get their work done and they can do this in a very efficient way, they can do this in a very efficient way, they can do this in a very efficient way, they tend to stick with the project, they tend to be productive and the project is is a success. If they can't, then it's a failure. And so, this kind of we'll we'll talk will feed into our active inference formulations and some of our reinforcement learning formulations as well. There's always a learning curve in any organization. You have to learn new tools. You have to learn new procedures. And that learning curve, if your learning curve is too steep, a lot of people won't bother to join the project. Sometimes projectwide compliance is important. If you have to comply with different uh regulations, sometimes that's a good thing and sometimes that's ownorous things. So this can work both ways. There's of course attentional resource managements. Do you force people to multitask all the time? Or do you have a work environment where people are doing one thing at a time? Because multitasking can be bad from the standpoint of focusing on any one set of issues. Multitasking can limit your productivity. There also has to be intrinsic motivation and reinforcement. Of course, we're going to address this directly with reinforcement learning. There's infrastructural lock in. Often times that's indirectly cognitive, but it can affect your ability to get your work done. Expertise or mastery. So if you're an expert in a certain area or if you're an expert with a certain tool, it makes it more likely that you can achieve what you want to achieve in your project. And then the coordination of goals and the coordination of goals is I want to be able to coordinate my goals across the team. I don't want to just do things on my own and then someone else do things on their own and then combine those at the end. We want to coordinate our efforts in an active way. We want to be able to u you know have this sort of collective cognition where we can an environment where we can do that. So when we define our different um terms here in terms of what we think about how we think about open source projects. So to define our different terms about what we think of when we think about open source projects we've identified three things. Uh there's the project their agents and their tasks. And this diagram shows like a project leader, a maintainer kind of contributor. And those are different states of agent. So if we're thinking about this grid and we're thinking about this agent-based model, we have different uh classes of member of the team, we have a project leader, a contributor, and maintainer. And there are multiple types of agents. And those agents can coordinate their efforts or not. And so we're going to model it in an agent-based setting. about this you know relates to team members. So a project is defined as a set of tasks and agents with different competencies. So you have agents that are contributors, maintainers project leaders. You also have tasks that are specific to contributors specific to project leaders, specific to maintainers, specific to other types of people, maybe technical writers or something like that. When we think of agents, we think of individuals and those agents usually have some sort of cognitive state. They also have maybe a task or an ascribed role. And agents uh see this division of labor in a project and have specific roles in a project like contributors and maintainers. And then tasks are defined as units of work. Sometimes we have issues which are just kind of single tasks that are you know clearly defined very tightly defined to what we call in our simulations MS Jupiter shop projects which are these large multi-stage projects where you have to complete one part of the project before you know what the next part of the project will look like. So we have these tasks, these agents and these projects and now we're going to model them and measure them in some to some degree. We're going to create metrics and then but we're going to mainly focus on simulation to see what this looks like. So again, we're going to turn to agent-based models or agent-based uh environments where we have and of course in an agent-based model, you have this grid world and on the grid world, you have agents that can occupy individual cells and those cells can be occupied by agents that have different roles, different internal states, and they interact with each other through rules. And you know, if you have agents next to you, you're you know, you can you can share information with them and they create these structures across the grid that are these collectives. And so, you know, we're going to be looking at collective behavior across the grid. We're going to be looking at individual behavior on the grid. And we're going to see how that evolves over time. So, our agent is in a certain position. it'll produce a spatial a bunch of agents will produce a spatial distribution with specific states. So we have a red state, a blue state and a gray state here. And so that's great. Uh we can do that and we can run the simulation. But then when we get our simulation results, we see that there's some patterns in here and and you know when you look at patterns with your eyes, sometimes you see spurious patterns, sometimes you see patterns that are meaningful. So the problem we have is how do we interpret macrolevel emerging patterns. So that's going to be a problem that we're going to have and we'll we'll come back to that later. So you know we want to think about other kinds of processes to model not just uh agents that live on a grid world but those agents living on a grid world do something. So ultimately these agents are trying to solve different collective action problems. So you know there are different aspects to this uh and we but we don't want to just model collective action for the sake of modeling collective action. We want to map this to some process in open source. And so here we have uh branch development in git and we have pull requests. So we have different branches where different branches are created for different purposes by different agents and they're all kind of uh pushing their work to a main branch uh where they're you know they're kind of taking the work from the main branch creating their own branch finishing their work and then issuing a pull request back to the main branch. So this is collective action because all the agents have to coordinate their activities so that at the end the main branch has like a coherent project and it isn't just you know everything that's pushed smooshed together and it's not workable. So you know there are a bunch of things that drive collective action in this in this context. So the first is internal motivation. A contributor wants to work on an issue. So an if a contributor doesn't want to work on an issue, they won't bother making a branch and they certainly won't bother making a pull request. There also has to be cooperation. So sometimes agents can get together, make their own branches and use a divide and conquer strategy to sort of address the available tasks and address what needs to be done. They coordinate their efforts through looking at these diffs uh which are the differences between different branches and addressing those accordingly. And then that leads to this finished pro project. So you have this sort of technological layer that allows us to make these branches, compare these diffs and do this all in an automated fashion which helps with the cognitive load. But it's also important to have the sort of internal motivation and this cooperative aspect of the behavior. And then of course there's the external motivation which is sort of addressing a technical layer which is that we have to make contributing to this project seamless and meaningful. So if the project is not you know easy to contribute to and it isn't really about anything then people won't bother to do any of this. Another process we can model is the issue queue or the cananband board. So canband boards are used ubiquitously in project management. Issue cues are also used to store issues to move issues through um different types of different levels of priority from beginning to completion. And so there are a lot of things that uh drive this u at the individual level at the team level and then sort of at the issue creation level itself. So individual prioritization can be characterized as what issues contributors should address first, later and last. So when we build a conveying board, we usually create these categories of you know uh to-do, in progress, urgent and finished. And we have to decide as individuals where to put those issues. Where what's what's the state of our of state of completion for an issue and where do we put it? So, you know, maybe we don't want to work on certain issues at all. Maybe we find other issues to be very appealing or maybe we take all of our issues and we move very slowly through them. Maybe we move faster and but we have to have this means of sort of addressing issues. Sometimes we procrastinate on them and you know we have to have this model of moving on things so that things get completed. Then we have team prioritization. So in team prioritization we can ask what issues are at what stage in project development. So if we have a team reviewing a canvan board that team will look at all the individuals and their collective prioritization techniques and then they will just basically say okay this is what the state of the project is and I'm going to give feedback to the individuals to or we're going to give feedback to the individuals to say what they need to do to improve these and then issue creation is basically what is the fundamental unit of work. So all of our individuals must have be in agreement on the same roughly the same fundamental unit of work. How big are your issues? And the team also has to be in agreement or else you have a lot of conflict. You have a lot of fighting over you know what exactly needs to be done, how much work needs to be done and what's the priority. And the third process we can model are units of contribution over time. So this is a GitHub dependency graph for branches and forks. If you look at this, you'll see that they're different owners and these lines kind of sometimes they kind of join together, sometimes they branch apart, sometimes they're maintaining themselves in parallel. And these are all sort of a record of pull requests and commits within a project. And so we can look at different workflows or different sort of uh work processes where multiple people are contributing to different things over time. and actually producing an outcome. And so we we're interested in that because we're interested in knowing how maybe a certain thing like a Docker file gets passed through many different community members or or contributors, you know, it gets passed through many different contributors on its way to completion. So here we have a number of different on this workflow network cycle. We have all these nodes and we have a number of connections between the nodes. The con these four connections define this network cycle. So the first uh connection are generic changes that were contributed by two uh contributors and then merged by a third. The second is creating a docker file which was contributed by one uh team member and merged by another. there was another clone created at three and then four was a fork made by another uh contributor. So you have a number of people working on this, a number of contributors, you have a number of actions being taken and you can analyze those as sort of getting work done and making the project you know basically sustainable. Now when we we do this in terms of simulation, we have this problem of indirect measurements. That is we're measuring these things but we're not actually measuring things like cognitive state. We're not so we're not like putting people in a scanner to measure cognitive state. We're looking at their outputs on GitHub or their outputs on Discourse or something like that. And so we we can't say for sure or we don't really give them formal behavioral tests to to look at their say motivations. So we don't know exactly why they're doing these things. We just know that they're doing a lot of them. So we have this problem of indirect measurements skewing our simulation output. GitHub contributions are an indirect measure of motivation. So we can say that people are motivated if they contribute to GitHub a lot. We can just make that assumption, but we don't know the details of it. You know, maybe they're doing it for a class project, which is motivation, but it's not the same as being like kind of motivated by the problem. And that's the problem. We don't really have a lot more information to go on than that. And so simulations based on these indirect measurements may not accurately reflect the dynamics we find in actual open source projects. So we can only assume that if they're contributing a lot to GitHub, their motivation is high. And that motivation carries over across, you know, time. So they're always highly motivated essentially. Um, we also have a problem with performing parameter inference on complex nonlinear models. So the problem with agent-based models, and we're going to use agent-based models as sort of like um, uh, a baseline here. We're going to always be modeling our agents on agent-based modeling um surfaces in a in frameworks and we're going to be using things like reinforcement learning and active inference to model these cognitive So we have this we're still working with agent-based models. We're just doing different things with them. We're we're building these hybrid models essentially. So the likelihood or probability functions for most agent-based models can rarely be evaluated in practice. So we can build likelihoods of motivation, but we don't know what the conditions are. We couldn't build like a model with conditional probabilities unless we went out and did more experimentation with people who work with open source projects. And even then, it wouldn't be complete in terms of getting data for, you know, across a wide range of projects. So we have to keep that in mind when we we're talking about these models is that they're incomplete in terms of their ability to characterize every behavioral state. So these are our Google Summer of Code scholars. So this project was built around Google Summer of Code getting students over the summer having them produce. We've done this over uh the last four seasons from 2022 to 2025 and we have uh had six people in our team who have uh come in through this route and each of them have contributed something different. So Brian Mccoral his pictures right there on the top he worked on active inference and collective active intelligence thinking about active inference and collective behavior. Pam Shogul who's next to him there um he learning and uh building Q-learning/ agent-based models as hybrid models. So he was interested in agent-based Q-learning. Harvey Roger Gopolon uh implemented agent-based modeling in MISA which is a Python package which allows us to do more than say working with something like uh Net Logo. uh you know just having that framework where we can actually build uh more sophisticated tools. Shupam Sony built uh did some more work on uh multi-agent reinforcement learning in this Q-learning agent-based hybrid model. Sarah Bastawala who's right there next to her name uh developed the uh LLAM OSC which is large language models with agent-based models. So these are large language models that create sort of tasks for agents in an agent-based model. And the tasks of course are conversational and sort of textbased. So if you want to create a bunch of issues for your agents, you create them with the large language model. You create conversations with the larger language model and then the agents work from that. So it's very interesting work, very exciting work. Um and then Vidi Rodria uh she's worked on reinforcement learning but a different sort of approach to it than the multi- aent approach. She worked on what they call SARSA which is a a model of inputs and outputs and then that merging that with agent-based models. So it was a hybrid model. These students used a number of tools to implement this stuff. uh they for the multi-agent reinforcement learning at the upper left they use temporal difference learning Q-learning and SARSA we'll get into those details later um uh some of our students also used MISA/Net logo this is at the upper right so MISA is implemented directly in Python net logo is a classic uh package that has a lot of prepackaged models that you can implement um and then you can build from there So you know this offers a lot of flexibility for implementing the guts of the agentbased model. Uh lang chain of course is a large language modeling tool or toolbox. Um and that's at the lower left and then the lower right is uh pymdp which is of course the tool box for active inference. So let's talk about the OSDC first. This is a model of collective active intelligence. So this is our active inference approach to open source sustainability and in this case we're characterizing the activity of our computational agents using a markoff decision process in Python and that's our PIMDP or MDP in Python. Uh this uh model implements a multi-arm bandit. So it focuses on explore exploit problems and as we'll see the SARSA approach also explores multi or explores this sort of uh collective behavior is a multi-arm bandit problem. Basically a multi-arm bandit the name of a multi-arm bandit comes from when people pull the lever of a slot machine. So you're like priming this stochastic process and you'll have like in front of you if you've ever gone to a casino you'll see people sometimes put like five slot machines at once and they'll pull the arms down and then see what they get. That's what you're doing in a multi-arm bandit and you're trying to figure out whether you should explore a space or exploit a space. So if you think about that in terms of issues, should you explore an issue deeply or should you explore uh more issues more uh sha in a more shallow way? Uh we implement this in different ways so that we can look at that as a as a sort of a factor for driving open source project behavior. The second point is building a statistical script description of the mind. So this of course is the uh the the internal um components or the internal model of each agent. So we want to build a statistical description of that mind. We want to give it beliefs and we want to have you know have each agent minimize its own free energy. Um, and so this is something that of course we also want the agents to sort of be able to figure out what other agents are thinking to so that they can match their beliefs and minimize their free energy together. So this is kind of where we're going with this in terms of the collective part of it. And then of course when we reduce our error, it reinforces causal relationships that are necessary to complete the project. of getting the agents to see what the causal relationships are between different parts of the project. If we can reduce their inferential error, then we can get all the agents to move in the same direction. So this was implemented using in interactively. So this is the link interactively. Uh it was used to do two things. Number one is to implement decision problems in a single agent as we said. So each of these agents have some decision process going on inside of them. And then two is to implement a theory of mind to infer the state of neighboring agents. So the agent on the left is trying to characterize the state of the agent on the right because the agent on the left doesn't know what the agent on the right is thinking. And if they're both thinking the same thing, they're able to maximize utility in the same way, then they'll be able to cooperate better and uh do things like divide and conquer. So this is where you have uh this agent is asking this other agent is this what you're thinking and they say yes. So that's that's the way this project worked. Um and it has ties to the uh reinforcement learning that we'll talk about next. So in the in the context of an agent-based model this is an agent based grid. Again the activity of one agent influences activities of their neighbors. This is typical of an agent-based model. So you know we expect those uh those internal states to spread as they're able to infer each other's state and we get these blocks where all the agents are doing the same thing. But we also get this spatial distribution of states and because this is a spatial array we have things like adjacency and distance that we can characterize the sort of spread of these ideas. So you know sometimes you'll have a block of contributors do thinking one way and another block thinking another way. It's either thinking the same way about a problem or having sort of the same level of motivation or whatever. If theory of mind is achieved, two neighboring cells should co-activate. So that's where they share covariance in their activity. And these are, you know, pairs where you can see that they're kind of activated together. Another thing that um Brian Mccoral worked on the active influence part. So, another thing I think Brian Mccor was interested in and it's kind of on the cutting room floor now, but I bring it up because it really does have relevance to sort of these internal states in the mind and that is Mick Ashby's ethical regulator. So, Mick Ashb is the grandson of WR Ashb who came up with the ever good regulator theorem in cybernetics and Mick has been working on this ethical regulator and super ethical systems idea. um he published a paper in 2020 on this and he has this whole model of how we can encourage ethical behaviors and this goes well beyond active inference but it shows you how like we can take a simple active inference model and it kind of think more deeply about things like ethics in projects. So do people act ethically in the project? Sometimes projects get blown up because people act unethically. They steal credit for ideas. they cut other people out of of the project or they, you know, they're not honest about what they're doing. Uh or, you know, sometimes they try to make money off of an open source project when the license explicitly states otherwise. So, we want to encourage ethical behaviors and we want to make sure that those are all there in place. And we coordinate our division of labor and resources amongst contributors so that everyone's acting ethically. And as a model of cooperation, we have to take into account fairness and efficiency. So when we talk about regulation, then if we look at our agent-based model, we have these different types of uh regulatory behaviors. So if if one agent influences another, we can have the zeroth order behavior. If agents influence each other and they have this theory of mind, there's a first order behavior. And if multiple agents are influencing each other uh collectively, that's a second order behavior. So there's also the sustainability auditing tool which was worked on after the project in 2022. This was implemented in net logo web. So this is an agent-based model there. And and this was kind of developed out of the active uh approach. And so in this case we're modeling two variables. We're modeling churn and we're modeling maximum sensory probability. So churn of course is the turnover in the number of agents. High churn suggests that the community has a low degree of cooperation, a low degree of theory of mind. Then maximum sensory probability. Remember we need to have some sensory information to make active inference work and that is the probability of a given signal from other agents or the environment. So we have to have a signal from other agents of a signal from the environment and then that's what activates sort of the agents cognitive state. So this is the agentbased model for our multi-agent reinforcement learning approach. This is modeled in net logo. We have our agents as stick figures and they have uh multi-agent reinforcement learning you driving their behavior their internal state. Those have these squares which are the resources of the project and we can populate this with a large or a small number of contributors. We can then divide them up into categories new maintainers contributors and then have solved issues and unsolved issues and the idea is that they solve all the issues to maintain a sustainable project. And then we have these different issue sizes. So earth shot, moonshot and Mars Jupiter shot. These are just different sizes of issues. Earth shot issues being single issues, moonshot issues being a collection of issues that are interdependent. Emerged Jupiter shot issues that are things that are kind of like multiple different sub projects basically. So a little bit about multi-agent reinforcement learning. This is a a simulation where we use multi-agent reinforcement learning. It's a population of agents acting on a common grid world. The agent is motivated by its owner words. Its actions advance its own interests. So in indiv in individual terms, it's self motivated. But then sometimes the of course those agents have to interact with their neighbors with other uh agents in the population. And those interests can either be coordinated or mutually opposed. So we can see here that we have these two different colored agent populations and they're kind of mixing over time. We can model different scales of project size using multi- aent reinforcement learning. So this shows the prevalence of earthshot issues, moonshot issues, Mars Jupiter shot issues and accepted pull requests over time. And you can see that they kind of increase with time. And this is just one run of the simulation. So again, our Earthshot issues are single issues. Our moonshot issues are issues that can be broken down to multiple single issues. And then our MS Jupiter shot is a problem that can't be broken down directly into multiple issues but need like you know different sub projects to be completed before the next sub project and as such they're sort of modular. So we use that to look at multi-agent reinforcement learning. Um this is where we also used um multi-agent reinforcement learning with Q-learning but also sustainhub which is this uh reinforcement learning with a SARSA model. So SARSA is state action reward state action. So it describes you monitor the state you take an action you get a reward you monitor the state you take an action. And so that's kind of the difference. We have Q-learning for multi-agent and the SARSA for the sustain hub or the single reinforcement learning. And so this is another simulation. Um this I think is also from uh Shupom where he's simulating Earth issues, moon issues, Mars issues and then accepted pull requests. And the accepted pull requests are always sparser than the other issues being solved. So that's, you know, it's kind of typical because you're combining all your issues into a single pull request. So this is uh kind of a way to monitor or at least model things going on in a community. The question is is, you know, is it something that is sustainable or not? And that's a little bit harder to determine. So let's walk through some of these methods um for reinforcement learning to see maybe how they compare with um active inference. So we start with Q-learning. So for any markov decision process, we identify an optimal action action selection policy. So what we want to do with that action selection policy is maximize the expected value of the total reward. Okay. So then this determines the quality of a given policy and action a subt with respect to state s oft provided a learning rate of alpha. So we have this equation down here where we have these different components. we have alpha times Q uh where we're evaluating the action and the state uh the quality of each action and state that evaluates the current action selection value. We add that to this estimate of optimal future values or the state of highest quality. And so we're looking for the maximum quality and then that's our kind of the way the direction we're going in Q-learning. trying to find that maximum uh future value the maximum quality of a given set of policies. Now the multi-arm bandit is also used then in reinforcement learning like we saw with active inference. It's used for task allocation role matching given prior performance. It encourages specialization while allowing for flexibility and it utilizes what we call on policy temporal difference learning which we'll get into next. And so it's it's important to recall the comm different community member types maintainer contributor innovator knowledge curator and we can use this to think about how to allocate different tasks to these different classes of contributor or these different classes of project member. So temporal difference learning can be stated as a state value function of a finite state markov decision process under a policy pi. Uh v subpi is equal to the state value function of the markoff decision process with states rewards and a discount rate importance which states the importance of future rewards. Uh so we have instead of our learning rate we have a discount rate and then we estimate uh v pi s which is the function for the state value and we evaluate this policy pi we obtain the reward are and we update our value function. Now some of the metrics that were created in the reinforcement learning um work and this was done by VD Rahira um we he she developed five different metrics that work in this in this uh context. So the first is the harmony index. So remember I said there are different things we want to measure and then there are different things that we can sort of have as like an output. So the first is the harmony index. This captures workload balance and task success rate. This is basically measures how well balanced workflow workloads are, how much cognitive load there is for each agent and then the success of agents at those tasks. Then there's the resilience quotient which measure measures adaptability during disruptions such as dropouts and workload spikes. means basically how resilient are you to different types of um you know disruptions in your work in the project. There's reassignment overhead which tracks in efficiency caused by reassigning tasks. There's initial an initial density of agents. So basically it's uh where if you have a dense setting of agents to start with, it's going to give you a very different result than a very a very small number of agents. And we can use that to sort of normalize a lot of the uh heterogeneity in open source projects. And then the contributor maintainer vision. So if we don't have a good vision, agents won't know if there's anything for them to do. But if the vision is there, then at least they have in their mind sort of a mental model that they have something to do. And so they go to find specific tasks to do to solve specific issues. The next thing I want to talk about is how large language models were used. So um this was Sarasawala of course uh this is Llama OSC. This is an open source large language model uh based on the llama platform and it's used to generate project tasks and issues. And so she basically developed this large language model agentbased model hybrid platform or this hybrid model. And that gives us a model of how to act upon tasks and project structure. So you would generate like all these different issues using a large language model. You generate conversations and then the agents would know how to act from that and give the project structure. Uh the workflow is in lang chain that's in Python and this allows for large language models to be used as a reasoning engine where you can have sequences of actions where agents decide what tools to use and what order to use them in. So and and in this work you know she had like two basic models of open source communities. So instead of thinking them as very heterogeneous structures or thinking them as idealized structures, she thought of them as either a benevolent dictator model or a meritocratic model. And so in the benevolent dictator model which is characterized by Linux and the Linux kernel, you have some single person who drives the project forward with their deep technical skills. So Linux Torvalds is the benevolent dictator in the Linux example. And then the meritocratic example is indicative of the Apache Foundation where anyone can engage at any level that they want. And so there's no leader, there's no dictator. It's just kind of like people do what they can in the project. I want to get into the way we're thinking about integrating a lot of these models together. We want to bring them together in the next phase of this project. So you know one of the things we need to understand are the dynamics or what does a team look like over the course of a project. So sometimes you know people leave projects sometimes people join projects. We have these sort of lack of altera dynamics in our agentbased model and that's you know kind of an interesting thing because none of those are described by any of the models that we've we've talked about. Um so this is a a point for future work. Uh we also know that like some of these dynamics because you get a lot of people leaving a lot of people joining a lot of flux and churn that it can create vulnerabilities as well as opportunities. And so if we want to know how to sort of build a model that can help us understand maybe what those opportunities are, those vulnerabilities are there are some ways forward. So one way is to build these sort of what we call metabrain models where you have different you take different models and you put them together uh different layers of different types of representations put them together as a structure that sort of you know communicates selectively between the models but processes different types of information. So in this case we think about the large language model as handling semantically defined tasks. Active inference is handling sort of the internal brain state of an agent. And then the multi-agent reinforcement learning as the motivational structure which means that you know you have this brain state and motivational structure working together on tasks that are semantically defined and then maybe an agent-based model as sort of the collective driver of collective behavior. So you have this agent-based model sort of coordinating all these different brains composed of large language models, multi-agent inference models. So I want to finish with sort of a thought from Jesse Parent and he's another person in our lab who's contributed to this work. Um, and he's been thinking a lot about project management and a lot about how we often create tools that don't go anywhere or we have like initiatives that kind of die because no one finds them relevant. So he says sustainability translates to problems in academic communities. Sometimes if you go to an academic community, you find that someone earns like a grant for tool building. So they get funding for tool building. They build these great tools and then the funding runs out and so they have this open source tool that they put out in the world. The funding runs out and it becomes an unmaintained tool eventually to die uh a premature death. So that's one path that's uh suboptimal I think from the standpoint of open source sustainability where you you get money to build something you build it and then it dies. The other example that he gives is that someone builds a tool and because it's not really relevant to anything, the tool is ignored and not maintained. And so then that's that's bad. But it gets worse because then someone else builds a similar tool. And that dies because it's it's ignored and not maintained. So you end up with this graveyard of tools that are all maybe pretty decent, but because no one's finding them relevant and no one's putting the money into maintenance, you people keep reinventing the wheel and then the wheel breaks apart. So that's not good. Um, so you know, Jesse makes this observation that academic communities more generally will often build out infrastructure which goes nowhere and is disconnected from real world impact. So that's something to think about if we're thinking about the active inference institute or our own lab or other institutions where people build infrastructure. It goes nowhere or it's not connected to the real world in in in relevant ways. And sometimes you just can't, you know, you can't avoid it because you don't necessarily know what people will find relevant. But that's part of sustainability that we haven't addressed and maybe is sort of the secret sauce in this whole equation. So, thank you for your attention. Next up is going to be Andrea Hotz collidoscopic cognition. How the brain can be modeled and use modeling without having a model. It is a 52minute video. Here we go. Good afternoon everyone. Today I want to share a paper with you radical embodied relation at any scale from remembering to navigating. My name is Andrea Hayyatt and this is for the active inference institute symposium for 2025. So this is a bit of a radical embodied understanding of mine and it works at any scale as the paper says. So it's supposed to expand some of our fundamental assumptions about brains, memories, minds, and bodily movement and how all those relate. So the title of this talk is kaleidoscopic cognition. How the brain can be modeled and use modeling without having a model. I know that sounds paradoxical and if you know me, you know I like paradoxes. Holding paradox is sort of the point of a lot of this work, but that title might seem impossible actually. So, how can the brain model without having models? And how can we use representations without them being stored somewhere? These questions have been really important in neuroscience and philosophy. Some might even say they've been plaguing neuroscience and philosophy of mind lately. Uh like we just can't move past them. But there's recent research on the hippocampus, which is the part of the brain that I studied. little seahorse shaped structure on both sides of your head in both loes and it's um forcing us to rethink some basic assumptions about a lot of these words that I've already mentioned. So what we're discovering is that the brain or I would rather actually talk about the central nervous system and the bodily sensory movement of it all. But let's just say the brain doesn't work like a computer with shared stored files. Instead, it operates more like a kaleidoscope. And I know that might sound crazy, but actually it's a very helpful strange new vision that we kind of need right now. And that it feels strange is probably a good thing. So it's this idea of creating shifting patterns through different but direct refraction of experience. So if you think of the encounter and then the body and then whatever that encounter is becoming as part of the body and sort of going back into the encounter as then you can think of it more as something prismatic or kaleidoscopic. So this framework is coming from neuroscience at least as I read certain studies and as I've been part of certain areas of that field and especially grid cells play cells these kinds of things which are in the hippocampus and I think it has profound implications for how we understand these words that we've been taking for granted like memory thinking but also navigation and everything that we've ever thought about as cognition. not only in humans but when we try to understand this activity that unites us uh across different life forms. I'm going to be talking about the hippocampus as a kaleidoscopic function, a way of trying to visualize some of this new research which is moving us beyond either or patterns that we assume and that I think we're all sort of stuck in at the moment in brain research even though we say we're not. I want to push us a little bit towards this kaleidoscopic or prismatic idea of cognition and hopefully that'll help us move beyond these representation wars and of talking that have sort of paralyzed us in cognitive science. The idea is to synthesize insights from four key papers that when read together will reveal something pretty extraordinary about cognition. And so these are the four papers and the first one is my paper, the one that's kind of framing this and I guess the sort of foundational text for it. And in there I argue that recent hippocample research forces us or allows us to reconceive what a neural representation is and to think of it as a communication communicative assessment rather than something stored or findable or some sort of hieroglyph hieroglyph in the brain. And I introduced the concept of way making which is just a way of talking about this process that we call cognition or this process that we call locomotion but on the same track. There's also a brand new nature neuroscience paper by Penn and colleagues which came out around the same time which talks about well it's called path integration in the hippocampus by shifting reference frames and I really see this as just crucial empirical work and just great timing that it comes out now because it has a lot to do with what I'm trying to discuss here and it's showing us that grid cells don't maintain stable patterns but are constantly shifting between reference frames even though there is a stability. So there's a paradox built into this, but it gives us a good way to understand this notion of kaleidoscopic modeling that I'm trying to introduce here. And then we have the Lens and Clark and Fristristen paper from 2017. Sorry if I said somebody's name wrong there. The active inference approach to ecological perception because I think there's a lot of ideas in there that fit very well with this. And also just the mathematical and theoretical framework needs to be built up um for how we might do this. And I think active inferences probably and especially also basian ideas in machine learning and so on have already started figuring out ways to maybe think about layered mathematics in the way that I'm talking about here. But also just the ecological side of it is uh very fitting that organisms engage in modeling behaviors without storing models. And of course sometimes we slip up and say models are in the brain but if you really look at what is actually being said it's not that the models are in the brain stored somewhere. It's that the whole body is modeling. This last paper too which inspired my first paper and it's from Yakob Belmont and the Doer Lab which is where I worked at the Max Blanc Institute. It's called navigating cognition spatial codes for human thinking and it shows us that the mechanics we use for physical navigation are the same ones we use for abstract thought. Even though thinking and geographical navigating can't be resolved as like the same thing, we can use the same metrics to understand them. But those metrics cannot be based on these either assumptions um that you know like we've been sort of using whether it's like global and local or even mental and physical like conceptual space, physical space, memory, navigation those um ways of talking and our use of the word representation as we're talking about them in cognitive neuroscience has to shift. So all these papers together point towards a unified understanding of cognition as kaleidoscopic or at least as constellatory is another word I use demanding we find ways to model it kaleidoscopically rather than binary and and these binary assumptions that frame so much of the way we talk about all of these uh topics the way we even use the words and the way we model them even if we don't realize that's happening. So by introducing this kaleidoscopic cognition, I'm trying to push us a bit into a dynamic immediate irreducible understanding um of under of what we mean by representation which is actually a communication. So we also understand that we're going to do this through communicating about the patterns of regularity and that this communication isn't findable but it is representational. So um that's a little tricky. This holding paradox idea is really good and also this kaleidoscope where you have similar patterns but based on the input of the light you can even think of it as prismatic I guess uh you're going to get different results um but those will all be tied to similar patterns. So again I've mentioned paradox quite a few times and you know the main paradoxes that motivated motivates this framework is representations without representations. Models without models or modeling without models or representing without representations maybe is a better way to say it. But traditional neuroscience always assumes that cognition requires stored models. Internal representations that the brain builds and maintains. So when you remember something from your childhood, your childhood home, you know, there must be some neural structure that is appearing in your brain that represents that home, right? So when you navigate through a city, you have some kind of map in your head. This is just the way we talk about it. Most people of course don't actually think you can, you know, pull a little map out of your brain or actually find some kind of structure in the brain that's going represent um home in some kind of onetoone way. However, we still talk about it that way in our papers and we still assume this language. So the reality is that organisms are navigating these complex environments without anything that looks like explicit stored representations. And in fact, you know, there's so much going on all the time that any of our actual experiments is just looking at a tiny bit of it. So if we find some brain activity that corresponds with a particular memory that we have, um that's actually just one little snapshot in a way uh of something that we were directly looking for. And this is still on this flatland kind of way of thinking about what we're doing in these experiments and also what the body is doing in relation to the brain, the environment and all these um ongoing movements that [clears throat] we are. So animals find their way home. They recognize patterns. They make predictions you know including humans of course. And yet we observe them and we watch the activity of you know rats who've been so helpful for us for example in neuroscience. Then the only way we can really communicate about what's happening there is through creating these representations, images or um you know that we we record the cell activity and then we see pictures of it. Um and all of that is representational but it's not inside the rat. It's us creating things to understand this ongoing movement and that gives us a visual, a video, a screen image, whatever. And those are the representations and those are the communications and that is a communication. I mean the the screenshot or the recording is meaningless except as communication and it's important. We're doing that obviously in order to better understand this whole process and ourselves and it's uh a very real thing but it's not findable somewhere. So the hippocampus is really central to making more scientific rigorous sense of what I just said and getting over this assumption and use of language that's binary and that does talk about representations as if they're inside brains rather than as if there are communications towards understanding brains. So for a long time we actually literally looked for Ingrams in the brain stored memories and um that was the hippocampus was like you know the the sort of heart of where all that came in the 50s looking for for Ingrams and memories because of HM and so on and then in the 70s we started thinking of the the same area of the brain as the brain's GPS because John found cells there that fired in relation to what um the animal was moving through. So you know you got play cells and grid cells and things like this but this recent research is even showing us that actually these cells don't maintain stable patterns. They're always shifting remapping reanchoring. We don't have like a representation in the brain that corresponds to something outside. Instead we have this ongoing communication this process this something like light coming into a prism and being refracted. You know that's where this kaleidoscopic comes from. That shifts as the encounter takes place. So all of these cells are shifting, remapping, re-anchoring as the encounter is changing. Just as if light comes into a prism, it's going to change in different ways without any kind of, you know, stopping and starting there. Even though when we study it, we have to look at it that way. So this is where this idea of kaleidoscopic cognition comes in. And it does sound fantastical, maybe psychedelic, but it's actually a really accurate way to begin to envision this. And that it feels strange and even hard is actually part of that because what's more like the vision, and again, we're talking a bit in metaphors here, but it's more like something is refracting between and as the encounter and the organism, which includes this whole nervous system. And that is the guiding action. That is the perception and action. And obviously this is really hard to talk about. Um, but that's what I'm trying to do here. I think we have to find ways to sound strange, be strange, and move towards finding other ways to really understand what's happening here. So, we create recordings and photos and videos and representations to understand this there. Those are real and communicative. But how could we create more kaleidoscopic ones? How can we begin to use our technology in ways that are more like this process so that we get used to thinking like this instead of uh creating things that are still stuck in this um binary sense of as if we've taken a picture of some kind of whole thing that is mental that is a memory. So before we get to that, let me just talk about this one question about representation which I've brought up. But how can the brain have representations and not have them at the same time? How do we hold this paradox without falling into something like computationalism or just total relativism which is you know the always the trap here? So that's where this hippocampus paper comes in which uh you can find easily and read and it's arguing that representations are communicative acts that there are ways we assess and communicate about these unbroken intimacies not things we like find stored in brains and I think that's something that's pretty obvious uh in neuroscience but we don't talk about it as if it is and that is a very big obstacle actually and of course there's been tons of representation debates I mean we've really wasted or maybe not wasted but so much energy and time has gone into getting stuck and going round and around in loops over this idea you know even with active infrance I think we all know this idea of talking about models is in the in the head becomes a trap and uh we just it's time to get out of it so that's why I'm uh not in this paper but in this presentation presenting this idea of kaleidoscopic or constellatory cognition to kind of just try to pop us out of that way of um positioning ourselves in cognitive neuroscience and in the way that we do our models. So we can understand these immediiacies like are refracting differently according to the position from which they're measured and they're not in the thing that's being measured. So weather maps are a really simple good example. We create a weather map. It represents weather patterns, but the map doesn't contain the weather. Even though it's tricky how we confuse those things with what weather is, but you know, there is no like stop and start to a storm in the sense of you can just pluck it out of of all the rest of the ongoing atmosphere or something like that. Uh it's you know and we need and create these representations of things like that so that we can communicate about them so that we know okay a big hurricane is coming and um we need to you know evacuate or whatever. These are really important representations that we create. But the map is the communicative tool that we use to understand and p predict those atmospheric dynamics. The the the dynamics of course they're dynamic. They're go ongoing. They're unlocatable. You don't find partly cloudy somewhere. You don't find 70° somewhere. You know that those are things we rep those are representations. And that's the same thing that we're talking about when we're looking at brains. It's something like 70° when when we're talking about a representation in the brain of memory or anything else. So that process that we call weather refracts differently from every position on the planet too. We could think about that and how depending on where you are, you're going to see that refraction differently. it's going to be assessed differently even though of course it's real what what is being assessed and it's also ongoing and also your assessment of it on one particular day from one particular position is going to be different than someone else from another in another day and that doesn't mean any of it's less real and in fact it'll still probably mean there's a whole lot of regularities that are shared what you're calling a storm and what they're calling a storm will have a lot of overlap but it won't be the same thing and it won't be something that we uh you know have found and then move on from. Uh so neural representations work that way too. When we look at an fMRI image or we record neurons, we're not seeing stored memories or cognitive maps inside the brain. We're creating representations, measurements, assessments that help us communicate about this ongoing central nervous system bodily environmental dynamic which is pretty wonderful actually and which we really do want to understand and that's why what we're doing is so important in all these fields. But the hippocample literature shows us that cognition and navigation, for example, which have kind of been fit into these different sides of a binary of mental and physical, for example, just to kind of caricature it a little bit. We've assumed for a long time that those needed to be reconciled somehow, that they were somehow different things and how could the same part of the brain be be doing them? But recently, we've started to understand, okay, they're they're just like refractions of the same process. At least that's how I'm putting it here. There are different assessments of this embodied way making that is happening uh all the time and depending what we're looking for or where we're standing or what day it is, it's going to be a little bit different. But also there are going to be regularities that we can [clears throat] uh that that remain that help us understand it better if we communicate about it through representations. So when you remember a past event, you're kind of moving through or navigating conceptual space. You could think of it like that. you're using the same mechanisms, right, to move through that sort of space that that happens when you're moving through geographical space, but you don't those aren't the same thing. It's just that, you know, there's similar regularities there in terms of what is happening. So, I use this word way making, which is come comes from Dao's philosophy to capture how that cognition and locomotion can be continuous, can be the same process. Even though when we're talking about something like remembering our childhood and when we're talking about something like uh walking to the park, we're talking about different things and there's no need to try to resolve those. The same way we wouldn't try to resolve wavelengths, you know, that have been refracted from the same light source or the same Yeah, we could think of it somehow as multiensory prisms if we can, but you know, we can't measure and find that in some absolute sense that there is a memory or there is a a little diagram in the head that's going to be um found every time we go to the grocery store. So anyway, the hippocampus doesn't store maps. It's operating kaleidoscopically. It's refracting sensory encounter into bodily action. At least that's the kind of wild idea I'm throwing in on this presentation. I haven't tried to put it in a paper, of course, but at least not yet. But I do think we make these maps, these representations to try to understand what is multi-layered, multi-dimensional, refracted in our measurements of it, even if in itself it's not. So just, you know, thinking of it like that can help us kind of avoid some of these traps of arguing about pure computationalism versus pure relativism more.